STATE OF ENVIRONMENT REPORT
MAKINDYE SSABAGABO MUNICIPALITY

2025-2027
JUNE 2025





Approved by
The Natural Resource Committee
under Minute Number ENRC/004/06/2025 ,
MSMC/COU/03/26/06/25

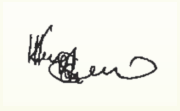
FOREWORD

Makindye Ssabagabo Municipality stands at the forefront of Uganda's rapidly urbanizing landscape, reflecting national trends where urbanization rates are among the highest in Sub-Saharan Africa. Formerly a cluster of vibrant townships, Makindye Ssabagabo has transformed into a dynamic municipality characterized by expanding administrative boundaries, substantial population growth, and increasing demand for public services. Established under the Local Governments Act, 1997 (Cap. 243) to empower local governments to manage resources and deliver key services, Makindye Ssabagabo Municipality bears the responsibility to plan for and protect its environment as an integral part of local development.

The Municipal State of the Environment Report (MSOER) provides a timely and comprehensive assessment of the state and trends of the municipality's social, environmental, and natural resources. This report explores critical sectors ranging from demographic dynamics and infrastructure to land use, water and wetland management, biodiversity, energy, waste, and atmospheric conditions. Special attention is given to challenges such as rapid population influx, informal settlement expansion, waste management constraints, wetland degradation, air quality concerns, and increasing pressure on land and ecosystems. At the same time, Makindye Ssabagabo's location within the Greater Kampala Metropolitan Area presents considerable opportunities with emerging investments in new industries, potential for eco-friendly urban planning, and prospects for harnessing underutilized land for organized development.

Preparation of this report has involved thorough analysis, scenario exploration, and widespread participation of stakeholders from government, civil society, and the private sector. The findings reinforce the urgency for integrated solutions that respond not only to local environmental challenges but also align with Uganda's Vision 2040, National Development Plan IV (2025/26–2029/30), and international commitments such as the SDGs. Decentralized planning, as prioritized in the NDP IV, underscores the vital role of municipalities like Makindye Ssabagabo in translating national strategies into locally relevant action.

Addressing today's pressing realities, ranging from climate variability, changing population structures, to shifting development pressures, will require collaborative partnerships and a shared vision among all stakeholders. This MSOER sets out actionable recommendations tailored to address the municipality's unique circumstances and prospects. Makindye Ssabagabo Municipality welcomes all partners, investors, and citizens to actively engage in implementing these solutions for a more resilient, inclusive, and sustainable urban future.



Ssemwanga Godfrey
His Worship, The Mayor
Makindye Ssabagabo Municipality

ACKNOWLEDGEMENT

The preparation of the Makindye Ssabagabo Municipality State of the Environment Report (MSOER) 2025 has been a collaborative effort, made possible through the commitment, expertise, and support of numerous individuals and institutions.

We express gratitude to the World Bank, through the Greater Kampala Metropolitan Urban Development Program (GKMA–UDP), for the support in producing this report.

We gratefully acknowledge the valuable contributions of the Municipal Natural Resources Committee and the various sectoral departments whose insights informed the compilation of this report.

Special thanks go to our development partners, private sector actors, and academic institutions that actively participated in consultations, shared information, and offered technical expertise.

We also recognize the dedicated community members and local leaders whose local knowledge and field observations enriched the findings of this report.

Finally, we appreciate the unwavering dedication of Ardent Services International Ltd, which coordinated data collection, analysis, and report writing. We hope that the collective effort demonstrated in producing this report will be sustained in its implementation, fostering a cleaner, healthier, and more resilient environment for present and future generations.



Richard K Monday
Town Clerk
Makindye Ssabagabo Municipality

EXECUTIVE SUMMARY

Uganda is experiencing an unprecedented urbanization, averaging a rate of 5% annually. As part of the Greater Kampala Metropolitan Area, Makindye Ssabagabo Municipality has rapidly emerged as Uganda's fastest-growing municipality. The municipality is made up of three divisions- Bunamwaya, Masajja, and Ndejje, which have become both a critical residential hub and a focal point for commerce, trade, and emerging industries. This explosive growth, combined with internal migration and peri-urban expansion, has put immense pressure on Makindye Ssabagabo's physical environment, demanding urgent and strategic municipal responses.

This 2025 Municipal State of Environment Report (MSOER) is produced to support evidence-based planning in line with the National Development Plan IV (2025/2026–2029/2030), which prioritizes sustainable industrialization, inclusive growth, and resilience-building. The report is the result of broad collaboration, coordinated by technical experts, municipal authorities, and direct input from community members, ensuring a holistic and representative analysis of Makindye Ssabagabo's environmental and socio-economic status.

This Report mirrors the municipality's realities, beginning with the population dynamics, land use transitions, and the environmental challenges that shape everyday life. It proceeds to a detailed methodology, sector profiles, and then explores Makindye Ssabagabo's unique biodiversity, land cover evolution, climate patterns, and the governance frameworks guiding municipal action.

Key Messages and Policy Implications

Population Growth and Urbanization

Makindye Ssabagabo Municipality has experienced dramatic population growth, doubling from 298,800 in 2015 to over 563,900 by 2025. This reflects an average annual growth rate far above the national urban average and signals powerful demographic momentum driven by both natural increase and migration. The population is notably youthful, presenting both an economic opportunity and a challenge for education, employment, and urban management.

The implications are profound:

- Explosive demand for land and housing is resulting in micro-subdivision of plots, a rush for land purchases, and a steep increase in informal settlements and peri-urban sprawl.
- Congestion and service deficits are common, especially in fast-growing zones such as Bunamwaya and Ndejje, where schools, healthcare, and utilities face stress. Without robust planning, congestion, public health risks, and unemployment will intensify.

Land Use, Biodiversity, and Ecosystem Health

- Land use transitions over recent decades have been dramatic. Built-up areas expanded from just 6.6% in 1986 to 35.2% in 2025, a 435% increase. This expansion has occurred at the direct expense of cropland, grassland, and, most concerningly, wetlands and forests. Wetlands in Makindye Ssabagabo Municipality, which include Kaliddubi, Kyetinda, Lufuka, Mayanja–Lufuka, and the Lubigi Channel, have shrunk by more than half, now covering only 6.5% of the land area.
- Degradation of wetlands and forests is leading to a loss of vital ecosystem services, including water purification, flood control, and wildlife habitat.
- Encroachment, conversion for settlements, and clay/sand mining are the principal drivers, with certain wetland systems (like Lubigi Channel) losing nearly all their natural vegetation in just the past five years.

- Biodiversity loss is escalating, undermining resilience to floods, drought, and climate extremes. Local communities have observed heightened flash flood frequency and declining water quality as a direct result.

Climate Variability and Environmental Risks

- Makindye Ssabagabo's rainfall regime is increasingly variable, with decadal analysis showing alternating periods of drought and intense rainfall. The average annual precipitation has increased, with recent years registering up to 1,760 mm-making the municipality relatively wet, but highly prone to inter-annual extremes.
- Increased flood frequency and stormwater surges are linked directly to wetland loss and unplanned drainage. Soil erosion and land instability pose additional risks, with infrastructure like roads and houses affected by frequent flooding and cracking.

Air, Water, and Noise Pollution

- Rapid urbanization and industrialization have led to widespread environmental health hazards:
- Air quality is severely compromised, with PM2.5 and PM10 levels exceeding safe limits due to emissions from traffic (boda bodas, taxis, aging trucks), open burning of waste, and small-scale industry.
- Noise pollution is pervasive, especially along main transport corridors, near industrial sites, entertainment venues, and places of worship.
- Water resources are at risk; many households still depend on unprotected sources, while wetlands and streams are increasingly contaminated by solid waste, industrial effluent, and sewage.

Waste Management

- Makindye Ssabagabo faces a mounting waste management crisis. Less than 60% of solid waste is collected, with frequent illegal dumping in wetlands, drains, and open spaces.
- Lack of a dedicated local landfill after the closure of Kiteezi has resulted in unregulated disposal practices, higher transportation costs, and environmental health risks.
- Poor waste sorting and inadequate infrastructure exacerbate the challenge, especially in informal settlements.

Socioeconomic and Service Sector Trends

- Education, health, energy, and transport services are under pressure from the rapid growth:
- Education: High primary school enrollment (~95%) is offset by severe overcrowding and low secondary school transition rates.
- Health: Good physical access is undermined by overburdened facilities and frequent shortages.
- Energy: Most urban households have grid electricity, but peri-urban and poor communities still depend on charcoal and firewood, fuelling deforestation.
- Transport: Road networks are expanding, but congestion and maintenance challenges remain acute.

Policy Recommendations and Outlook

- Implement rigorous land use planning and enforcement, focusing on curbing unplanned settlements, preserving remaining wetlands and green spaces, and promoting high-density, mixed-use development.

- Strengthen wetland and forest conservation via buffer zones, community-based restoration projects, and stricter control over sand/clay mining and backfilling activities.
- Upgrade waste management infrastructure-including investment in transfer stations, recycling, and community sensitization to achieve universal collection and safe disposal.
- Enhance urban infrastructure and service provision, targeting underserved areas for health, education, sanitation, and transport improvements.
- Promote climate adaptation and disaster risk reduction-including urban greening, improved drainage, and real-time environmental monitoring.
- Intensify public engagement and capacity building, empowering citizens, particularly youth and women, to participate in environmental protection and monitoring.
- Leverage partnerships under the GKMA-UDP and with the private sector and NGOs for integrated urban development.

This MSOER aims to provide a comprehensive overview of the municipality's environmental status. The information presented is designed to equip decision-makers with the evidence they need to make informed and effective choices. The technical team is confident that by implementing the proposed solutions, the municipality will progress towards a healthier environment and foster inclusive urbanization that benefits all residents.

Table of Contents

FOREWORD ---- i

ACKNOWLEDGEMENT ---- ii

EXECUTIVE SUMMARY ---- iii

List of Tables ---- x

List of Figures ---- xi

List of Acronyms ---- xii

1. Chapter One: Background ---- 1

1.1 Introduction ---- 1

1.1.1 Objectives ---- 1

1.1.2 Purpose of the report ---- 2

1.2 Makindye Ssabagabo Municipality profile ---- 2

1.2.1 Geographical location ---- 2

1.2.2 Municipal structure ---- 2

1.3 SOER Preparation Process ---- 3

1.4 The Legal and Policy Framework Governing Environmental Management ---- 4

1.5 Institutional Arrangements in Environmental Management ---- 9

2. Thematic Assessments ---- 10

2.1 Water Resources Management ---- 10

2.1.1 Introduction ---- 10

2.1.2 State and trend ---- 10

Surface Water ---- 10

Groundwater Resources (Springs and Boreholes) ---- 12

2.1.3 Drivers of water resources degradation ---- 14

2.1.4 Impact of water resources degradation ---- 16

2.1.5 Responses ---- 17

2.1.6 Recommendations ---- 19

2.2 Biodiversity ---- 21

2.2.1 State and trend ---- 21

2.2.2 Drivers of Biodiversity Degradation ---- 25

2.2.3 Impact of Biodiversity degradation ---- 27

2.2.4 Responses to Biodiversity degradation ---- 28

2.2.5 Recommendations ---- 29

2.3 Wetlands ---- 30

2.3.1 State and trend ---- 30
2.3.2 Drivers of Wetland Degradation ---- 34
2.3.3 Impact of Wetland Degradation ---- 36
2.3.4 Responses to Wetland Degradation ---- 37
2.3.5 Recommendations ---- 39
2.4 Waste Management ---- 41
2.5 Climate Change ---- 51
2.6 Air Quality ---- 59
2.7 Land Use and Soils ---- 67
2.7.1 State and trend ---- 67
2.7.2 Drivers of land cover/use changes ---- 68
2.7.3 Impact of Land use/cover changes ---- 70
2.7.5 Responses ---- 71
2.7.6 Recommendations ---- 73
2.8 Urbanization and settlement ---- 74
2.8.1 Introduction ---- 74
2.8.2 State and trend ---- 74
2.8.3 Drivers of urbanization and settlement ---- 75
2.8.4 Impacts of urbanization and settlement ---- 76
2.8.5 Responses ---- 77
2.8.6 Recommendations ---- 78
2.9 Socio-economic status ---- 79
2.9.1 State and trend ---- 79
3. Chapter Three: Integrated Assessment ---- 86
3.1 Environment and Agriculture ---- 86
3.1.1 Impacts of Agriculture on the Environment in Makindye Ssabagabo ---- 86
3.1.2 Recommendations to environmental challenges from Agriculture in Makindye Ssabagabo ---- 86
3.2 Environment and Industry ---- 87
3.2.1 Impacts of Industrialization on the Environment in Makindye Ssabagabo ---- 87
3.2.2 Recommendations to environmental challenges associated with industrialization in Makindye Ssabagabo ---- 87
3.3 Environment and Energy ---- 88
3.3.1 Introduction ---- 88
3.3.2 State and Trends ---- 88

3.3.3 Drivers---89
3.3.4 Impacts of Energy resources on the environment in Makindye Ssabagabo ---89
3.3.5 Recommendations to environmental impacts of energy resources in Makindye Ssabagabo ---90
3.4 Environment and Transport---90
3.4.1 State and Trend---90
3.4.2 Impacts from the Transport Sector on the Environment ---90
3.4.3 Recommendations to Environmental Challenges from the Transport sector---91
3.5 Environment and Education ---91
3.5.1 Introduction ---91
3.5.2 Distribution and Access to Primary Education Services---91
3.5.3 Distribution and Access to Secondary Education Services---91
3.5.4 Environmental protection opportunities by educational entities in Makindye Ssabagabo ---92
4. Chapter Four: Environment and Planning---93
4.1 Stakeholder Involvement in Environment and Planning---93
4.2 Environmental policy implementation and performance---94
4.2.1 Makindye Ssabagabo's performance on waste and circular economy---94
4.2.2 Makindye Ssabagabo's performance on air quality---95
4.2.4 Makindye Ssabagabo's performance on climate change---95
4.3 Improving environmental performance and policy implementation ---95
4.4 Challenges in Environmental Management ---96
4.4.1 Structural challenges ---96
4.4.2 Financial-related---96
4.4.3 Equipment related ---96
4.5 Framework for integrating environmental considerations into local development planning and decision-making processes---96
4.5.1 Monitoring and Evaluation Framework ---96
4.5.2 Indicator development and regular assessments---97
4.5.3 Feedback and Adaptive Management---98
5. Chapter Five: Future Outlook---99
5.1 Outlook components ---99
5.1.1 Identifying trends ---99
5.1.2 New developments, and potential risks or opportunities--- 100
6. Chapter Six: Recommendations --- 103

6.1 Conclusion ---------- 103
6.2 Policy recommendations---------- 103
Priority actions on Climate change---------- 103
Priority actions on biodiversity ---------- 103
Priority actions on waste management ---------- 104
Priority actions on Water resources ---------- 104
References ---------- 105

List of Tables

Table 1: Frameworks, Policies, Plans, and Acts relevant to the Environment in Makindye Ssabagabo Municipality——4
Table 2: Surface water quality status in Makindye Ssabagabo——**Error! Bookmark not defined.**
Table 3: Wetland Water Quality Status in Makindye Ssabagabo——**Error! Bookmark not defined.**
Table 4: Summary of the DPSIR Framework for Biodiversity——30
Table 5: DPSIR for wetlands in Makindye Ssabagabo——39
*Table 6: Perceived causes of climate change in Makindye Ssabagabo Municipality***Error! Bookmark not defined.**
Table 7: Perceived impacts of climate change on local livelihoods in Makindye Ssabagabo Municipality——**Error! Bookmark not defined.**
Table 8: Perceived impacts of climate change on ecosystems——**Error! Bookmark not defined.**
Table 9: Air quality ranges adopted from global standards——**Error! Bookmark not defined.**
Table 10: Projected population in Makindye Ssabagabo Municipality——79
Table 11: The Demand for Education in Makindye Ssabagabo——92
Table 12: Stakeholder framework for Environmental management in Makindye Ssabagabo——93
*Table 13: Policies, strategies, and plans on waste and circular economy***Error! Bookmark not defined.**
Table 18: Emerging aspects on the existing and future scenarios for adaptive capacities on environmental management——99

List of Figures

Figure 1: The DPSIR framework. Source: Burkhard, B. & Müller, F. (2008). ——— *3*
Figure 2: Water and Environment Institutional Framework ——— *10*
Figure 3: Surface water coverage in Makindye Ssabagabo ——— *11*
Figure 4: State of wetlands in Makindye Ssabagabo ——— ***Error! Bookmark not defined.***
Figure 5: Ground sources in Makindye Ssabagabo ——— *13*
Figure 6: Crested cranes and egrets in a wetland, dumped with plastics and rubber ***Error! Bookmark not defined.***
Figure 7: Wetlands in Makindye Ssabagabo Municipality ——— *31*
Figure 8: Coverage of wetlands by division in Makindye Ssabagabo Municipality ——— *32*
Figure 9: Coverage of intact wetlands in Makindye Ssabagabo Municipality in 2005, 2015, and 2025 *32*
Figure 10: Wetland use/cover changes between 2005 and 2025 in Makindye Ssabagabo Municipality *33*
Figure 11: Wetland seasonality in Makindye Ssabagabo Municipality ——— *34*
Figure 12: Urban extension into a wetland section in Makindye Sssabagabo ——— *35*
Figure 13: Bridge construction across a wetland section in Makindye Ssabagabo to control flooding. ——— *38*
Figure 14: Waste generation by source ——— ***Error! Bookmark not defined.***
Figure 15: Waste generation per waste type ——— ***Error! Bookmark not defined.***
Figure 16: Waste generation per division ——— ***Error! Bookmark not defined.***
Figure 17: Household waste generation in lower, middle-, and higher-income areas ***Error! Bookmark not defined.***
Figure 18: Projected generation of organic waste in Makindye Ssabagabo Municipality between 2022 and 2030 ——— ***Error! Bookmark not defined.***
Figure 19: Waste management funding in Makindye Ssabagabo Municipality between 2020/2021 and 2024/2025 ——— ***Error! Bookmark not defined.***
Figure 20: Mean annual rainfall and mean annual temperature in 2024 ***Error! Bookmark not defined.***
Figure 21: Indicators of climate change in Makindye Ssabagabo Municipality ***Error! Bookmark not defined.***
Figure 22: Observed and projected mean temperatures for Makindye Ssabagabo municipality from 1994 to 2030 ——— ***Error! Bookmark not defined.***
Figure 23: Observed and projected rainfall for Makindye Ssabagabo municipality from 1994 to 2030 ——— ***Error! Bookmark not defined.***
Figure 24: Homes affected by floods in Namasuba ward, Makindye Ssabagabo Municipality ——— ***Error! Bookmark not defined.***
Figure 25: Household responses to climate change in Makindye Ssabagabo Municipality ——— ***Error! Bookmark not defined.***
Figure 26: Barriers to climate change adaptation in Makindye Ssabagabo Municipality ——— ***Error! Bookmark not defined.***
Figure 27: State and trend of land use/cover in Makindye Ssabagabo (2015-2025) ——— *68*
Figure 28: Urbanization trends between 2015 and 2025 in Makindye Ssabagabo ——— *75*
Figure 29: Contribution of Industry to Employment, % ——— ***Error! Bookmark not defined.***

List of Acronyms

AFOLU	Agriculture, Forestry and Land Use
CBD	Convention on Biological Diversity
CDO	Community Development Officer
CHIRPS	Climate Hazards Group InfraRed Precipitation with Station dat
CO	Carbon monoxide
DEAPs	District Environment Action Plans
DMCs	Dangerous Mechanical Condition
ENRC	Environment and Natural Resources Committees
ESIA	Environmental and Social Impact Assessment
FGDs	Focus Group Discussions
FY	Financial Year
GDP	Gross Domestic Product
GHGs	Greenhouse Gases
GKMA	Greater Kampala Metropolitan Area
ICT	Information and Communication Technology
KCCA	Kampala City Council Authority
LC1	Local Council 1
LPG	Liquified Petroleum Gas
MDAs	Ministries, Departments, Agencies
MOUs	Memorandum of Understanding
MSOER	Municipal State of the Environment Report
NDP IV	National Development Plan IV
NEMA	National Environmental Management Authority
NGO	Non-Governmental Organization
Nox	Nitrogen Oxides
NPHC	National Population and Housing Census
NRM	Natural Resources Management
OPM	Office of the Prime Minister
PM10	Particulate Matter
RCPs	Representative Concentration Pathways
RUSLE	Revised Universal Soil Loss Equation
SACCOs	Savings and Credit Cooperative Organizations
SDGs	Sustainable Development Goals
SOER	State of Environment Report
SPI	Standardized Precipitation Index
TVETs	Technical Vocational Education and Training

UBOS	Uganda National Bureau of Statistics
UEGCL	Uganda Electricity Generation Company Limited
USGS	United States Geological Survey
WASH	Water, Sanitation and Hygiene

1. Chapter One: Background

1.1 Introduction

The National Environment Act, Cap 181 assigns urban councils the responsibility of promoting sustainable environmental and natural resource management. This is achieved through the formation of multilevel Environment and Natural Resources Committees (ENRCs) in each urban council. These committees are composed of administrators, elected leaders, and technical experts to ensure broad representation and effective functionality. According to Section 28 (Section 46) of the Act, ENRCs are required to compile a State of Environment Report (SOER) every two years. However, the production of these reports is sometimes limited by insufficient funding. The SOER acts as a key reference document for tracking environmental changes and provides essential information to support environmental planning and policy decisions.

In this context, the Municipal State of Environment Report (MSOER) is a vital statutory instrument used to monitor, evaluate, and communicate the status of the environment and natural resources in Makindye Ssabagabo Municipality. It is prepared in accordance with the National Environment Act (2019) and the guidelines set by the National Environment Management Authority (NEMA). The report provides evidence-based insights to support policy development, strategic planning, and public engagement. The MSOER assesses environmental conditions, trends, underlying causes, and pressures, while identifying opportunities and challenges through a participatory process. It incorporates biophysical, social, and economic dimensions and involves a wide range of stakeholders. The report underscores the critical importance of environmental sustainability in urban development and community well-being.

Ultimately, the MSOER contributes to Makindye Ssabagabo's vision to deliver a well-planned, Clean, and Prosperous Municipality by 2030, aligning with Uganda Vision 2040, the Fourth National Development Plan (NDP IV), and the United Nations Sustainable Development Goals (SDGs). It promotes transparency and collective responsibility in working toward a sustainable future.

1.1.1 Objectives

The key objectives of the MSOER are:

1. To assess the current condition and trends of the environment and natural resources in Makindye Ssabagabo Municipality, considering biophysical, economic, social, and demographic dimensions.
2. To identify and analyze key opportunities, risks, and challenges that influence environmental sustainability in the municipality.
3. To evaluate the performance and impact of current environmental policies, laws, and management strategies in addressing the municipality's environmental issues.
4. To determine the direct and underlying drivers and pressures contributing to environmental change and assess their effects on human health, livelihoods, and ecosystem services.

5. To provide evidence-based and actionable recommendations for the sustainable management of the environment and natural resources, aimed at guiding policymakers and stakeholders at all levels.
6. To promote inclusive stakeholder engagement and public awareness, highlighting the importance of environmental stewardship in advancing development and improving quality of life.

1.1.2 Purpose of the report

The Makindye Ssabagabo Municipal State of Environment Report (MSOER) 2025 is intended to provide policymakers, stakeholders, and the public with a comprehensive, evidence-based evaluation of the current state, trends, and emerging environmental issues within Makindye Ssabagabo Municipality. This report examines the key drivers, pressures, and impacts of environmental change, assesses the effectiveness of existing policies, and identifies both opportunities and challenges for sustainable environmental management. The goal of the MSOER is to support informed decision-making, promote environmental accountability, and encourage collective action toward resilience, inclusive growth, and long-term sustainability in the municipality.

1.2 Makindye Ssabagabo Municipality profile

1.2.1 Geographical location

The Makindye Ssabagabo Municipality is located in Kyadondo County, Wakiso District, Central region of Uganda, bordering Kampala in the north, Mukono District in the east, Kajjansi Town Council in the south-west, Lake Victoria in the south, and Nsangi Town Council in the north-west. Makindye Ssabagabo is between coordinates N 000°14'34.0", E 32°33'36" off the Equator with an average elevation of 1,432 meters above mean sea level. It covers a total land area of 8,473.4 ha (84.734 km², of which 1,761.8 ha (17.618 km²) is covered by water and 6,711.6 ha (67.116 km²) is free land. Makindye Ssabagabo lies along the latitude 32.640557 and longitude 0.397239 and covers 98.83 km².

The municipality has a warm tropical climate, with temperatures ranging from 25°C to 29°C (77°F to 84°F). There are two wet and two dry seasons. The dry seasons run from December to February and June to August, while wet seasons are from March to May and September to November. March to May is wetter and warmer, while December to February is hotter and drier, when it can sometimes reach 36°C (96.8°F).

1.2.2 Municipal structure

Makindye Ssabagabo Municipality operates as an urban local government and is governed by the Makindye Ssabagabo Municipal Council. Its administrative structure is comprised of a Municipal Council supported by an Executive Committee and several sectoral committees. The Municipal Council serves as the primary policy-making authority and functions through three key committees: Social Services; Finance and Planning; and Production, Works, and General Purpose. These committees supervise the activities of nine technical departments led by the Town Clerk, as the chief technical officer.

The nine departments include: Administration; Finance and Planning; Council; Production and Marketing; Primary Health; Water and Sanitation; Education and Sports; Works and Water; Natural Resources; and Community-Based Services. Together, they form the Municipal Technical Planning Committee.

On a lower tier, the municipality is divided into three administrative divisions; Masajja, Ndejje, and Bunamwaya each functioning as a Local Council III (LC III) led by a chairperson and a team of elected councilors. Makindye Ssabagabo serves as a residential area for many people who work in Kampala, earning it the status of a dormitory town. The municipality is subdivided into eight wards: Bunamwaya, Mutundwe, Mutungo, Seguku, Ndejje, Busabala, Namasuba, and Masajja, and further broken down into 55 villages.

1.3 SOER Preparation Process

The Makindye Ssabagabo MOESR was compiled in close cooperation with stakeholders in the municipality. The overarching methodological framework, the Driver-Pressure-State-Impact-Response (DPSIR) Framework, was used to explore the relationships between people and the environment (Figure 1). The DPSIR framework provides a structure within which to present the indicators needed to enable feedback to policymakers on environmental quality and the resulting impact of the political choices made, or to be made in the future. The DPSIR framework also assumes a chain of causal links starting with 'driving forces' (economic sectors, human activities) through 'pressures' (emissions, waste) to 'states' (physical, chemical and biological) and 'impacts' on ecosystems, human health and functions, eventually leading to political 'responses' (prioritization, target setting, indicators). Establishing a DPSIR framework for a particular setting is a complex task as all the various cause-and-effect relationships have to be carefully described, and environmental changes can rarely be attributed to a single cause.
Through discussions at stakeholder meetings and consultations with people from the industry and academic field, broad thematic areas were identified and assessed.

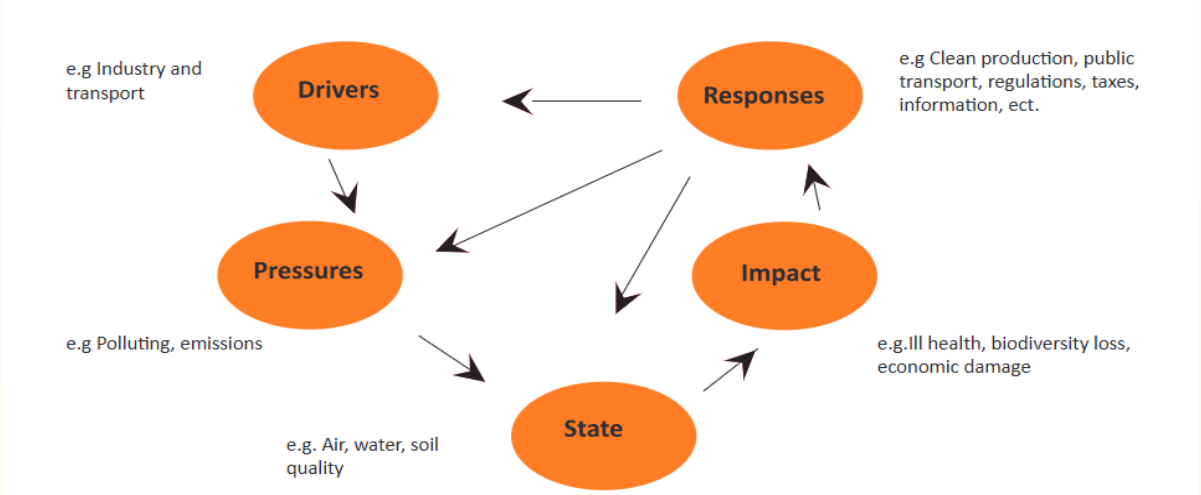


Figure 1: The DPSIR framework[1].

[1] Burkhard, B. & Müller, F. 2008. Drivers-Pressure-State-Impact-Response. Encyclopedia of Ecology. 967-970. 10.10

1.4 The Legal and Policy Framework Governing Environmental Management

Makindye Ssabagabo Municipality operates within a multi-layered legal and policy framework that guides its environmental management efforts. These frameworks range from international conventions to national regulations, each providing specific guidance and requirements that directly influence how the municipality addresses environmental challenges. The following table outlines the key frameworks relevant to environmental management in Makindye Ssabagabo and explains their specific relevance to the municipality's context.

Table 1: Frameworks, Policies, Plans, and Acts relevant to the Environment in Makindye Ssabagabo Municipality

Frameworks, Policies, Plans and Acts	Relevance to Environment
International Frameworks	
The Ramsar Convention on Wetlands (1971)	The Convention on Wetlands of International Importance provides a framework for national action and international cooperation for the conservation and wise use of wetlands and their resources.
Basel Convention (1989)	The Basel Convention regulates transboundary movements of hazardous waste, protecting human health and environment. It promotes environmentally sound management of hazardous waste.
UN Framework Convention on Climate Change (UNFCCC) (1992)	The UNFCCC aims to stabilize greenhouse gas concentrations and prevent climate change. It sets the stage for international cooperation on climate change mitigation and adaptation.
Convention on Biological Diversity (CBD) (1992)	The CBD conserves biological diversity, promotes sustainable use, and ensures fair sharing of genetic resources. It protects ecosystems, species, and genetic diversity.
UN Convention to Combat Desertification (UNCCD) (1994)	The UNCCD addresses desertification and land degradation, promoting sustainable land management and restoration. It supports countries in developing effective strategies to combat desertification.
Paris Agreement (2015)	The Paris Agreement limits global warming to well below 2°C and pursues efforts to limit it to 1.5°C. It promotes collective action on climate change and encourages countries to transition to low-carbon economies.
Sustainable Development Goals (2015)	SDGs address critical issues such as climate change, biodiversity loss, and sustainable resource management, aiming to balance economic development with environmental protection. Achieving these goals is essential for protecting the planet's ecosystems, promoting sustainable development and ensuring a livable future for generations to come.
United Nations Environment Programme (UNEP)	UNEP provides global guidance on environmental issues, including climate change, biodiversity conservation, and pollution control.

UN 2030 Agenda (2015)	The UN Agenda for sustainable development emphasizes the importance of environmental protection and sustainable use of natural resources to achieve a more equitable and prosperous world. The Agenda's Sustainable Development Goals (SDGs) include specific targets related to environmental conservation, climate action, and sustainable development, such as SDG 13 on climate action and SDG 14 on life below water. By achieving these goals, the agenda aims to promote sustainable development, reduce environmental degradation, and ensure a healthy and thriving planet for future generations.
Regional Planning Frameworks	
African Union Climate Change Strategy (2022)	This strategy supports climate-resilient development in Africa, enhancing climate change adaptation and mitigation. It promotes regional cooperation and knowledge sharing.
East African Climate Change Policy (2011)	This policy guides climate change adaptation and mitigation in East Africa, promoting regional cooperation and climate-resilient development. It supports countries in developing effective climate change response strategies.
Eastern African Regional Framework Agreement on Air Pollution (2008)	Focuses on mitigating the adverse impacts of air pollution from various sources including waste disposal and incineration. By promoting cleaner practices, and reducing emissions, it supports sustainable waste management, minimizing harm to human health and the environment.
African Convention on the Conservation of Nature and Natural Resources (1968, Revised 2017)	The convention promotes sustainable management and conservation of natural resources, and aims to protect the environment from pollution and degradation but indirectly supports waste management practices that minimize harm to ecosystems and human health by emphasizing the importance of environmental protection, thereby contributing to a more sustainable approach to waste management in Africa.
National Planning Frameworks	
Uganda Vision (2040)	Uganda Vision 2040 aims to transform Uganda into a modern, prosperous nation while promoting sustainable development and environmental conservation. The vision emphasizes the need for green growth, sustainable natural resource management, and climate resilience to ensure a healthy environment and sustainable development. By 2040, the vision seeks to achieve a balanced and sustainable development that enhances the country's natural capital and mitigates environmental degradation.
Fourth National Development Plan (NDP IV) 2025/26-2029/30	The Fourth National Development Plan (NDP IV) for Uganda prioritizes sustainable environmental management and natural resource conservation, aiming to balance economic growth with environmental protection. The plan

	promotes green growth, climate resilience, and sustainable land use, while addressing environmental challenges such as pollution, deforestation, and climate change.
National Policies and guidelines	
The 1995 Constitution of Uganda (1995)	The constitutional framework provides a foundation for environmental governance and ensures that environmental considerations are integrated into national development planning and decision-making processes. Article 237(2) and Article 245 call for the protection and preservation of the Environment from abuse, pollution and degradation. It stipulates management of the environment for sustainable development and promoting Environment awareness.
The National Environment Management Policy (1994)	Provides a framework for environmental conservation in Uganda, promoting sustainable development and responsible use of natural resources. The policy plays a crucial role in protecting Uganda's environment, enhancing livelihoods, and ensuring a healthy and sustainable future for its citizens. the second objective requires entities to integrate environmental concerns in all development-oriented policies, planning, and activities at national, district, and local levels, with participation of the people.
National Wetlands Policy (1995)	This policy conserves and manages wetland resources sustainably, protecting wetland ecosystems and biodiversity. It promotes the wise use of wetlands.
The National Water Policy (1999)	The policy provides a framework for water resource management and development. It emphasizes the importance of protecting water resources from pollution, including waste-related contamination, to ensure sustainable management. By promoting water conservation and pollution prevention, this policy supports waste management efforts, particularly in reducing waste that can harm water sources and ecosystems.
Electronic Waste (E-Waste) Management Policy for Uganda (2012)	This policy provides a framework for managing electronic waste, promoting environmentally sound disposal, and encouraging recycling. It aims to mitigate environmental and health risks associated with e-waste, while promoting sustainable practices in the electronics sector. It also promotes the implementation of regulations and standards for the management of e-waste, encourages the adoption of environmentally friendly technologies, and fosters public awareness and participation. By setting out clear objectives and actions, the policy seeks to improve the handling of electronic waste, reduce its adverse impacts, and support sustainable development goals while safeguarding public health and the environment in Uganda.
National Climate Change Policy (2015)	This policy provides a framework for climate change response measures in Uganda, supporting the country's transition to a low-carbon economy. It enhances climate resilience and promotes sustainable development.

Guidelines for E-Waste Management in Uganda (2016)	Provides a framework for proper handling, recycling, and disposal of electronic waste reducing environmental and health risks associated with improper disposal. They outline best practices and standards for e-waste management which supports sustainable waste management practices and promote environmentally responsive electronic waste disposal in Uganda.
National Acts	
The Water Act ,1995 (Cap 152)	The Act provides a framework for the management, use, and conservation of water resources in Uganda, promoting sustainable utilization and protection of water resources. The Act aims to ensure that water resources are managed in a way that minimizes pollution, conserves water, and protects the environment, ultimately supporting sustainable development and public health. It calls for the formation of water user associations geared towards the sustainable management of water and water resources.
National Water and Sewerage Corporation Act, 1995(Cap 317)	This Act establishes NWSC as a statutory body responsible for providing water and sewerage services in Uganda, promoting efficient management of water resources and environmental protection. By regulating water supply and sanitation services, the Act aims to ensure sustainable use of water resources, minimize environmental harm, and promote public health.
The Local Governments Act, 1997(Cap 243)	The Act empowers local governments to manage and regulate environmental issues within their jurisdictions, promoting sustainable development and environmental conservation. By decentralizing decision making and service delivery, the Act enables local governments to address environmental concerns, such as waste management and natural resource conservation, in a more effective and responsive manner.
The Land Act,1998(Cap 227)	This act regulates land use and management to prevent degradation and promote sustainable practices. By overseeing land tenure and use, the Act aims to conserve natural resources, minimize environmental harm, and ensure that land is utilized in a way that supports sustainable development.
National Forestry and Tree Planting Act,2003 (Cap 246)	This act promotes the sustainable management and conservation of forest resources in Uganda, while also encouraging tree planting and sustainable land use practices. The Act provides a framework for protecting forests, regulating forest activities, and promoting community involvement in forest management, ultimately contributing to environmental conservation and sustainable development.
National Environment Act ,2019 (Cap 181)	This Act provides a legal framework for environmental management and conservation in Uganda, promoting sustainable development and protection of natural

	resources. The Act emphasizes environmental conservation, addresses climate change, and promotes sustainable use of resources, ensuring accountability and compliance with environmental regulations through enhanced penalties for offences.
The Uganda Wildlife Act, 2019 (repealing Cap 200)	This Act focuses on sustainable wildlife management, conservation and protection of wildlife in Uganda. This Act establishes the Uganda Wildlife Authority to oversee conservation efforts, regulate wildlife user rights, and manage wildlife protected areas, ultimately promoting environmental conservation and sustainability.
The Public Health Act, Cap 281	The Act plays a crucial role in protecting the environment and public health by regulating sanitation, waste management, and pollution control. The Act empowers authorities to take measures to prevent the spread of diseases and promote a clean environment, thereby contributing to environmental health and wellbeing. By enforcing provisions related to environmental sanitation and hygiene, the Act helps mitigate environmental health risks and promotes a healthier environment for Ugandans.
The Occupational Safety and Health Act (2006)	The Act protects workers' health and safety in various workplaces, which indirectly contributes to environment protection by reducing occupational hazards and pollution. The Act promotes safe working conditions, proper handling of chemicals, and waste management thereby minimizing environmental risks and ensuring healthier workplace. By enforcing occupational safety and health standards, the Act helps prevent work related injuries, illnesses, and environmental degradation.
National regulations	
The Petroleum (Exploration and Production) (Conduct of Exploration Operations) Regulations (1993).	It provides guidelines on ensuring environmental management in the petroleum sector with respect to prevention and control of pollution, ensuring health and safety during petroleum activities
The National Environmental Impact Assessment (EIA) Regulations (1998)	The regulations offer guidelines on the minimization of negative environmental impacts from projects through aspects like project briefs and EIA reports. Further stipulations on the, approval or rejection of projects or cancellation of EIA certificates are provided.
The National Environment Waste Management Regulation (1998)	Encompasses all types of hazardous and non-hazardous waste along the value chain including storage, dumping, transportation, disposal, reducing, reuse and recovery (3Rs).
The National environment (Wetlands, River Banks and Lake Shores Management) Regulations (2000)	Provides guidelines for the conservation and wise use of wetlands, riverbanks and Lakeshores and the resources therein and in line with the public trust dogma clause 2 of article 237 of the Constitution (public trust doctrine).
The National Environment (Hilly and Mountainous Areas Management) Regulations (2000)	The regulations aim to conserve and protect Uganda's hilly and mountainous areas from degradation, promoting sustainable land use and environmental protection. They

	help mitigate environmental risks such as soil erosion, landslides, and loss of biodiversity in these ecologically sensitive areas.
National Environment (Standards for Discharge of Effluent into Water or Land) Regulations (2020)	These regulations set standards for effluent discharge to prevent water pollution. Their effectiveness relies on regular monitoring and enforcement to ensure compliance.

1.5 Institutional Arrangements in Environmental Management

Makindye Ssabagabo Municipality's institutional arrangement for environmental management is part of the Water and Environment Sector Institutional framework.
The Water and Environment Sector consists of two sub-sectors: the Water & Sanitation (WSS) sub-sector and the Environment & Natural Resources (ENR) sub-sector. The Water and Sanitation Sub-Sector comprises Water Development, Water Resources Management, Rural Water Supply and Sanitation, Urban Water Supply and Sanitation, and Water for Production. The Environment and the Natural Resources Sub-Sector comprises Environmental Management; management of forests and trees; management of Wetlands and aquatic resources; and Meteorology, Weather, and Climate Change.

The Water and Environment Sector Working Group (WESWG) provides policy and technical guidance for the sector. It comprises representatives from all key sector institutions (Government, Development Partners, and NGOs).

The Ministry also works closely with other Government Ministries, Local Governments where Makindye Ssabagabo Municipality is located, the Private Sector, and Non-Governmental Organizations (NGOs), as well as other stakeholders at the Community, District, Regional, and National levels.

The Sector's institutional framework is set out in the figure below:

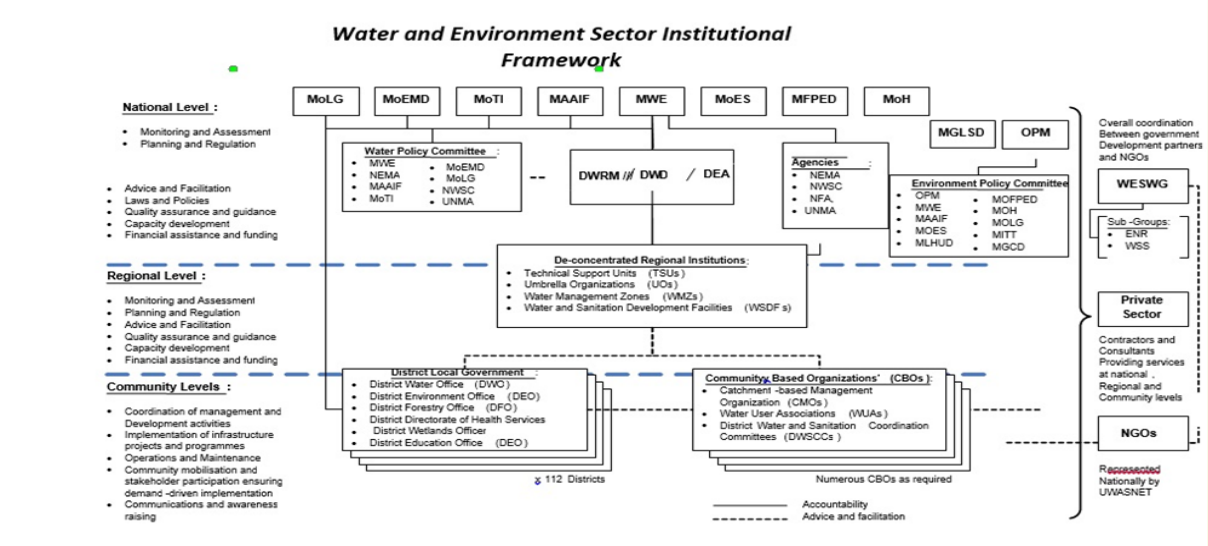


***Figure 2: Water and Environment Institutional Framework*[2]**

At the municipal level, Makindye Ssabagabo has established its own environmental governance structure that aligns with the national framework. The municipality's Environment and Natural Resources Committee (ENRC), as mandated by the National Environment Act (2019), coordinates environmental management activities. This committee works closely with the Natural Resources Department, which implements day-to-day environmental programs. The municipality also collaborates with NEMA's regional office for technical guidance and compliance monitoring. Additionally, the municipality engages with local communities through LC structures and community-based organizations to ensure bottom-up participation in environmental management initiatives. This multi-level governance approach ensures that national environmental priorities are localized and implemented effectively within Makindye Ssabagabo's unique context.

2. Thematic Assessments

2.1 Water Resources Management

2.1.1 Introduction

Makindye Ssabagabo is endowed with diverse water resources that cater to the needs of domestic, agricultural, and ecological use. However, the challenges of urban population pressure, waste management, and weak enforcement of land use have contributed to the degradation of water resources. Water resources in Makindye Ssabagabo include rivers, wetlands, and groundwater sources.

2.1.2 State and trend

The water resources in Makindye Ssabagabo can be broadly categorized into surface water (including rivers and streams), wetlands, and groundwater sources.

Surface Water

Makindye Ssabagabo is traversed by several small rivers and streams, including Mayanja Kato, Kajjansi, Lufuka, Mayanja, Walugogo, and Mazimaka (Figure 3).

[2] Source: https://www.mwe.go.ug/mwe/institutional-framework-water-and-environment

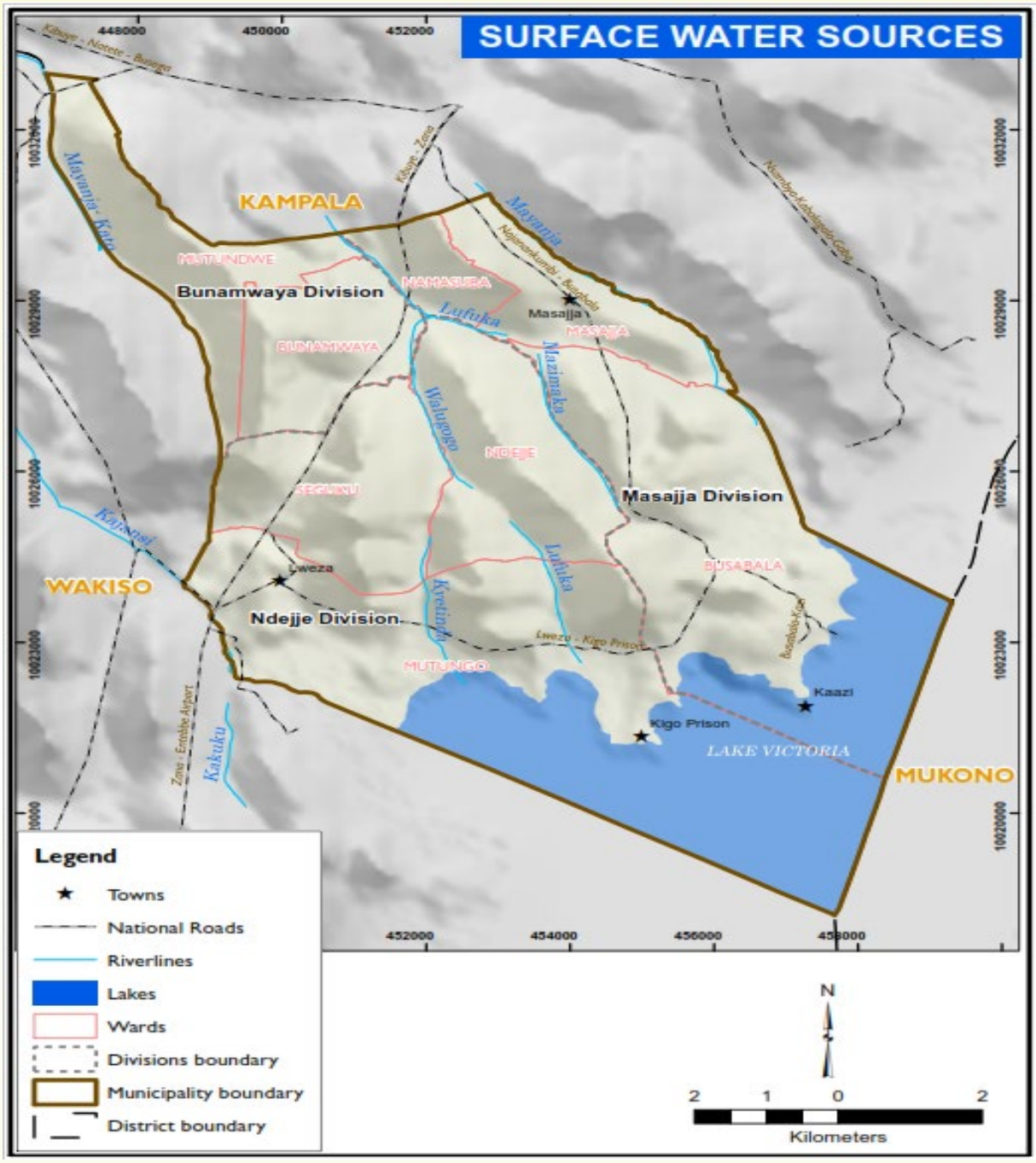


Figure 3: Surface water coverage in Makindye Ssabagabo

These water sources play an important role in agriculture, drainage, and maintaining ecological balance within the Municipality. However, they are increasingly facing serious environmental damage. For example, Mayanja Kato was severely affected by illegal settlements, prompting the National Environment Agency (NEMA) to demolish illegal structures in 2023. Lufuka River, which flows through Zana and Namasuba and into Lake Victoria, is also affected by frequent flooding due to blocked waterways and unregulated construction activities. The Kajjansi and Mayanja rivers also show signs of pollution pressure due to increased urban development.

In addition, rivers such as Walugogo and Mazimaka are located near rapidly urbanizing areas such as Zana and Kikajjo and are particularly vulnerable to soil erosion and pollution. Urban expansion,

improper waste disposal, and inadequate environmental legislation continue to threaten the sustainability of these surface water resources. Without urgent action, these systems could lose their ecological function and endanger human and ecological well-being in Makindye Ssabagabo.

Table xx: Water Quality sampling of the key water courses in Makindye Ssabagabo Municipality

No.	Sampling Location	BOD (mg/l)	Water Temperature (^{0}C)	Soluble Salts (g/l)
1.	Kyetinda (450277 23966)	0.8	26.6	0
2.	Kyetinda (449433 22537)	0.3	22.1	0
3.	Mazimaka-Kaliddubi (454010 28206)	0.7	24.7	0
4.	Mazimaka-Kawaga (456828 24273)	0.3	23.1	0
5.	Mayanga Kato-Lubigi channel (451189 29258)	0.4	27.4	0
6.	Lufuka (450827 30130)	0.5	29.5	0
7.	Lufuka (451436 29284)	0.4	30.6	0
8.	Mayanja (456025 28066)	0.8	25.5	0

Source: Field assessments (2025)

Groundwater Resources (Springs and Boreholes)

In Makindye Ssabagabo, groundwater sources, including protected springs, shallow wells, and boreholes (Figure 1), among others, are an essential component of the water supply, especially in peri-urban and low-income communities where piped infrastructure is inadequate or nonexistent. The main threats to groundwater resources are unregulated land use, illegal waste dumping, and unplanned settlements. In low-lying areas of Salama, residents have reported an increase in typhoid and diarrhea cases, especially after heavy rains. These health risks are often associated with pit latrines, leaking septic tanks, and the proximity of illegal dumping sites to groundwater sources. This indicates faecal contamination, likely due to poor wastewater management and shallow underground flow paths in the region's red soil.

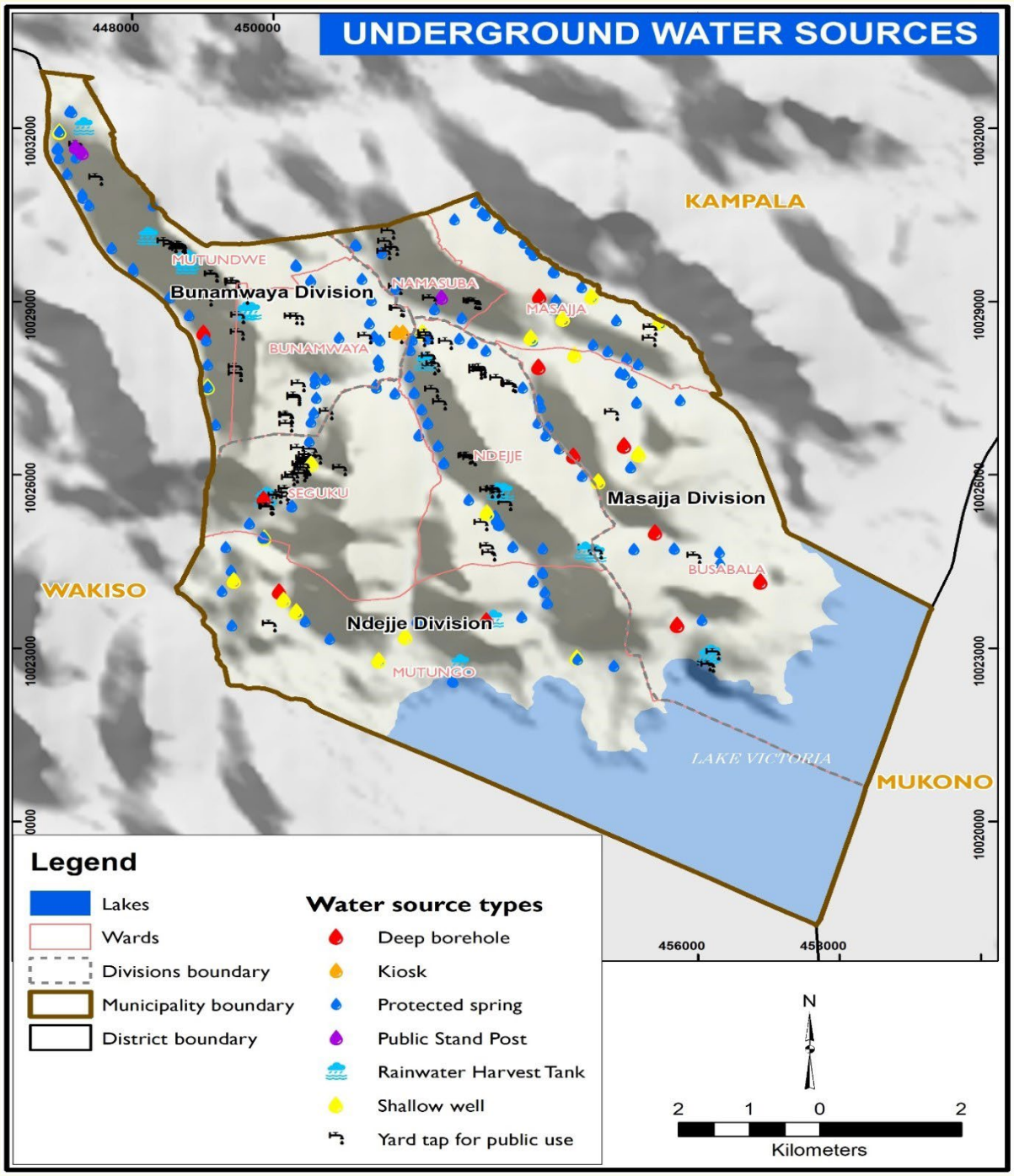


Figure 4: Ground sources in Makindye Ssabagabo

In addition, due to population growth and increased groundwater extraction, water volume and pressure have declined in some boreholes in Ndeje-Lubugumu and Zana, indicating groundwater overload and insufficient groundwater recharge. The lack of proper land use laws and weak enforcement of protection zones around water sources further increases the vulnerability of these water sources.

2.1.3 Drivers of water resources degradation

Water extraction for agricultural use and industrial use is a major driver of water resources degradation, particularly in wards like Mutungo, and Kigo, where peri-urban farming occurs near rivers and wetlands such as Mayanja-Kato, Walugogo Stream, and Kyetinda Wetland (Wetlands Management Department, 2020). The use of chemical fertilizers and pesticides contributes to nutrient loading, with rainfall washing these pollutants into nearby water bodies (WaterAid Uganda, 2020). This runoff leads to eutrophication, reduced water quality, and ecosystem disruption. In areas like Ndejje, elevated nitrate levels in groundwater suggest fertilizer leaching, while streams like Walugogo exhibit signs of pollution such as algal growth and decreased clarity due to inadequate buffer zones and poor agrochemical regulation (NEMA, 2021).



Chemical pollution from Boda-boda and car washing in Lufuka Stream

Unregulated sand and clay mining in Makindye Ssabagabo, especially in Kajjansi Ward, Seguku Central, and Mutungo, directly contributes to water source degradation by destabilizing riverbanks and wetland systems (MWE, 2021). This activity around key water bodies like Kajjansi Stream, Mayanja–Lufuka Wetland, and Lubigi Channel leads to erosion, heavy sedimentation, and the disruption of natural water flow (NEMA, 2021). As a result, water sources become turbid, less effective in natural filtration, and more prone to flooding and pollution. The removal of vegetation further exposes catchment areas, allowing contaminants to infiltrate both surface and groundwater sources, accelerating the decline in water quality (Wetlands Management Department, 2020) across the municipality.

Uncontrolled urbanization and encroachment are the main causes of the deterioration of water resources in Makindye Ssabagabo, especially in the rapidly developing wards of Namasuba-Kikajjo, Mutundwe, Zana, Seguku Central, Lubowa, and Ndejje. In these areas, informal settlements have massively expanded into the fragile ecosystems, especially around the Lufuka River, the Lubigi River (Makindye extension), the Walugogo River, and the Kaliddubi, Kawaga, and Kyetinda wetlands (NEMA, 2021). In Zana, people have constructed houses within the Lufuka River corridor, narrowing the river channel and causing increased flooding during rains (KCCA, 2022). In Seguku Central, parts of the Lubigi Channel have been blocked by solid waste and illegal drainage structures, disrupting water flow (Wetlands Management Department, 2020). Similarly, in Mutundwe, the Walugogo River has been diverted and partially filled in to construct roads and market access, limiting its water storage and filtration functions (MWE, 2021). In addition to physical development, many urban areas lack adequate sewage treatment systems, resulting in the direct discharge of greywater, sewage, and solid waste into nearby wetlands and streams (WaterAid Uganda, 2020). In Ndejje, leaks from pit latrines and septic tanks enter shallow aquifers, while in Kikajjo and Lubowa, rainwater mixes with domestic waste before flowing into wetlands such as Kyetinda and Kaliddubi, leading to increased nutrient loading and bacterial contamination (NEMA, 2021). These activities degrade water quality, causing eutrophication, turbidity, and loss of aquatic life, and compromise the natural filtration, flood control, and groundwater recharge of these ecosystems. Extensive encroachment and pollution indicate that poorly planned urban development is accelerating the degradation of surface and groundwater resources in Makindye Ssabagabo (KCCA, 2022; MWE, 2021).

Poor solid waste management in Makindye Ssabagabo, especially in densely populated wards like Kikajjo, Bunamwaya, Namasuba, and Mutundwe, has significantly altered the quality of water in nearby rivers, streams, and wetlands. With insufficient waste collection services, residents dispose of plastics, food remains, and other household waste directly into Walugogo Stream, Mazimaka Stream, Kaliddubi Wetland, Kawagga Wetland, and Lubigi Channel (Makindye extension) (KCCA, 2022; NEMA, 2023). These wastes block natural drainage channels, altering flow patterns and reducing the capacity of these ecosystems to regulate and filter water (Mugerwa et al., 2021). As rainwater washes accumulated garbage into these water bodies, it introduces leachates rich in nutrients, microplastics, heavy metals (like lead and zinc), and pathogens from decomposing organic matter (Tumwebaze & Nyenje, 2020; UN-Habitat, 2022). This has led to increased turbidity, high biochemical oxygen demand (BOD), and low dissolved oxygen levels, which harm aquatic life and render the water unsafe for domestic use (Wandiga et al., 2021).

Industrial and car wash wastewater is increasingly contributing to the deterioration of water quality in Makindye Ssabagabo, especially in semi-industrial areas such as Mutundwe and Bunamwaya East. In these areas, there are many car washing areas, auto repair garages, and small workshops that discharge untreated wastewater directly into the drainage channels (Mugerwa et al., 2021; Tumwebaze & Nyenje, 2020). This wastewater contains motor oil, detergents, grease and heavy metals such as lead, zinc, and cadmium, which eventually flow into important water bodies such as the Lubigi, Kajjansi, and Lufuka rivers (NEMA, 2021). In the absence of treatment facilities or sedimentation traps, these pollutants can seriously affect water quality by increasing the Chemical Oxygen Demand (COD) and toxicity (Wandiga et al., 2021; KCCA, 2022). The wastewater not only destroys aquatic habitats but also compromises the availability of clean quality

water for domestic and agricultural purposes, thereby increasing environmental pressure on the already fragile urban water supply system in Makindye Ssabagabo (UN-Habitat, 2022).

2.1.4 Impact of water resources degradation

Surface and groundwater contamination from untreated wastewater and industrial effluents has resulted in frequent outbreaks of waterborne diseases such as cholera, typhoid, and diarrhea, particularly in densely populated and low-income areas of Makindye Ssabagabo like Mutundwe and Masajja (WaterAid Uganda, 2021; NEMA, 2021). These neighborhoods often lack adequate sanitation infrastructure, leading residents to dispose of greywater from kitchens, bathrooms, and schools directly into roadside drains or open stormwater channels (KCCA, 2022). These channels typically feed into major watercourses such as the Lufuka River and Lubigi Channel, where the mixing of domestic effluents with untreated industrial waste significantly elevates microbial and chemical contamination levels (MWE, 2021). The situation is worsened during the rainy season, when overland flow rapidly transports pathogens and organic pollutants into both surface and shallow groundwater systems, increasing exposure risks for communities that rely on these sources for drinking, cooking, and hygiene (NWSC, 2023; WHO, 2020).

Degradation of wetlands such as Kyetinda, Mayanja–Lufuka, and Kaliddubi in Makindye Ssabagabo has led to a severe decline in native biodiversity, particularly affecting wetland-dependent flora and fauna (NEMA, 2021). These wetlands, once rich in papyrus vegetation, aquatic plants, frogs, and marshland bird species, have been increasingly encroached upon due to rapid urban expansion. In Kigo, for example, wetland areas have been reclaimed for real estate developments and banana plantations, leading to habitat fragmentation and the displacement of species such as the grey crowned crane (*Balearica regulorum*), Uganda's national bird. Similarly, in Kajjansi, the conversion of wetland buffer zones into vegetable gardens and informal settlements has disrupted fish breeding grounds vital to local fisheries, especially affecting native tilapia and catfish species (Wetlands Management Department, 2022).

In Makindye Ssabagabo, streams like Mayanja–Kato and Lufuka have suffered severe pollution due to siltation, low oxygen levels, and chemical runoff, leading to a sharp decline in fish species diversity. This has particularly affected artisanal fishers in Kigo and Kajjansi, where reported fish catches have dropped significantly since 2019. As a result, many have abandoned fishing, worsening unemployment and food insecurity in low-income wards such as Kabuuma and Lufuka. The degradation of aquatic ecosystems not only undermines livelihoods but also reduces access to affordable protein for vulnerable communities (NEMA, 2021; NPA, 2022; Wetlands Management Department, 2022).

In Makindye Ssabagabo, uncontrolled vegetation clearing along riverbanks in areas like Kyetinda and Bunamwaya has led to soil erosion, causing heavy sedimentation in rivers and wetlands. This reduces water flow capacity, clogs drainage systems, and increases water turbidity, making the water less suitable for domestic and agricultural use. These effects are clear signs of water source degradation, which threatens local water availability, increases flood risk, and compromises ecosystem functions essential for water purification and storage (NEMA, 2021).

With shallow wells and springs contaminated or drying up, many residents are forced to buy water from private vendors at inflated prices. This is especially prevalent in Mutundwe, Lufuka, and Masajja, where boreholes have been found to contain high levels of nitrate and iron, indicating both agricultural leaching and waste infiltration. In 2020, water prices in parts of Makindye Ssabagabo doubled during the dry season due to reduced local supply (WaterAid Uganda, 2021). This issue is especially pronounced during the dry season, when surface water sources dwindle and groundwater levels recede. For example, in 2020, local reports indicated that water prices in neighborhoods like Mutundwe nearly doubled, rising from UGX 200 to UGX 500 per jerrycan, due to reduced availability from traditional sources and increased dependence on vendors (Makindye Ssabagabo Division Council, 2021)

Degraded water sources like wetlands have intensified flooding, prompting frequent emergency interventions by the municipal authorities. In April 2023, after flooding from the River Lufuka affected households in Namasuba and Bunamwaya, Makindye Ssabagabo's municipal government had to deploy water pumps, temporary latrines, and health outreach services, straining its limited budget (National Planning Authority, 2022). These recurrent costs divert funds from long-term development projects.

The continued encroachment on wetlands despite existing laws reflects poor enforcement and weak institutional capacity, hence eroding public trust in governance. This undermines the credibility of agencies like NEMA, the Wakiso District Environment Office, and KCCA, resulting in public apathy and non-compliance. In 2022, residents in Bunamwaya staged protests against the construction of a fuel station near a wetland, blaming the municipality for failing to act on illegal approvals (National Planning Authority, 2022).

2.1.5 Responses

As a response measure to the worsening wetland degradation and flooding in Makindye Ssabagabo, the government, through the National Environment Management Authority (NEMA) and the Wakiso District Natural Resources Office, undertook the demarcation and restoration of critical wetlands such as Kyetinda, Mayanja–Lufuka, and the Lubigi Channel (Makindye Extension). These actions were implemented in reaction to escalating encroachment, blocked drainage, and ecosystem collapse that had already triggered flooding and biodiversity loss in areas like Bunamwaya, Kigo, and Kajjansi (NEMA, 2021). In 2022, as part of the emergency response to these environmental impacts, wetland boundaries were physically marked with concrete pillars to halt further degradation, and eviction notices were issued to illegal settlers and developers, particularly in Bunamwaya and Kabuuma Wards, where informal construction had spread into protected wetlands. Therefore, the boundary demarcation and Evictions served as a post-impact response, aimed at containing further damage, restoring lost functionality, and enabling future rehabilitation efforts (National Planning Authority, 2022).

As a response measure to ongoing wetland degradation and illegal land use, authorities in Makindye Ssabagabo, working with the National Environment Management Authority (NEMA), have undertaken targeted enforcement operations to reclaim encroached wetlands. These actions were not preventive (mitigation) but rather reactive responses to already visible environmental damage, such as blocked drainage channels, disrupted water flow, and declining wetland function.

In 2020 and 2022, joint operations led to the demolition of unauthorized structures in environmentally sensitive areas, particularly within the Kajjansi Wetland and parts of Mutungo, where illegal settlements and sand mining had caused extensive ecological harm (National Planning Authority, 2022). The enforcement aimed to immediately halt ongoing destruction, recover damaged wetland zones, and reinforce legal compliance following community resistance and prior inaction. These interventions served as post-degradation responses, triggered by reports of flooding, sedimentation, and biodiversity loss in affected areas. In several cases, reinstatement of natural wetland vegetation followed the demolition activities, helping restore water retention and filtration functions of the ecosystem.

As a response measure to the worsening water pollution and drainage degradation, which leads to water source degradation, the Makindye Ssabagabo Municipal Council, in collaboration with the National Water and Sewerage Corporation (NWSC), implemented targeted wastewater management interventions. These actions were taken after significant environmental damage had already occurred, particularly in high-risk zones like Masajja, where untreated greywater and effluents from car wash bays were being directly discharged into open drains and water channels (NEMA, 2021). This led to increased turbidity and chemical contamination in nearby streams such as River Lufuka and the Lubigi Channel, prompting the need for immediate corrective actions. In response, sewerage lines were extended to these hotspots to divert waste from the open environment into proper treatment systems. Additionally, in 2021, a review and revalidation of permits for car washing bays was conducted to enforce environmental standards and stop illegal discharge practices (KCCA, 2021). These measures were not part of proactive planning but were implemented in reaction to observed pollution, public complaints, and environmental monitoring reports (National Planning Authority, 2022; Ministry of Water and Environment, 2022). They serve as critical response interventions aimed at restoring degraded water quality and reducing health risks for downstream communities.

As a response measure to recurrent flooding and pollution outbreaks, the Makindye Ssabagabo Municipal Council implemented targeted infrastructure upgrades in 2023. This action followed multiple cases of stream overflows and wastewater contamination in vulnerable areas like Mutundwe and Bunamwaya East, where inadequate drainage infrastructure had previously failed during heavy rainfall events. In response, the council launched the rehabilitation of critical drainage channels, involving the clearing and widening of blocked drains, relining of rainwater pathways, and the installation of sedimentation traps and culverts in flood-prone zones, especially near the River Lufuka, which had flooded homes in Namasuba and its surroundings in 2022 and 2023 (WaterAid Uganda, 2023; Makindye Ssabagabo Municipal Council, 2023). This rapid-response intervention aimed to provide short-term flood relief and reduce waterborne health risks exacerbated by urban runoff and wetland degradation.

As a response to ongoing water source degradation caused by illegal wetland encroachment and sand mining, Makindye Ssabagabo authorities, in collaboration with NEMA, conducted enforcement operations in 2020 and 2022. These operations involved demolishing unauthorized structures in environmentally sensitive areas such as Kaliddubi Wetland and Mutungo. The goal was to halt further ecological damage and begin the process of restoring wetland functions critical for water regulation and quality (National Planning Authority, 2022).

In 2022, boreholes and protected springs in Busabala, Kabuuma, and Mutundwe were rehabilitated through donor-supported programs to improve water quality. These included fencing, construction of drainage aprons, and protection from nearby latrines (National Planning Authority, 2022). WaterAid and other NGOs have conducted Water, Sanitation, and Hygiene (WASH) campaigns targeting schools and households in Mutundwe and Kabuuma to promote safe waste disposal and protect springs. These efforts have increased awareness and reduced indiscriminate dumping near water points (WaterAid Uganda, 2023).

Community-based organizations like Friends of Nature and local LC1 committees have participated in tree planting and re-vegetation around Kyetinda Wetland and Walugogo Stream. In 2021, youth groups in Kigo restored 2 hectares of wetland buffer using bamboo and vetiver grass to control erosion and runoff (WaterAid Uganda, 2021).

Local environmental watchdog groups have been set up in wards like Bunamwaya East and Namasuba to monitor illegal waste disposal and report water pollution events. This has led to faster municipal responses and the penalization of polluters (NEMA, 2021).

2.1.6 Recommendations

One of the most effective strategies to reduce surface water degradation in Makindye Ssabagabo is the enforcement of wetland and riparian buffer zones along key rivers and streams such as River Lufuka, Lubigi Channel, and Mayanja–Lufuka. The municipality should demarcate and protect 30–100-meter-wide buffers using legal frameworks, preventing further encroachment by real estate developers and farmers. Restoration efforts should prioritize the replanting of native vegetation like vetiver grass, bamboo, and indigenous trees, which naturally filter runoff and stabilize soil to prevent erosion. Additionally, community wetland monitors can be empowered to report illegal activities in buffer zones, enhancing enforcement and accountability. These measures align with NEMA's guidelines for wetland conservation (NEMA, 2021; National Planning Authority, 2022).

Despite the existence of national effluent standards, enforcement in Makindye Ssabagabo remains weak, especially in semi-industrial zones. The Municipal Environment Office and NEMA should conduct regular inspections and enforce permits for all car washes, garages, and workshops. Facilities that discharge oil, grease, and detergents into storm drains should be fined under the Polluter Pays Principle, and their operations suspended until compliance is achieved. Strengthening enforcement through training and the use of mobile monitoring tools will enhance compliance and reduce the pollution burden in streams such as Walugogo and Mayanja–Kato (NEMA, 2021; WaterAid Uganda, 2023).

Degraded and undersized drainage systems in flood-prone wards such as Kabuuma, Mutundwe, and Namasuba must be rehabilitated to handle increased runoff volumes caused by urbanization and wetland loss. The municipality should install sedimentation traps, stormwater retention ponds, and percolation pits to manage surface runoff and reduce pollution during heavy rains. GIS tools should be used to prioritize critical hotspots for infrastructure upgrades based on hydrological modeling and land use change. This approach not only reduces flood risk but also enhances water recharge and prevents contamination of rivers like Lufuka (National Planning Authority, 2022).

Establishing local groundwater quality monitoring groups can enhance early detection of pollution and support municipal planning. Selected water sources can be fitted with simple tools like manual nitrate strips, pH meters, and water level sensors to track changes in quality and quantity. Trained community water user committees in Mutundwe, Namasuba, and Kabuuma can maintain records and share data with the local health and water departments for timely interventions. These participatory monitoring systems have been recommended by WaterAid Uganda and shown to be effective in enhancing local accountability (WaterAid Uganda, 2023).

Localized treatment systems can reduce the volume of untreated wastewater entering surface water sources. In areas such as Masajja, Mutundwe, and Bunamwaya East, decentralized systems like sedimentation tanks, bio-digesters, and constructed wetlands should be installed to treat greywater from car wash bays, workshops, and households before discharge into drainage systems. These treatment units offer low-cost, low-maintenance solutions suitable for semi-urban communities and can reduce pollutant loads in streams like Lubigi and Kajjansi. Treated water can also be reused for landscaping or cleaning, easing pressure on freshwater sources. Such models have proven successful in peri-urban Uganda and are promoted under WaterAid's sustainable WASH programs (WaterAid Uganda, 2023).

Fertilizer leaching from farms in Kigo, Mutungo, and Kabuuma poses a serious threat to groundwater, especially during rainy seasons. To counter this, farmers should be trained on integrated nutrient management and encouraged to adopt organic farming practices. The use of vegetative buffer strips and contour farming can also help retain nutrients and prevent infiltration into groundwater. Extension workers should support farmers with soil testing and tailored fertilizer application to avoid overuse. NEMA has previously identified agrochemical runoff as a major driver of nitrate contamination in peri-urban aquifers (NEMA, 2021).

Transitioning from traditional pit latrines to ecological sanitation systems can reduce the risk of groundwater contamination. In high-density settlements near shallow aquifers, like Kabuuma and Ndejje, eco-toilets such as urine-diverting dry toilets (UDDTs) and lined ventilated improved pits (VIPs) should be promoted. These toilets reduce leachate infiltration and enable nutrient recovery. Community education and training for masons on the construction of improved sanitation facilities is essential to enhance adoption and ensure proper implementation (WaterAid Uganda, 2021).

To safeguard underground water sources, protection zones should be legally enforced around springs and shallow boreholes, especially in areas like Ndejje, Mutundwe, and Kabuuma. These zones should prevent the construction of pit latrines, septic tanks, and solid waste dumps within at least 30 meters of any groundwater source. In areas where this proximity exists, efforts should be made to relocate or seal high-risk sanitation structures. Proper fencing and management of boreholes will also help prevent direct contamination from animals and humans, in line with best practices outlined by the Ministry of Water and Environment (National Planning Authority, 2022).

2.2 Biodiversity

2.2.1 State and trend

Makindye Ssabagabo, a rapidly urbanizing municipality in the Kampala Metropolitan Area of Uganda, is endowed with an array of diverse ecosystems that play a vital role in maintaining ecological balance and supporting the livelihoods of its residents. These ecosystems ranging from inland wetlands and flowing rivers to patches of remnant forests and vegetated landscapes harbor significant biodiversity. However, recent assessments highlight a worrying trend of degradation, particularly within aquatic habitats, with potential long-term consequences for both biodiversity and human well-being.

According to Langdale-Brown et al (1964), the vegetation characterizing the Makindye Ssabagabo Municipality area is predominantly a Combretum savanna mosaic of forestry remnants, grassland, and shrubland. Wetland vegetation is largely characterised by *Cyperus papryfera, C. latifolius, and Pernisetum purpurem.* Impacts from degradation for urbanization, agricultural development, fuel harvesting, and artisanal sand mining have further modified the original forest cover. Intense evaporation together with the erratic nature of rainfall results in the land being covered with bushes, and any annual grasses which may appear in the wet season. The dominant grasses are *Imperata cylindrica, Hyparrhenia filipendula,* and *Cymbopogon afronardus.* Invasive species of *Parthenium spp, Brousonetia papryfera, and Lantana Camara* were common in disturbed habitats across the municipality.

Flora diversity assessment in Makindye Ssabagabo

Field biodiversity assessments near and within wetland areas of the municipality revealed that most of the species belonged to the families *Poaceae, Acanthaceae,* and *Fabaceae*. The least common families included *Apocynaceae, Ebenaceae, Asclepiadaceae, Burseraceae,* and *Flacourtiaceae. Stachytarpheta jamaicensis, Leonotis leonurus, Eucalyptus grandis, Cynodon dactylon, Ricinus cuminis, Terminalia superba, Markhamia lutea, Psidium guajava, Solanum incanum, Tithonia diversifolia, Euphorbia tirucalli, Acacia hockii, Hyparrhenia filipendula, Brachiaria decumbens, Imperata Cylindrica* define the various habitat and vegetation types in the sampled areas. Wetland plant species were characterised with dominant vegetation varying from *Cyperus Paparus* in Lubigi Channel and Kawaga (Busabala); *Cyperus latifolius-Pennisetum purpureum* in Lufuka, Kyetinda, Kaliddubi, and Mayanja-lufuka wetlands. Other wetland species included

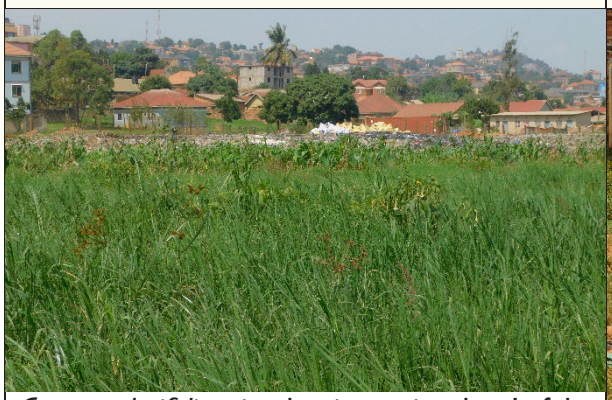
Makindye Ssabagabo is largely modified with dotted large mature native, fruit trees and ornamental plant species such as *Mangifera indica, Artocarpus heterophyllus, Ficus Benjaminii, Duranta erecta,* and the ashok tree. Modified agricultural and agroforestry landscape vegetation is characterised by Eucalyptus and pine woodlots especially along Kalidubi wetland. Maize and Bananas are the major crops grown in the municipality. Thus, a total of 256 species across 168 genera were recorded. *Milicia excelsa* was the only species of conservation concern recorded commonly around homesteads.



Cyperus papyrus is dominant in Kyetinda wetland



Pennisetum purpureum dominant vegetation in the Lufuka wetland



Cyperus latifolius is dominant in the Lufuka wetland



Invasive *Parthenium hysterophorus* in the Colocasia-Cyperus degraded section of Kaliddubi Wetland

Common wetland plant communities in Makindye Ssabagabo Municipality

Faunal Diversity Assessment in Makindye Ssabagabo

The general landscape is uneven, with land gently sloping to moderately steep, dominated by grassland savanna, wetland, and woodland vegetation neighbouring subsistence farmland. These varying environments attract different grassland, woodland, and wetland fauna that easily find, create, or adapt to the habitats. Due to the heavily modified landscape, field assessment of faunal diversity was limited to avian communities near and within the wetland ecosystems in Makindye Ssabagabo.

Birds: The savanna grassland, woodland, and wetland mosaic and subsistence farmland, attracts largely seed-eating and wetland birds such as cranes, Ibises, Storks, Egrets, Herons, Weavers, Finches, Swallows, and Doves. The most common species of birds recorded were the Grey crowned cranes, Hadada Ibis, Marabou Storks, Pied Crows, Cattle Egrets, Rock Pigeons, House sparrows *(Passer griseus),* Sunbirds *(Cinnyris spp.),* Grey-headed Kingfisher *(Halcyon leucocephala)*, Grey Heron *(Ardea cinerea)*, Crowned Lapwing (*Vanellus coronatus*), Ring-necked Dove (*Streptopelia capicola),* Red-rumped Swallow (*Cecropis daurica*), and Dark-capped Bulbul (*Pycnonotus tricolor*), among others. The Lubigi channel has the largest biodiversity of avian communities. Lufuka wetland has the largest population of Grey crowned cranes (Nationally endangered). Further studies are recommended as to whether the Lufuka wetland is a breeding site. The detailed species list is shown in Table 2.



Grey crowned crane



African wattled lapwing



Cattle egret



Pied kingfisher



African open bill



Common hammerkop

Table 2: Species of birds recorded in Makindye Ssabagabo

Common name	Species name	Abundancy	IUCN Status
African Fire finch	*Lagonosticta rubricata*	*2*	*Least Concern*
African Green-Pigeon	*Treron calvus*	*7*	*Least Concern*
African mourning dove	*Streptopelia decipiens*	*8*	*Least Concern*
African pipit	*Anthus cinnamomeus*	*7*	*Least Concern*
African Sacred Ibis	*Threskiornis aethiopicus*	*11*	*Least Concern*
African Thrush	*Turdus pelios*	*2*	*Least Concern*
Angola Swallow	*Hirundo angolensis*	*5*	*Least Concern*
Banded Martin	*Riparia cincta*	*7*	*Least Concern*
Black winged Kite	*Elanus caeruleus*	*1*	*Least Concern*

Brown Babbler	*Turdoides plebejus*	*9*	*Least Concern*
Barred Warbler	*Sylvia nisoria*	*50+*	*Least Concern*
Cattle Egret	*Bubulcus ibis*	*7*	*Least Concern*
Grey Heron	*Ardea cinerea*	*2*	*Least Concern*
Grey crowned crane	*Balearica regulorum Gruidae*	*+50*	*Endangered*
Croaking Cisticola	*Cisticola natalensis*	*4*	*Least Concern*
Crowned Lapwing	*Vanellus coronatus*	*29*	*Least Concern*
Great white Egret	*Ardea alba*	*7*	*Least Concern*
Grey-backed Fiscal	*Lanius excubitoroides*	*20+*	*Least Concern*
Grey-headed Kingfisher	*Halcyon leucocephala*	*1*	*Least Concern*
Laughing Dove	*Spilopelia senegalensis*	*5*	*Least Concern*
Northern Grey-headed Sparrow	*Passer griseus*	*7*	*Least Concern*
Northern Puffback	*Dryoscopus gambensis*	*1*	*Least Concern*
Plain-backed Pipit	*Anthus leucophrys*	*2*	*Least Concern*
Pied Crow	*Corvus albus*	*3*	*Least Concern*
Pin-tailed Whydah	*Vidua macroura*	*10*	*Least Concern*
Purple-banded Sunbird	*Cinnyris bifasciatus*	*43*	*Least Concern*
Red-chested Sunbird	*Cinnyris erythrocercus*	*42*	*Least Concern*
Red-eyed Dove	*Streptopelia semitorquata*	*21*	*Least Concern*
Red-faced Cisticola	*Cisticola erythrops*	*5*	*Least Concern*
Red-headed Cordon-bleu	*Uraeginthus cyanocephalus*	*20+*	*Least Concern*
Red-headed Weaver	*Anaplectes rubriceps*	*2*	*Least Concern*
Red-rumped Swallow	*Cecropis daurica*	*50+*	*Least Concern*
Ring-necked Dove	*Streptopelia capicola*	*7*	*Least Concern*
Rufous Sparrow	*Zonotrichia capensis*	*2*	*Least Concern*
Rufous-naped Lark	*Mirafra africana*	*1*	*Least Concern*
Ruppell's Starling	*Lamprotornis purpuroptera*	*30+*	*Least Concern*
Senegal Lapwing	*Vanellus lugubris*	*19*	*Least Concern*
Speckled Mouse bird	*Colius striatus*	*17*	*Least Concern*
Spot-flanked Barbet	*Tricholaema lacrymosa*	*16*	*Least Concern*
Tawny-flanked Prinia	*Prinia subflava*	*6*	*Least Concern*
Trilling Cisticola	*Cisticola woosnami*	*8*	*Least Concern*
Tropical Boubou	*Laniarius aethiopicus*	*2*	*Least Concern*
Vitelline Masked Weaver	*Ploceus vitellinus*	*2*	*Least Concern*
Wahlberg's Eagle	*Hieraaetus wahlbergi*	*1*	*Least Concern*
White-browed Robin-chat	*Cossypha heuglini*	*6*	*Least Concern*
White-headed Barbet	*Lybius leucocephalus*	*1*	*Least Concern*
White-Headed saw-wing	*Psalidoprocne albiceps*	*6*	*Least Concern*
Yellow White-eye	*Zosterops senegalensis*	*12*	*Least Concern*
Yellow-billed Kite	*Milvus aegyptius*	*2*	*Least Concern*



Papyrus harvesting at Kalidubi wetland



Aquaculture in Kyetinda wetland



Making of clay stoves in Kyetinda



Roadside tree nurseries extract water from Kyetinda wetland

Ecosystem services provided by wetlands

2.2.2 Drivers of Biodiversity Degradation

The biodiversity of Makindye Ssabagabo is increasingly threatened by a combination of human-driven forces that have intensified over the past decade. Among the most significant of these are rapid urbanization, unsustainable agricultural practices, extractive industries, pollution, and poorly regulated infrastructural development. Each of these drivers contributes uniquely and cumulatively to the degradation and loss of biodiversity in the region.



Brick laying in Lufuka wetland



Backfilling with Muram at Mayaja-lufuka



Construction in Kyetinda wetland



Chemical pollution from Car washing at Lufuka

Biodiversity degradation in Makindye Ssabagabo

Rapid Urbanization

One of the most significant pressures on biodiversity in Makindye Ssabagabo is rapid urbanization, fueled by a staggering population increase of 88.7% between 2015 and 2025. This urban growth has triggered widespread land conversion, particularly targeting wetlands and riparian zones, which are often perceived as vacant or underutilized land. These ecosystems, however, are essential habitats for numerous plant and animal species and play a critical role in water regulation and purification. The expansion of residential and commercial infrastructure into these sensitive areas fragments habitats, reduces ecological connectivity, and directly leads to species displacement or loss. The transformation of natural landscapes into built-up environments represents not only a loss of biodiversity but also a weakening of the municipality's natural resilience to climate change and flooding.

Sand and Clay Mining

Another major driver of biodiversity loss is sand and clay mining, particularly in the Kajjansi and Kaliddubi wetland systems. These extractive activities, often unregulated or poorly managed, result in the physical removal of topsoil and vegetation, leading to severe habitat destruction and erosion. Mining operations alter the natural hydrology of wetlands, creating artificial depressions and changing water flow patterns. The disruption of these systems affects aquatic organisms and plant communities, while sedimentation downstream can smother habitats and reduce water quality. Additionally, the noise and activity associated with mining displace wildlife and hinder natural ecological processes such as seed dispersal and pollination.

Chemical pollution from car washing and waste dumping

Chemical pollution from car washing through the introduction of heavy metals, oils, and fuels, particularly in Kyetinda and Lufuka wetlands, can impact biodiversity negatively. The discharge of untreated waste into wetlands and rivers introduces harmful substances such as heavy metals, pathogens, and excessive nutrients. These pollutants compromise water quality, leading to toxic conditions for fish, amphibians, and macroinvertebrates. The accumulation of waste also contributes to habitat degradation, particularly in urban wetlands that serve as informal dumping sites. The biological impact is evident in declining species richness, reduced reproductive success, and the local disappearance of sensitive species. Furthermore, polluted wetlands lose their capacity to provide ecosystem services such as water filtration, thereby posing health risks to surrounding communities.

Infrastructural Development

Infrastructural development particularly construction of residential houses is proceeding at a rapid pace in Makindye Ssabagabo, often without adequate environmental safeguards. Construction within wetlands and floodplains interferes with natural water flow, leading to altered drainage patterns, flooding, and habitat loss. Many of these developments are approved or implemented without comprehensive environmental impact assessments or enforcement of mitigation measures. The result is a permanent conversion of ecologically valuable land into impervious surfaces, which not only accelerates biodiversity loss but also increases vulnerability to climate-induced hazards such as urban flooding. This unchecked expansion reflects a gap in land use planning and regulatory enforcement, undermining long-term sustainability goals.

2.2.3 Impact of Biodiversity degradation

Habitat Loss and Fragmentation

One of the most immediate outcomes of ecosystem degradation is habitat loss and fragmentation. As wetlands and riparian zones are drained, filled, or built upon, many species lose their natural habitats or find themselves isolated in small, disconnected patches. The habitat for the grey crowned crane (*Balearica regulorum*) has been reduced through infrastructure development and industrialization, especially at Lufuka wetland. Once a common sight in the municipality, its population is now dwindling due to the destruction of nesting and foraging areas. Similarly, fish breeding grounds have been destroyed or severely degraded, especially in areas where wetlands have been converted for agriculture or infrastructure. The fragmentation of habitats not only reduces species numbers but also impairs ecological processes such as migration, breeding, and feeding.

Loss of Ecosystem Services

Ecosystem degradation has seriously undermined the services that natural systems once provided to the municipality:

Flood Regulation: Wetlands serve as natural buffers that absorb excess rainwater and regulate stream flows. For example, the overflow of River Lufuka in 2023 resulted in damage to homes, roads, and crops

Water Purification: Wetlands and vegetated riparian buffers naturally filter pollutants from runoff and waste discharge. But with increasing siltation, nutrient overload from fertilizers, and the loss of filtering vegetation, this function is breaking down. Contaminants now flow more freely into open water bodies, leading to algal blooms, decreased oxygen levels, and unsafe water for both human and wildlife consumption. Water quality in wetlands is highlighted in section 2.1.

Public Health Risks: The stagnation of polluted wetlands has turned many of these once-productive ecosystems into public health hazards. Pools of contaminated water now serve as breeding grounds for mosquitoes and other disease vectors, contributing to rising cases of malaria. Similarly, poor sanitation and contamination of water sources have been linked to outbreaks of waterborne diseases such as cholera and typhoid, particularly in informal settlements.

2.2.4 Responses to Biodiversity degradation

Makindye Ssabagabo Municipal Council and its partners have undertaken a range of targeted interventions aimed at restoring ecological integrity and improving environmental governance.

Wetland Boundary Demarcation

One of the foundational steps in protecting wetlands from encroachment has been the demarcation of their boundaries. Concrete markers have been installed in key wetland systems such as Kyetinda and Lubigi, serving both as physical deterrents to illegal development and as visual reminders of protected zones. This intervention is critical in reinforcing the legal status of wetlands and preventing further encroachment. Demarcation also facilitates enforcement by clearly distinguishing conservation areas from land designated for other uses, helping to reduce conflicts between development and environmental protection goals.

Enforcement Operations

To complement boundary demarcation, the municipal council has conducted enforcement operations targeting illegal developments within ecologically sensitive areas. Notably, structures erected without authorization in Kajjansi and Mutungo wetlands were demolished in operations carried out in 2020 and 2022. These actions, though sometimes met with resistance, have sent a clear message about the municipality's commitment to upholding environmental laws. By reclaiming occupied wetlands and restoring them to their natural function, these operations contribute directly to the recovery of biodiversity and the restoration of vital ecosystem services such as flood regulation and water filtration.

Drainage Rehabilitation

With the increasing frequency and intensity of floods, particularly around the River Lufuka, drainage rehabilitation has become a top priority. Municipal authorities have undertaken the clearing of clogged drainage channels and the upgrading of stormwater infrastructure in flood-prone neighborhoods. These works are essential not only for improving water flow and reducing flood risks but also for curbing the spread of waterborne diseases often linked to stagnant water and poor drainage. Restoring natural hydrology through improved drainage also benefits surrounding ecosystems by reducing waterlogging and allowing wetlands to resume their natural absorptive functions.

Community Engagement

Community involvement has been central to the municipality's environmental interventions, recognizing that sustainable conservation requires local ownership and participation. Several grassroots initiatives have emerged:

- **Tree Planting in Degraded areas:** Community members have participated in tree planting campaigns aimed at restoring riparian zones and wetlands that have been cleared or degraded. These efforts help stabilize soil, enhance habitat for wildlife, and improve microclimatic conditions.
- **NGO-led WASH Campaigns:** Local and international NGOs have spearheaded Water, Sanitation, and Hygiene (WASH) campaigns, raising awareness about pollution prevention, proper waste disposal, and the importance of clean water. These campaigns often involve school outreach, women's groups, and youth-led activities, fostering a culture of environmental responsibility.

- **Local Watchdog Groups:** Community-based watchdog groups have emerged as key players in monitoring pollution and illegal activities. These groups report unauthorized developments, dumping of waste, and other violations, acting as a first line of defense for the environment. Their involvement not only strengthens enforcement but also enhances transparency and accountability.

2.2.5 Recommendations

The following actions can help support biodiversity conservation in Makindye Ssabagabo Municipality.

Policy and Governance

Effective conservation requires robust and enforceable legal instruments. In Makindye Ssabagabo, the priority lies in strengthening the legal framework and resolving longstanding land use conflicts.

- **Pass ordinances against further encroachment;** The municipality should pass ordinances as legal tools to prevent encroachment, particularly in high-risk floodplains and biodiversity hotspots.

Habitat Restoration

Rehabilitating damaged ecosystems is key to restoring biodiversity and reactivating lost ecosystem services.

- **Revegetate Buffers with Indigenous Species:** Strategic planting of native vegetation such as vetiver grass, bamboo, and other riparian species can stabilize riverbanks, reduce erosion, and provide critical habitat. These species are not only ecologically appropriate but also fast-growing and effective in filtering runoff.

Community Engagement

Sustainable conservation is impossible without the support and participation of local communities. Empowering residents through training and awareness is essential.

- **Train Local Water User Committees:** Community-based water users should be trained to monitor groundwater levels, stream flow, and water quality. These citizen-led groups can serve as the municipality's eyes and ears, reporting pollution incidents, illegal dumping, and wetland encroachment in real time.
- **Raise Public Awareness:** A major barrier to wetland conservation is the persistent perception that wetlands are "wastelands." Through school programs, media campaigns, and village dialogues, the municipality must promote awareness of the ecological and economic value of wetlands such as flood mitigation, water purification, and habitat provision. Changing mindsets is key to changing behaviors.

Sustainable Land Use

A transition toward sustainable land use practices can significantly reduce the stress on Makindye Ssabagabo's ecosystems.

- **Promote Organic and Climate-Smart Agriculture:** Encouraging organic farming practices and integrated nutrient management can drastically reduce fertilizer runoff, which is a major contributor to wetland eutrophication. Municipal support through training, input subsidies, and certification programs can drive adoption.
- **Upgrade Sanitation Infrastructure:** Poor sanitation near wetlands has contributed to pollution and disease outbreaks. Promoting eco-toilets, waste-separating latrines, and

safe sludge management will help prevent fecal contamination of wetlands and water bodies, especially in informal settlements.

Monitoring and Research

Science-based management requires real-time data and up-to-date ecological information.

- **Conduct Regular Biodiversity Surveys:** The current biodiversity database in Makindye Ssabagabo is outdated. Comprehensive surveys to update floral and faunal inventories are crucial for tracking conservation success and identifying emerging threats. Particular attention should be paid to indicator species such as amphibians, birds, and wetland plants.

Table 3: Summary of the DPSIR Framework for Biodiversity

Component	Details
Drivers	Urbanization, agriculture, mining, and industrial expansion
Pressures	Pollution, encroachment, deforestation
State	59.7% of wetlands degraded; declining biodiversity
Impact	Flooding, health risks, biodiversity loss, reduced livelihoods
Response	Demarcation, enforcement, rehabilitation, and awareness campaigns

2.3 Wetlands

2.3.1 State and trend

Wetlands are among the most critical ecosystems in Makindye Ssabagabo Municipality, playing a vital role in biodiversity conservation, water regulation, flood control, and supporting local livelihoods. However, over the last two decades, these natural assets have come under immense pressure due to urban expansion, unsustainable land use practices, and weak enforcement of environmental regulations. A closer examination of the current wetland coverage and condition in the municipality reveals a troubling pattern of decline and degradation, raising urgent concerns for sustainable urban planning and environmental management.

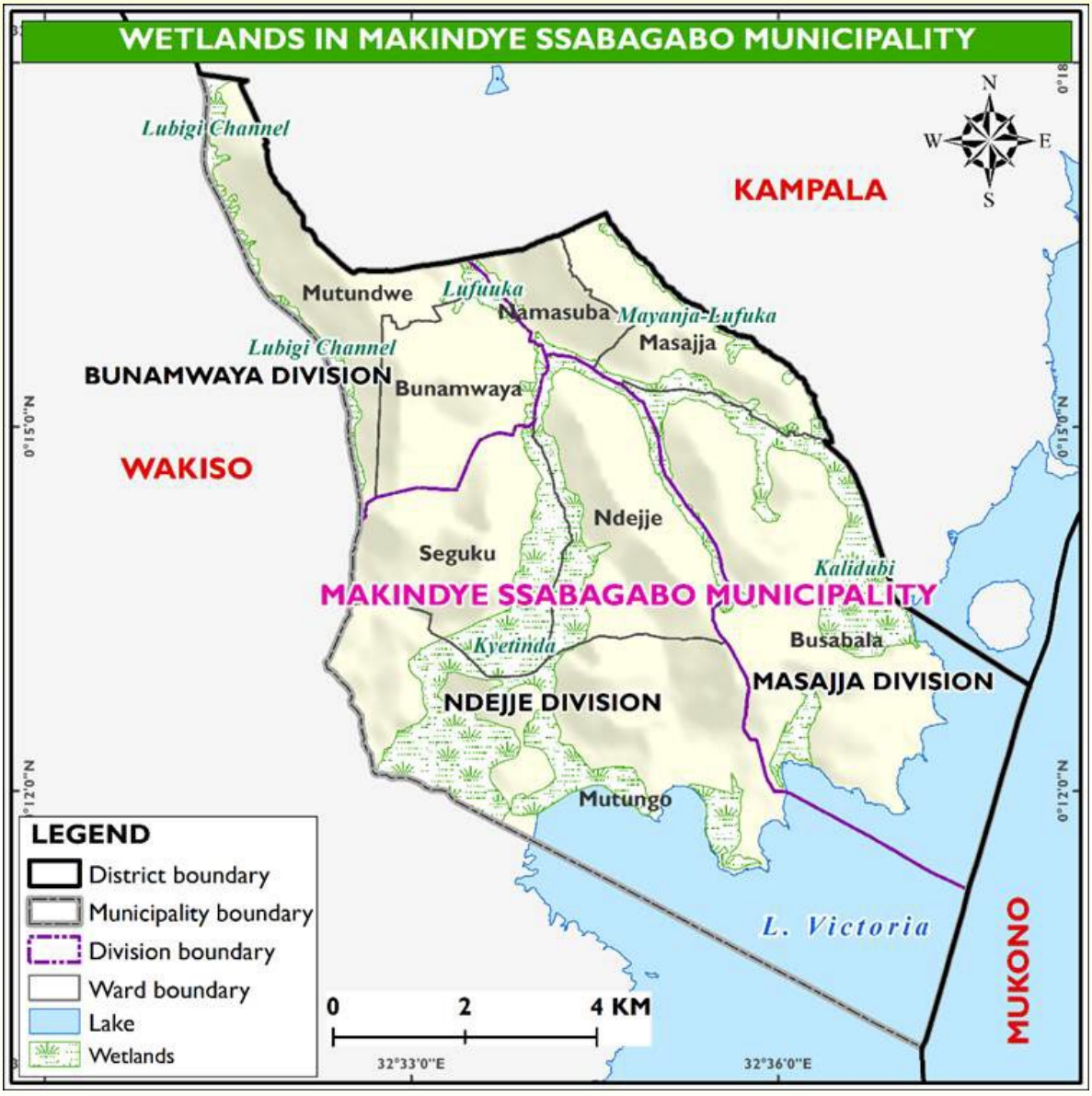


Figure 5: Wetlands in Makindye Ssabagabo Municipality

Wetland Coverage

As of 2025, the total wetland area in Makindye Ssabagabo stands at 1,600.8 hectares, which constitutes 18.9% of the municipality's total surface area. This figure includes both permanent and seasonal wetlands scattered across the divisions of Bunamwaya, Masajja, and Ndejje, with Ndejje hosting the largest share at 58% of total wetland area (Figure 12). These wetlands include key systems such as Kaliddubi, Kyetinda, Lufuka, Mayanja–Lufuka, and Lubigi Channel, which serve as habitats for diverse flora and fauna while offering essential ecosystem services such as water purification, groundwater recharge, and climate regulation.

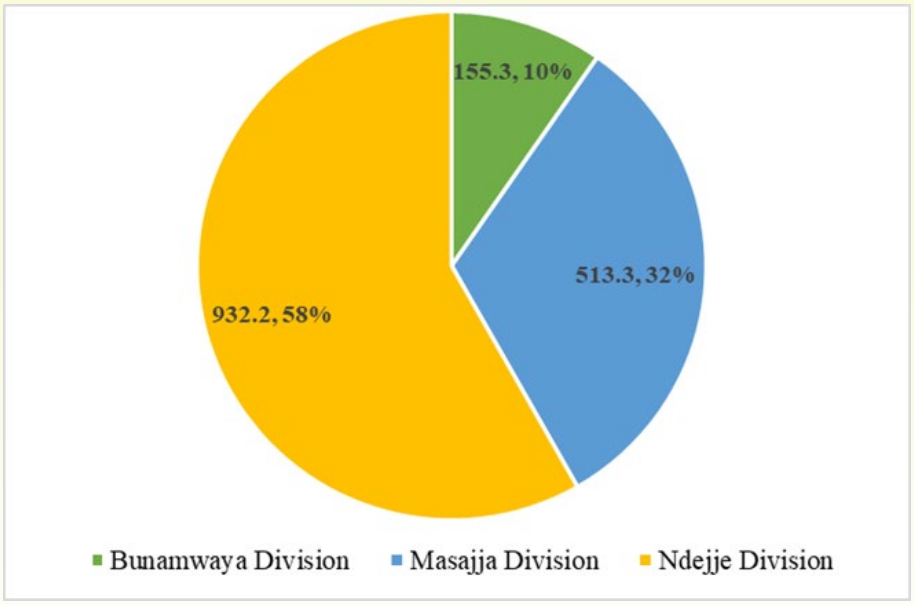


Figure 6: Coverage of wetlands by division in Makindye Ssabagabo Municipality

Wetland Condition

Despite their ecological and socio-economic value, the condition of wetlands in the municipality has deteriorated significantly. By 2025, 59.7% of the wetlands—equivalent to 947.6 hectares—are classified as degraded. These areas are characterized by conversion into built-up zones, croplands, sand mining pits, and areas of solid waste dumping. In contrast, only 640.9 hectares (40.3%) remain intact, mostly located in sections of Ndejje and Mutungo where encroachment is relatively lower (Figure 13). The degraded wetlands have not only lost their vegetative cover but also their natural hydrological and ecological functions, contributing to increased flooding, poor water quality, and reduced biodiversity.

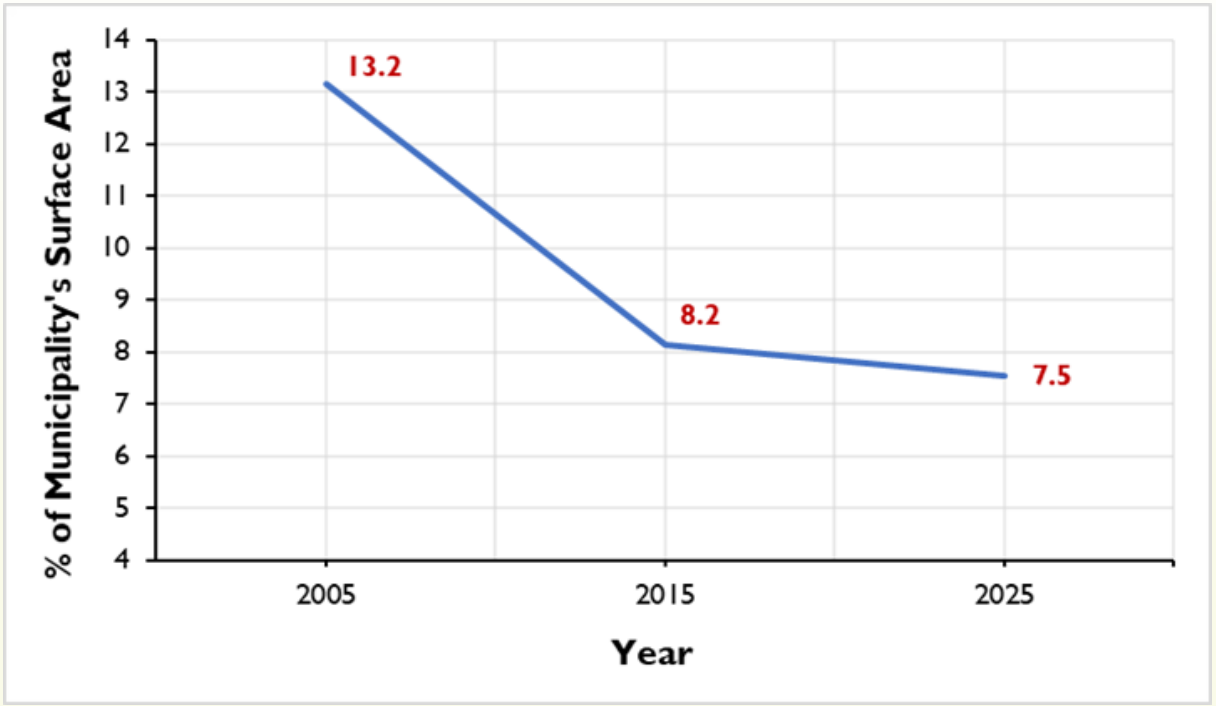


Figure 7: Coverage of intact wetlands in Makindye Ssabagabo Municipality in 2005, 2015, and 2025

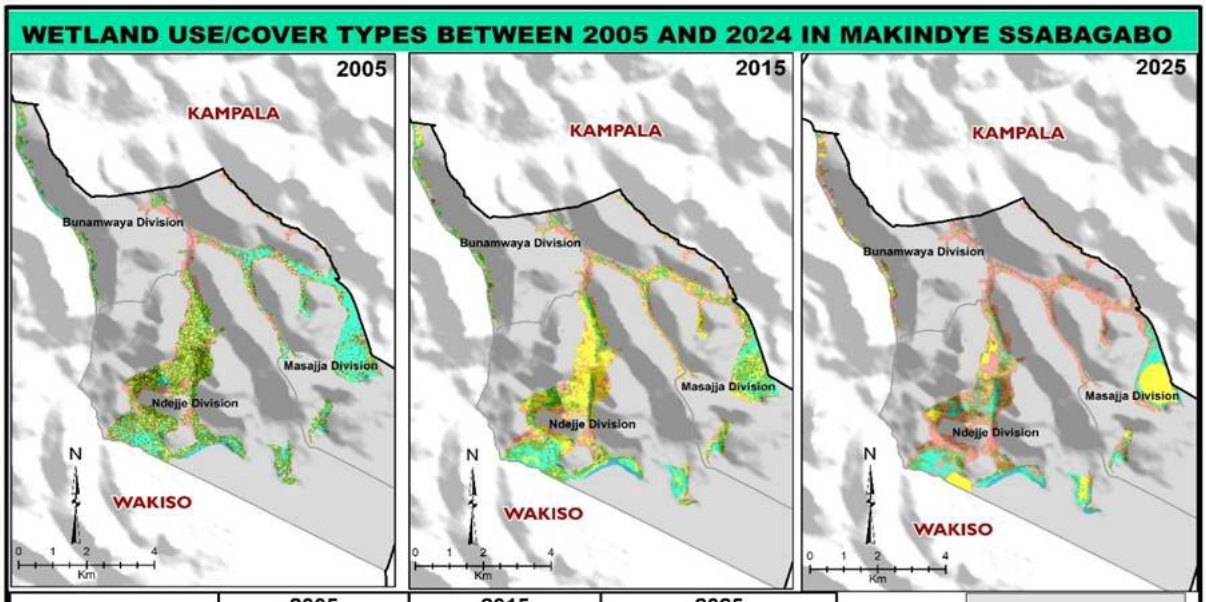
Long-term Wetland Loss: 2005 to 2025

An alarming trend is evident when examining the historical trajectory of wetland loss. In 2005, Makindye Ssabagabo had 1,118.7 hectares of intact wetlands. Over a 20-year period, that figure has fallen sharply to 640.9 hectares, representing a net loss of 477.8 hectares. This equates to a 5.6% reduction of the municipality's total land area and a 42.7% decrease in intact wetlands. The driving forces behind this loss include rapid population growth, informal settlements, infrastructure expansion, and land speculation, particularly in areas close to Kampala and along major road corridors.

Land Use and Wetland Cover Changes (2005 vs 2025)

The changes in wetland use and land cover between 2005 and 2025 further illustrate the growing pressure on these ecosystems: Built-up areas within wetlands have increased by 28.8%, becoming the dominant land use category by 2025 (Figure 18). This reflects widespread construction of homes, commercial buildings, and roads within former wetland zones. Croplands have expanded marginally by 0.7%. The combined area under papyrus vegetation, natural woodlands, and grasslands has declined by 33.3%, indicating a loss of biodiversity and native vegetation that previously supported ecological resilience.

Open water coverage within wetland zones has shown a slight increase, largely attributed to the artificial expansion of water surfaces for agriculture and drainage alteration for land development. By 2025, built-up areas account for 45% of the wetland zones, while croplands represent another 15%. This dominance of anthropogenic land uses underscores the extent to which natural wetlands have been converted into economically productive but ecologically damaging spaces.



Wetland use/ cover types	2005 Area (acres)	2005 %	2015 Area (acres)	2015 %	2025 Area (acres)	2025 %
Built-up areas	631.0	16.0	853.2	21.6	1,786.2	45.2
Croplands	561.1	14.2	1,387.9	35.1	585.8	14.8
Grasslands	1,035.2	26.2	825.0	20.9	465.4	11.8
Open water	22.8	0.6	44.5	1.1	169.3	4.3
Papyrus	985.0	24.9	621.3	15.7	488.4	12.3
Woodlands	720.5	18.2	223.7	5.7	460.6	11.6
Total	**3,955.7**	**100**	**3,955.7**	**100**	**3,955.7**	**100**

Figure 8: Wetland use/cover changes between 2005 and 2025 in Makindye Ssabagabo Municipality

Seasonal vs Permanent Wetlands

Makindye Ssabagabo's wetlands are composed of both permanent and seasonal systems: Permanent wetlands make up the majority, comprising 69.5% of the total wetland area (1,112.4 hectares). These areas are consistently waterlogged and perform vital year-round functions in water filtration, aquifer recharge, and habitat provision. Seasonal wetlands account for 30.5%, or 488.4 hectares, and they flood during rainy seasons (Figure 14). However, they are more vulnerable to conversion for short-term agriculture and settlement due to their dry-season accessibility. Seasonal wetlands are often undervalued and face higher degradation rates, despite their importance for buffering floodwaters and supporting seasonal biodiversity cycles.

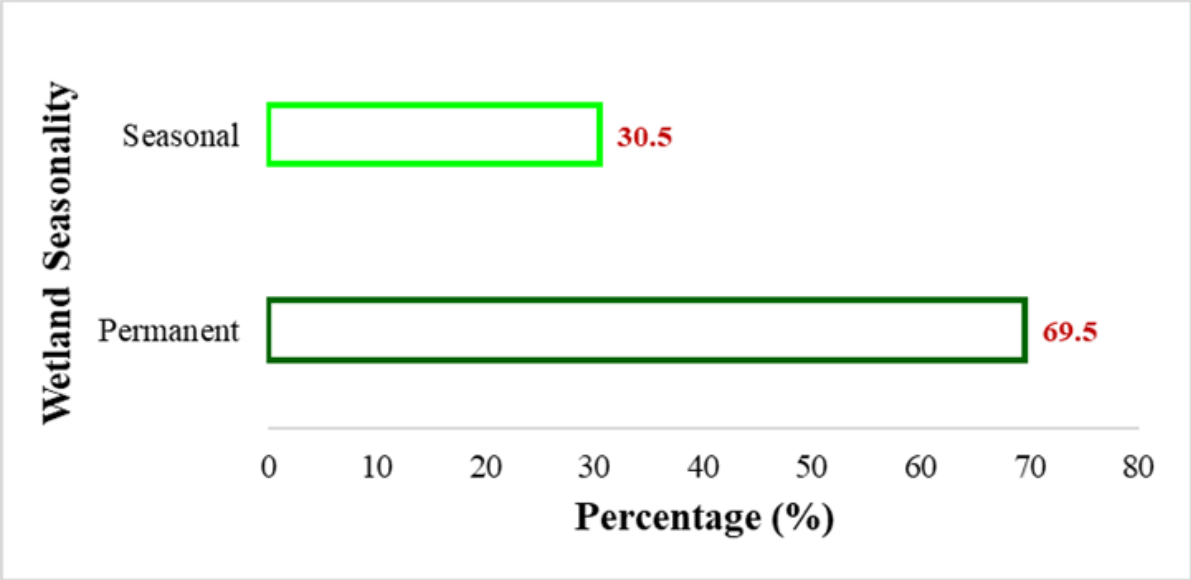


Figure 9: Wetland seasonality in Makindye Ssabagabo Municipality

2.3.2 Drivers of Wetland Degradation

The wetlands of Makindye Ssabagabo Municipality are undergoing significant degradation driven by a complex interplay of anthropogenic activities and institutional weaknesses. These drivers, both direct and indirect, have accelerated the loss of wetland area and quality, compromising their ecological functions and the services they provide to the urban population. Based on local assessments and stakeholder consultations, the primary drivers of wetland degradation in the municipality can be described in concrete detail as follows:

Urban Expansion and Infrastructural Development

The most prominent driver of wetland degradation is rapid urban expansion, accounting for 18.3% of all degradation drivers reported by surveyed households. As the municipality continues to urbanize, wetlands have increasingly become targets for real estate development, the construction of roads, fuel stations, and other forms of physical infrastructure. This development often occurs without proper environmental assessments or consideration for wetland boundaries. The divisions most heavily affected are Bunamwaya, Masajja, and Ndejje, where informal settlements, industrial parks, and roadworks have encroached into protected wetland areas. This has led to the direct loss of wetland vegetation, disrupted hydrological flows, and increased surface runoff, which in turn exacerbates flooding and sedimentation downstream.



Figure 10: Urban extension into a wetland section in Makindye Ssabagabo

Solid Waste Dumping

Illegal solid waste disposal into wetlands represents another significant threat, contributing 10.2% to the overall degradation. Communities in areas such as Kaliddubi, Kawagga, and Lubigi frequently dispose of household and commercial waste—including plastics, food remnants, and non-biodegradable materials—into nearby wetlands. This practice clogs natural drainage systems, alters the physical structure of wetlands, and introduces harmful substances such as microplastics, heavy metals, and pathogens into aquatic environments. In addition to diminishing water quality, waste dumping creates unsightly landscapes and contributes to the spread of waterborne and vector-borne diseases.

Sand and Clay Mining

Unregulated extraction of sand and clay, particularly in Kajjansi and Bunamwaya, has intensified wetland degradation, contributing 8.8% to the total drivers. These materials are extracted for use in construction and brickmaking, but the mining processes involve the clearing of vegetation, excavation of wetland soil, and compaction of land surfaces. As a result, wetlands in these areas suffer from severe erosion, increased sedimentation in nearby water channels, and the loss of microhabitats essential for aquatic and semi-aquatic species. Mining also undermines the structural integrity of wetlands, reducing their water storage and filtration capacities.

Brickmaking and Agricultural Encroachment

Wetlands have also been converted into agricultural plots and brick kilns, accounting for 8.8–9.9% of degradation. Residents utilize the fertile soils and constant moisture in wetlands to grow rice, yams, and vegetables, often without proper land use controls. Others extract clay from wetland soils for brick production, drying bricks directly within the wetland areas. While these activities are economically beneficial to local communities, they are ecologically unsustainable. Repeated tilling and burning of wetland vegetation, along with the diversion of water for irrigation or brick soaking, degrade the wetland's natural structure and disrupt wildlife habitats. In the long term, this leads to reduced biodiversity and altered seasonal wetland hydrology.

Lax Governance and Weak Law Enforcement

One of the most persistent underlying drivers is institutional weakness, which facilitates and sustains illegal activities in wetland zones. Reports indicate that some local council leaders are complicit in the illegal sale of wetland land, undermining official conservation efforts. Additionally, although national and local wetland protection laws exist, their enforcement remains weak. The

limited law enforcement allows continued degradation without consequence and erodes public trust in environmental governance institutions like NEMA and the Municipal Environment Office.

Industrial Effluents and Untreated Greywater

Finally, pollution from industrial and domestic sources is a growing concern. In semi-industrial zones such as Lubigi, Kajjansi, and Lufuka, garages, car washes, and workshops discharge untreated greywater containing oil, grease, detergents, and heavy metals directly into wetlands. These pollutants increase the chemical load of wetland waters, reduce oxygen levels, and lead to the death of aquatic flora and fauna. The toxic buildup also compromises the wetland's ability to purify water, which has implications for downstream users and ecosystems. Without pre-treatment systems or effective waste capture infrastructure, these pollutants continue to accumulate and intensify degradation.

2.3.3 Impact of Wetland Degradation

Wetlands are foundational to the ecological balance, water security, and livelihood stability of Makindye Ssabagabo Municipality. However, years of unchecked encroachment, pollution, and poor land use practices have severely undermined the capacity of these ecosystems to function effectively. The degradation of wetlands has triggered a cascade of both ecological and socioeconomic impacts, threatening biodiversity, public health, and urban resilience.

Ecological Impacts

Biodiversity Loss: One of the most immediate and visible consequences of wetland degradation is the loss of biodiversity. Native plant species such as Cyperus papyrus, previously dominant in many wetlands, are now declining due to physical destruction and competition from invasive species. The clearing of wetlands for farming and construction has also displaced key animal species, including frogs, fish, and bird species such as the grey crowned crane, Uganda's national bird. These developments have led to habitat fragmentation, reducing connectivity between wetlands and other ecological zones, which limits species movement and breeding. As these habitats shrink and degrade, populations become increasingly isolated, vulnerable to extinction, and less resilient to environmental changes such as floods or droughts.

Water Quality Deterioration: Another significant ecological impact is the decline in water quality within wetland systems. Chemical and organic pollution from solid waste, agricultural runoff, and industrial effluents has led to elevated levels of Total Dissolved Solids (TDS) and Electrical Conductivity (EC). For example, water in Lubigi Wetland shows a TDS of 1,141 mg/L and an EC of 1,993 µS/cm, both well above acceptable ecological thresholds. Physically, this pollution manifests in discolored water—often reddish or dark grey—alongside algal blooms and foul odors, all of which indicate eutrophication and excessive nutrient loading. The result is reduced oxygen availability, leading to the death of aquatic organisms and the collapse of fragile aquatic food chains.

Hydrological Disruption: The physical structure and flow of water in wetlands have also been severely disrupted. Wetlands that once acted as natural sponges—soaking up rainwater and slowly releasing it—are now choked by silt, vegetation overgrowth, and human debris. This has created sluggish or stagnant streamflow, particularly visible in Lufuka and Kaliddubi wetlands. The hydrological imbalance reduces the wetland's capacity to regulate floods, increasing the risk of flash flooding, especially in downstream areas. It also affects the natural process of groundwater

recharge, as the compacted and contaminated wetland soils lose their permeability and water-holding capacity.

Socioeconomic Impacts

Increased Flooding: The encroachment and degradation of wetlands have significantly worsened urban flooding in wetland-adjacent neighborhoods. Areas such as Namasuba, Kabuuma, and Bunamwaya frequently experience flash floods during the rainy season, as blocked wetlands and drainage channels can no longer absorb and regulate surface runoff. The consequences are severe as homes are damaged, residents are displaced, and essential services such as schools, shops, and markets are disrupted. Municipal emergency resources are increasingly diverted toward flood response, drawing funds away from long-term infrastructure and development projects.

Public Health Risks: With the decline of natural water purification functions, degraded wetlands have become breeding grounds for disease vectors, particularly mosquitoes and water snails. This has led to increased outbreaks of malaria, bilharzia, and cholera, particularly in densely populated low-income areas like Mutundwe and Masajja. Additionally, standing polluted water in wetlands contributes to foul smells, reduced air quality, and skin infections, further undermining the well-being of vulnerable populations, especially children and the elderly.

Livelihood Loss: Wetlands have traditionally supported a wide range of livelihoods in Makindye Ssabagabo—such as papyrus harvesting, fishing, clay and sand mining, and even herbal medicine collection. However, as biodiversity diminishes and access to wetlands becomes restricted or unviable, many of these activities are no longer sustainable. For instance, declining fish stocks have driven many artisanal fishers out of business, while papyrus and craft materials have become scarce due to vegetation loss. The reduction in natural resources also increases economic vulnerability and food insecurity, particularly among women and youth, who are disproportionately dependent on wetland-based incomes.

Decline in Ecosystem Services: Perhaps most critically, wetland degradation has led to the collapse of essential ecosystem services: water filtration functions are compromised due to pollution and vegetation loss; groundwater recharge is reduced by land compaction and siltation; and temperature regulation—once aided by the moisture-retaining properties of wetlands—has weakened, contributing to microclimatic extremes and urban heat pockets. The long-term consequences of losing these services are profound: they affect not just the environment but also urban sustainability, climate resilience, and public welfare.

2.3.4 Responses to Wetland Degradation

Makindye Ssabagabo Municipality has undertaken a series of interventions in response to the growing threat of wetland degradation. These actions—spanning policy enforcement, physical restoration, infrastructure improvement, and community engagement—are aimed at halting further destruction, reversing ecological damage, and building resilience within urban and peri-urban ecosystems. While many of these efforts have been reactive, they mark important steps toward a more sustainable wetland management approach.



Figure 11: Bridge construction across a wetland section in Makindye Ssabagabo to control flooding.

Policy and Enforcement Measures

Wetland Demarcation (2022–2024): A critical policy response was the physical demarcation of wetland boundaries, implemented between 2022 and 2024. Key wetlands such as Kyetinda, Mayanja–Lufuka, and the Lubigi Channel were marked with concrete pillars to legally and visibly outline protected zones. This initiative was spearheaded by the National Environment Management Authority (NEMA) in collaboration with the Municipal Environment Office and aimed to curb encroachment by making wetland limits unambiguous to developers and communities.

Eviction and Demolition Campaigns (2020–2022): To reinforce legal protections, evictions and demolitions were carried out in Kajjansi, Mutungo, and Bunamwaya, where illegal settlements and commercial structures had overtaken ecologically sensitive zones. These operations focused on high-risk areas, particularly where biodiversity loss, blocked drainage, and recurrent flooding had been reported. The aim was not only to remove encroachers but also to create room for ecological restoration and water flow improvement. These campaigns were often controversial and met with resistance but were deemed necessary to restore environmental order.

Drainage Rehabilitation (2023): In response to severe flooding, especially along the River Lufuka, the municipality embarked on a drainage rehabilitation program in 2023. This included widening and clearing blocked drains, installing sedimentation traps, and Re-lining rainwater pathways. These interventions were carried out in flood-prone zones, particularly in Namasuba, Mutundwe, and Bunamwaya, to enhance stormwater flow and reduce the impact of urban runoff on wetlands.

Spring and Borehole Rehabilitation: To improve access to safe water and reduce dependency on contaminated wetland sources, springs and boreholes in Ndejje, Kabuuma, and Mutundwe were rehabilitated. These upgrades involved fencing off water sources, constructing drainage aprons, and protecting from nearby latrines and waste dumps. These interventions helped minimize contamination risks and restore community trust in formal water systems.

Community-Based Restoration: Involving local communities in wetland restoration has been a key strategy. Initiatives include tree-planting campaigns in Walugogo and Kyetinda wetlands and vegetative buffer restoration using bamboo and vetiver grass, particularly by youth groups in Kigo,

who successfully restored over 2 hectares in 2021. These efforts enhance erosion control, promote community ownership, and restore biodiversity-supporting habitats.

Awareness and Community Engagement

WASH Campaigns: Organizations such as WaterAid Uganda have launched WASH campaigns targeting schools and households in environmentally vulnerable areas like Mutundwe and Kabuuma. These campaigns focus on proper sanitation practices, spring protection, and safe waste disposal. The objective is to change community behavior and build a culture of environmental stewardship.

Environmental Watchdog Groups: To sustain these efforts, environmental watchdog groups have been established in communities such as Bunamwaya East and Namasuba. These groups are trained to monitor illegal waste dumping, report encroachment, and work with municipal authorities to enforce wetland protection measures. This form of community-based monitoring enhances early detection of environmental violations and supports a more responsive municipal system.

Table 4: DPSIR for wetlands in Makindye Ssabagabo

Component	Key Highlights
Drivers	Urban growth, sand/clay mining, poor waste disposal, industrial effluents
Pressures	Encroachment, pollution, drainage blockage, and vegetation loss
State	59.7% degraded; 477.8 ha lost between 2005 and 2025
Impact	Biodiversity loss, flooding, water contamination, and livelihood disruption
Response	Boundary demarcation, enforcement, restoration projects, and community mobilization

2.3.5 Recommendations

To address the accelerating degradation of wetlands in Makindye Ssabagabo Municipality, a comprehensive and multifaceted strategy is required. The recommendations below propose concrete and actionable interventions aimed at strengthening conservation, restoring ecological function, protecting water quality, and engaging stakeholders at all levels.

1. Strengthen Wetland Protection and Regulation

- Conduct regular patrols and surveillance through the Municipal Environment Office and trained community monitors to detect illegal dumping, land clearing, and unauthorized construction early.

2. Implement Ecological Restoration Measures

- Restoring the physical and ecological integrity of degraded wetlands is crucial to reversing environmental damage and restoring ecosystem services.
- Revegetate degraded buffers using native plant species such as vetiver grass, papyrus, bamboo, and indigenous trees to stabilize soils and provide habitat.
- Clear blocked and silted wetland channels to restore natural hydrological flow, improving flood absorption and groundwater recharge.
- Prioritize riparian zone restoration to filter surface runoff, reduce sedimentation, and prevent erosion.

3. Improve Wastewater and Pollution Control

- Effective management of pollution sources is essential to safeguard wetland water quality.
- Construct decentralized wastewater treatment units, such as bio-digesters and constructed wetlands, in pollution hotspots like Masajja, Mutundwe, and Bunamwaya.
- Monitor and regulate industrial effluent discharges, especially from car wash bays, auto garages, and informal workshops in zones like Kajjansi.
- Enforce the Polluter Pays Principle by penalizing businesses that discharge untreated wastewater into wetlands and suspending repeat offenders until compliance is verified.

4. Enhance Community Awareness and Participation

- Local communities must be informed, engaged, and empowered to participate in conservation efforts.
- Launch targeted public sensitization campaigns that highlight the economic, cultural, and ecological value of wetlands.
- Counter the perception of wetlands as "wastelands" by demonstrating their role in flood regulation, clean water supply, and biodiversity conservation.
- Support environmental watchdog groups at the ward level to monitor wetland conditions and report violations to municipal authorities.

5. Strengthen Institutional Capacity and Coordination

- The sustainability of interventions depends on adequate institutional strength and interagency coordination.
- Hire and train technical personnel on wetland management at both municipal and division levels, addressing current staffing shortages.
- Increase municipal budget allocations for wetland-related activities, including enforcement, monitoring, and restoration.
- Improve coordination between NEMA and municipal structures, ensuring that enforcement actions (e.g., evictions) are collaboratively planned and communicated.

6. Control Agricultural and Resource Extraction Activities

- Wetland degradation due to unsustainable economic activities must be addressed through regulation and livelihood transition.
- Implement farming guidelines within wetland-adjacent areas, promoting buffer planting, contour farming, and organic nutrient management to reduce runoff.
- Ban or strictly limit sand and clay mining, brickmaking, and overgrazing in ecologically sensitive wetlands.
- Facilitate alternative livelihood programs, including ecotourism, aquaculture, and sustainable farming, to reduce dependency on destructive wetland uses.

7. Upgrade Urban Infrastructure to Support Wetland Health

- Urban planning must incorporate nature-based solutions that complement wetland ecosystems:
- Expand and maintain stormwater drainage infrastructure to reduce flooding and waterlogging in wetland zones.
- Install sedimentation traps, stormwater retention ponds, and percolation pits in high-runoff areas to prevent sediment and pollutant inflow.
- Use GIS mapping and hydrological models to identify restoration priority zones and guide infrastructure investments effectively.

8. Improve Groundwater Protection Near Wetlands

- Protection of groundwater sources adjacent to wetlands is vital for safe water supply and sanitation.
- Enforce sanitary protection zones (minimum 30 meters) around boreholes and springs, especially in vulnerable areas like Ndejje, Mutundwe, and Kabuuma.
- Promote the use of ecological sanitation systems such as urine-diverting dry toilets (UDDTs) in densely populated areas with shallow aquifers to reduce leachate pollution.

9. Promote Long-Term Monitoring and Early Warning Systems

- Building resilience requires proactive and ongoing data collection and risk preparedness.
- Establish community-based water quality monitoring programs, equipped with tools like manual nitrate test strips, pH meters, and simple turbidity tests.
- Set up early warning systems for flood risks and wetland degradation, leveraging mobile alerts and local radio communication to notify residents of imminent threats.

2.4 Waste Management

2.4.1 State and trend of waste management

Waste Generation in the Municipality

Solid waste generation in Makindye Ssabagabo Municipality is approximately 1,899 tonnes per day, totaling 693,128 tonnes annually. Household sources lead in waste generation, contributing 1,257 tonnes daily, which accounts for 98.9 per cent of the total daily waste produced within the municipality. Non-household sources (businesses, schools, restaurants, and health centers) contribute about 1 per cent (**Figure 1)**.

Organic waste leads with a daily generation of around 63%, followed by plastic waste at 25%. Paper and glass waste contribute 6.4% and 1.5%, respectively. Metallic and electronic waste contribute less than 1%, while other waste types, such as textile waste, rubber waste, ceramics, foam ash, and charcoal residues, contribute 3.2%, as shown in **Figure 2**. Masajja division leads in waste generation with an annual generation of 317,991 tonnes (45.9%). Bunamwaya and Ndejje contribute 230,251 tonnes (33.2%) and 144,884.6 tonnes (20.9%), respectively, as shown in **Figure 3**.

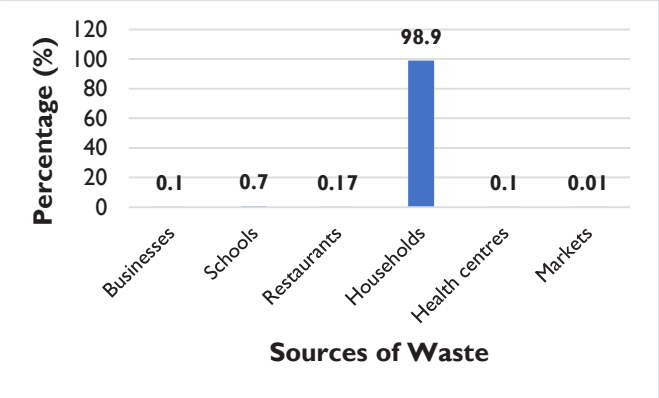


Figure 12: Waste generation by source

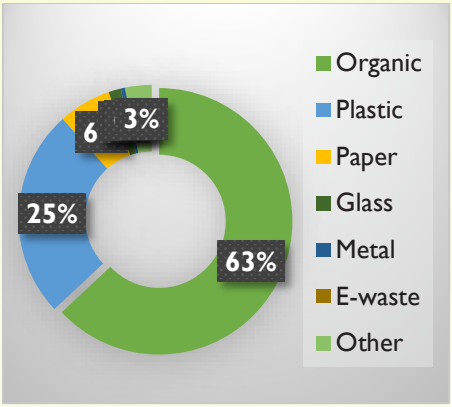


Figure 13: Waste generation per waste type

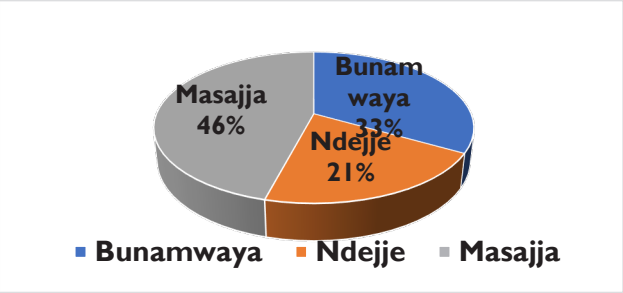


Figure 14: Waste generation per division

The pre-feasibility study of municipal organic waste recovery alternatives conducted by GIZ in 2022 indicated that low-income households generate the highest amount of food waste (around 150-175 tonnes/day) followed by middle income households (around 140 tonnes/day) and then high-income households (around 50 tonnes/day) (**Figure 4**). This shows that economically constrained households waste significant amounts of food possibly due to lack of proper storage (e.g., no refrigeration), buying in bulk due to cost, leading to spoilage, and limited knowledge or resources to preserve or repurpose leftovers. Organic waste combining food (kitchen/canteen) and green (garden/park) waste represents around 60-70% of the total household waste generated.

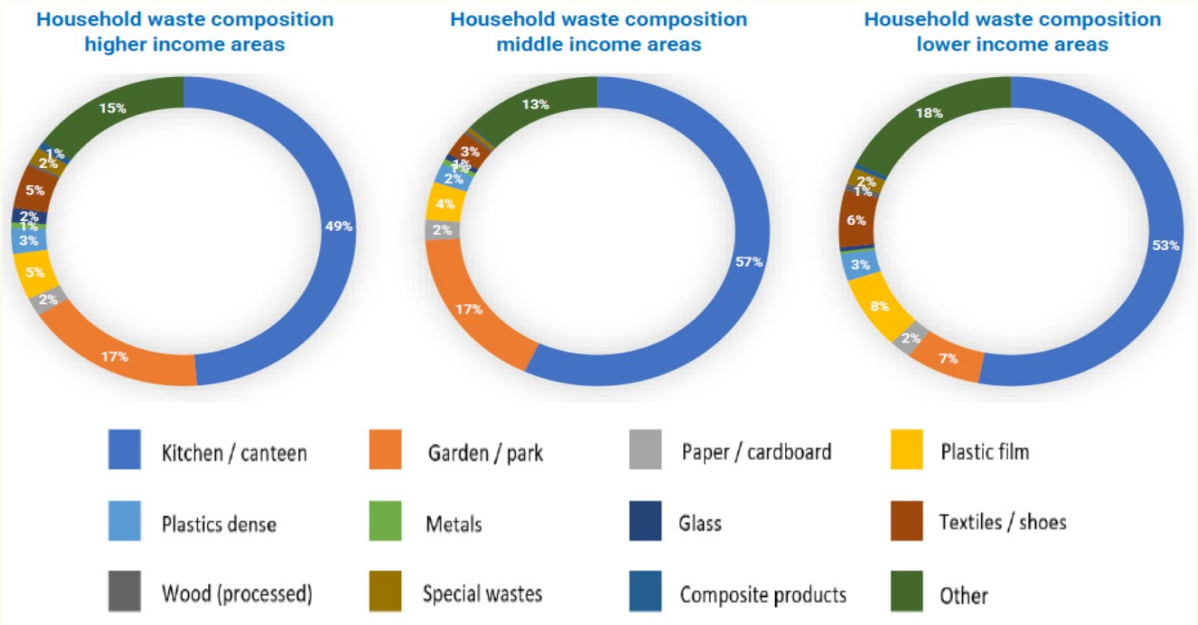


Figure 15: Household waste generation in lower, middle-, and higher-income areas
Source: GIZ, 2022

Between 2025 and 2030, generation of organic waste in Makindye Ssabagabo Municipality is projected to increase by 13,706 tonnes (12.7%) (**Figure 5**). An increase in organic waste generation in the municipality can have serious environmental, health, and economic consequences. Environmentally, it leads to higher greenhouse gas emissions particularly methane from landfills and increases the risk of water and soil contamination. Public health may also be affected due to the attraction of pests, unpleasant Odors, and unsanitary conditions in affected areas. The growing volume of waste places a strain on municipal waste management systems, resulting in higher collection and disposal costs and reducing landfill lifespan. Additionally, the improper handling of organic waste represents a lost opportunity to convert it into valuable resources like compost or biogas, which could also create jobs and support a circular economy. Ultimately, rising organic waste reflects inefficient resource use and highlights the urgent need for sustainable waste management practices.

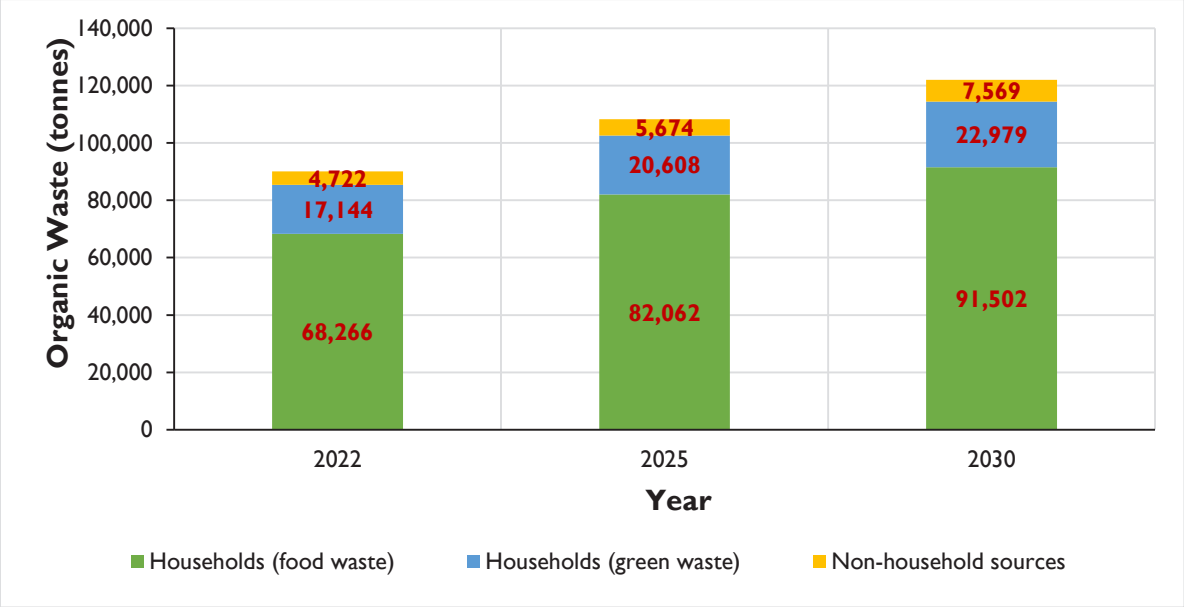


Figure 16: Projected generation of organic waste in Makindye Ssabagabo Municipality between 2022 and 2030

Data Source: GIZ, 2022

Waste Segregation

Waste segregation in healthcare facilities is carried out using colour-coded medical bins to ensure proper handling and disposal. Black bins are used to collect non-infectious waste such as food remnants, paper, plastics, and other general refuse. Yellow bins are designated for infectious waste, including used gloves, cotton wool, swabs, bandages, sliced cotton, and gauze. Red bins are reserved for highly infectious waste, such as anatomical waste, including amputated body parts and placentas. While non-infectious waste is collected, transported, and disposed of by domestic waste collection companies. Infectious and highly infectious waste is handled separately by specialised service providers.

Waste Sorting and Recovery

Outside of health facilities, there is no formal waste sorting at the source. All types of waste are typically deposited into a single bin, then transported and disposed of as mixed waste. The only sorting that occurs is done informally by individuals who collect plastic and metallic waste for

income. These collectors gather the materials indiscriminately from various locations and later sell them to designated plastic and metal waste collection centres.

Waste Treatment

Waste treatment is primarily done by large industries which produce liquid waste. Such industries treat their liquid waste using guidelines provided by NEMA, which also monitors the treatment quality and compliance with standards. According to the physical development plan of Makindye Ssabagabo municipality (2019-2040), no public sewerage systems exist in municipality except small treatment plants and waste stabilisation ponds owned and operated by estate developers and institutions namely; Lubowa Housing Estate (04 waste ponds) by NHCC and Serena resort (02 waste ponds) by Hotels and Cape villa homes.

Waste Collection, Transportation, and Disposal

Waste collection in the municipality is conducted by contracted and non-contracted service providers. The contracted service providers operate under formal agreements with Makindye Ssabagabo Municipal Council, while the non-contracted providers lack official arrangements with the municipality and instead have private agreements with individual households for waste collection services. The Council has contracted one company per division to handle municipal solid waste: A&M Executive Cleaning Services for Masajja Division, Dumpster Klin for Ndejje Division, and Clean Waste Management Limited for Bunamwaya Division. Non-contracted service providers for waste collection include Home Klin (U) Ltd, Bin It Services Limited, De-Waste (U) Ltd, Nabugabo Updeal Joint Venture, and Ark-Waste and Besta Waste Services. Besides the non-contracted service providers, individual waste collectors also operate within the municipality, mainly serving households. They transport waste using tricycles (**Plate 1**), bicycles, wheelbarrows, or by carrying it manually. Individuals commonly referred to as "Kawenjjas" also collect plastic waste from various sources, including households, businesses, illegal dumping sites, and streets.

The collected waste is transported using light-duty collection trucks and dispose of at a designated dumpsite located in Buyala, Mpigi district, along Mityana Road. However, due to the long distance of approximately 30 kilometers from the division to Buyala, each truck can only complete one full trip per day. As a result, the total amount of waste collected by the service providers contracted by the Municipal Council, is limited to about 3,000 to 3,600 kilograms daily, which is a small fraction (0.3%) of the over 1,277 tonnes of waste generated across the municipality each day. In addition, the individual waste collectors often lack the means to transport waste to the designated disposal site in Buyala. As a result, they are suspected of illegally dumping waste in wetlands (**Plate 2**) and open spaces.



Plate 1:An individual waste collector arriving to pick waste from a household



Plate 2: Solid waste disposal in Lufuka wetland, Mutundwe parish, Bunamwaya division

Funding Gap in Waste Management

Despite growing urbanization and increasing volumes of solid waste, the financial resources allocated to waste management remain inadequate to meet infrastructure, operational, and capacity development needs. **Figure 6** presents waste management funding in Makindye Ssabagabo Municipality between 2020/2021 and 2024/2025 financial years.

In all financial years, the realized amounts are significantly lower than the planned amounts, indicating consistent underfunding or revenue underperformance. Financial year 2022/2023 had the highest planned budget (~100 million UGX), but the realized amount was one of the lowest (~15 million UGX), producing the widest gap (~84 million UGX). Financial year 2020/2021 showed the best realized performance relative to its plan (~40% of the planned amount was realized). The funding gap increased between 2020/2021 and 2022/2023 and then reduced between 2022/2023 and 2024/2025.

The consequences of the funding gap include inadequate collection coverage, especially in informal and peri-urban areas; frequent breakdowns of equipment and limited replacement of aging fleet; lack of investment in recycling, composting, or waste-to-energy innovations; insufficient staff and resources for enforcement and public education; and continued reliance on illegal dumping and uncontrolled disposal sites.

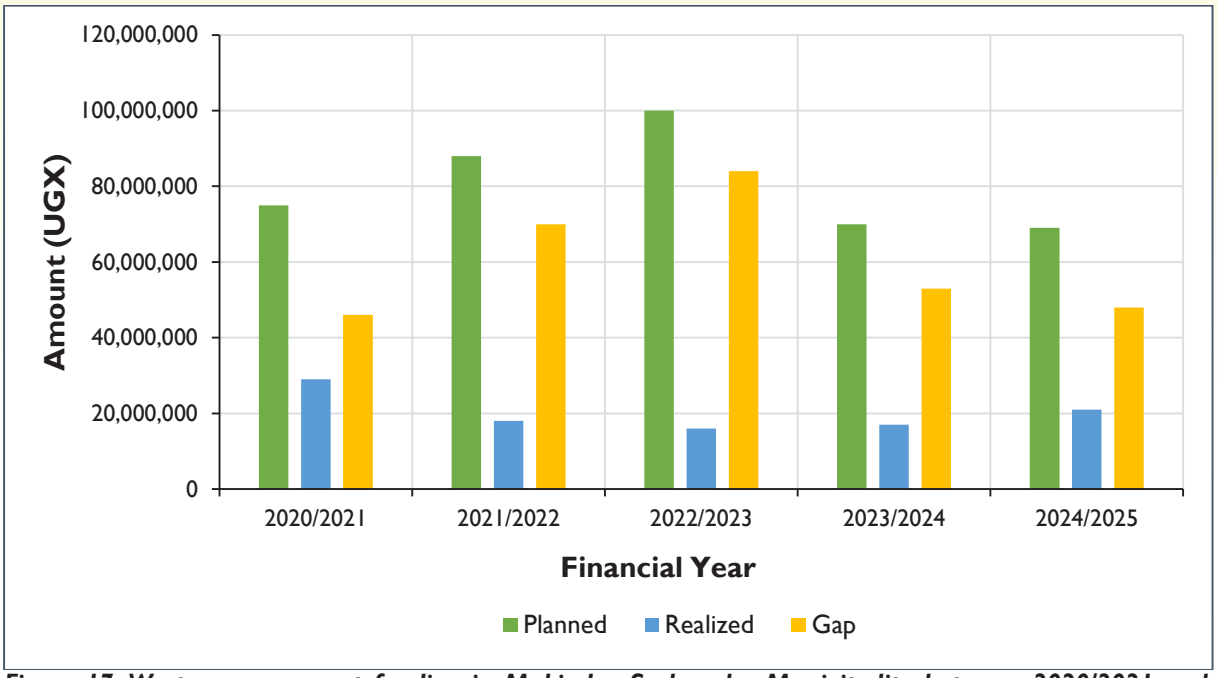


Figure 17: Waste management funding in Makindye Ssabagabo Municipality between 2020/2021 and 2024/2025

2.4.2 Drivers of poor waste management

Makindye Ssabagabo Municipality faces multiple challenges in achieving effective solid waste management, driven by a combination of structural, financial, institutional, and behavioral factors. The drivers of poor waste management in the Municipality are briefly described below;

Rapid urbanization and population growth: The municipality is experiencing fast-paced urban expansion and population growth. Between 2015 and 2025, the population in Makindye Ssabagabo municipality is projected to have increased by 256,100 people (88.7%) (UBOS, 2025). This has increased the volume of waste generated beyond the capacity of existing collection, transportation, and disposal infrastructure, resulting in overwhelmed systems and inefficient service delivery.

Inadequate financial resources: The funding gap for waste management in Makindye Ssabagabo Municipality rose from 46M UGX (61.3%) in FY 2020/2021 to 84M UGX (84%) in FY 2022/2023. Limited budget allocations limit the municipality's ability to invest in essential infrastructure development, enforcement personnel, equipment, innovations, and waste collection services.

Weak enforcement of regulations: Lax enforcement of waste management laws contributes to illegal dumping, open burning, and general non-compliance by residents and businesses. In addition,

the municipality lacks a byelaw on waste management. This creates a regulatory gap that hinders enforcement, accountability, and compliance with proper waste handling practices.

Limited capacity of contracted service providers: Each contracted waste service provider collects 0.3% of the total waste generated in the Municipality. This is attributed to inadequate equipment, trained personnel, and logistical resources, which leaves most waste uncollected in communities, leading to illegal dumping, accumulation of waste in public spaces, and increased health and environmental risks.

High waste collection fees: The monthly fee for contracted and non-contracted service providers typically ranges between UGX 10,000 and UGX 30,000, depending on the volume of waste generated. The low-income households that can't afford these fees resort to individual waste collectors, who have limited capacity to transport waste to Buyala. This leads to the dumping of waste in wetlands, open spaces, and drainage channels.

Long distance of disposal site in Buyala: This increases the time and cost associated with waste collection and transportation. When waste collection trucks must travel long distances, it results in delayed pickups, inefficient use of resources, and increased fuel consumption, which can strain the service providers' budgets. This often leads to irregular waste collection and the accumulation of waste in neighborhoods, contributing to illegal dumping and creating health and environmental hazards. Additionally, the lack of accessible disposal sites in close proximity discourages proper waste disposal practices among residents.

Low public awareness and participation: This hinders communities' understanding of proper waste disposal practices, such as waste segregation, recycling, and reducing waste generation. When people are unaware of the environmental and health impacts of improper waste disposal, they are less likely to engage in responsible waste management behaviors. This results in increased littering, illegal dumping, and contamination of waste streams, making waste management efforts less effective and straining municipal resources. Additionally, without public involvement in waste reduction initiatives, achieving long-term sustainability and community cooperation becomes difficult.

Poor infrastructure and logistical constraints: These limit the capacity to efficiently collect, transport, and dispose of waste across the municipality. Inadequate road networks, especially in informal or densely populated areas, hinder access for waste collection vehicles, resulting in uncollected waste and illegal dumping. A shortage of waste bins, collection points, and transfer stations further disrupts proper waste handling and storage at the community level. Additionally, insufficient or poorly maintained equipment, such as trucks and compactors, causes delays and breakdowns in service delivery. These logistical challenges create service gaps, contribute to environmental pollution, and undermine public confidence in municipal waste management systems.

Institutional weaknesses and poor coordination: Fragmented responsibilities among municipal departments and minimal collaboration with private and civil society actors reduce operational efficiency. When roles and responsibilities among municipal departments, service providers, and other stakeholders are unclear or poorly coordinated, efforts become fragmented and

inconsistent. This results in delayed decision-making, misallocation of resources, and failure to enforce regulations effectively. Moreover, the absence of integrated planning and collaboration limits innovation, reduces accountability, and undermines the development of comprehensive and sustainable waste management systems.

Limited private sector involvement: The private sector's role in waste collection, recycling, and resource recovery remains underdeveloped due to a lack of supportive policies and incentives. When the private sector is not actively engaged, there are fewer service providers, limited competition, and minimal advancement in technologies or practices such as recycling, composting, and waste-to-energy solutions. This places a heavier burden on under-resourced municipal systems, resulting in inefficient waste collection, transportation, and disposal.

Absence of waste segregation and recycling systems: This causes waste to be mixed together, making efficient handling, processing, and resource recovery difficult or impossible. Mixed waste contaminates recyclable materials, reduces the quality of compostable organic waste, and increases the volume of waste sent to landfills, accelerating their exhaustion. This not only raises environmental and health risks but also increases the cost and complexity of waste management. Without segregation and recycling, valuable materials are lost, and opportunities for circular economy practices, such as composting and reuse, are missed- undermining sustainability and efficiency in the waste sector.

2.4.3 Impact of poor waste management

Rapid urbanization, population growth, and limited institutional capacity in Makindye Ssabagabo Municipality have strained the existing waste management systems, resulting in various environmental and socio-economic challenges. The accumulation and improper disposal of waste have contributed to pollution, health risks, infrastructure strain, and reduced quality of life, particularly in low-income and high-density settlements. This section presents the key impacts of poor waste management within the municipality, highlighting how these issues undermine environmental sustainability, hinder equitable service delivery, and conflict with national and international policy frameworks.

Environmental pollution and ecosystem degradation: The indiscriminate dumping and unregulated disposal of waste particularly in wetlands, open spaces, and near water bodies; has led to widespread environmental degradation. These practices have led to contamination of soil and water resources, destruction of natural habitats, disruption of local ecosystems, and loss of biodiversity. This undermines National Environment Management Policy (NEMP) Objective 2.2.3, which promotes the sustainable use and conservation of natural resources, and obstructs progress toward SDG 6.3, which calls for improved water quality, and SDG 15.1 and 15.5, which focus on ecosystem conservation and halting biodiversity loss.

Public health risks: Accumulated waste in residential and commercial areas has become a breeding ground for disease vectors, contributing to the spread of typhoid, malaria, and respiratory infections. Open burning of waste further exacerbates air pollution and increases exposure to toxic emissions. These public health risks are in direct conflict with NEMP Objective 2.2.7, which aims to safeguard environmental health, and are also misaligned with SDG 3.3 and 3.9, which seek

to combat communicable diseases and reduce deaths from pollution, as well as SDG 11.6, which addresses the environmental impact of urban areas.

Increased incidence of urban flooding: The disposal of solid waste into drainage systems has resulted in frequent blockages, particularly during the rainy season. These blockages have led to localized flooding, causing property damage, disrupting transport and livelihoods, and exacerbating the spread of waterborne diseases. This challenge undermines disaster preparedness efforts under NEMP's integrated environmental planning objectives and hinders progress toward SDG 11.5, which targets urban resilience to disasters, and SDG 13.1, which emphasizes climate adaptation and risk reduction.

Deterioration of urban aesthetics and livability: The presence of scattered waste and illegal dumpsites has diminished the aesthetic quality of the municipality, deterring investment, tourism, and general urban livability. This deterioration counters NEMP Objective 2.2.10, which promotes sustainable urban development, and negatively impacts SDG 11.3, which calls for inclusive and sustainable urbanization, and SDG 8.9, which promotes sustainable tourism and local economic development.

Economic burden on municipal systems: The municipality faces financial and operational challenges due to limited waste infrastructure, inadequate resource recovery, and reliance on costly landfill disposal. These inefficiencies strain municipal budgets and accelerate the exhaustion of available disposal sites. This is inconsistent with NEMP Objective 2.2.12, which emphasizes building institutional capacity for effective environmental management, and stalls advancement toward SDG 12.5, which calls for waste reduction and recycling, and SDG 11.6, which promotes efficient urban waste management.

Widening social inequalities: Informal settlements and low-income communities experience the most severe consequences of poor waste management due to irregular or non-existent collection services. This contributes to environmental injustice, as vulnerable populations face disproportionate health and environmental risks. Such inequities contravene the NEMP's guiding principle of environmental justice and equal access to services, and obstruct progress toward SDG 1.4 (access to basic services), SDG 10.2 (social inclusion), and SDG 11.1 (safe and inclusive human settlements).

The impacts of poor waste management in Makindye Ssabagabo Municipality are widespread and interconnected. They compromise environmental sustainability, public health, economic development, and social equity. Effective interventions must be aligned with national policy objectives and global development goals to ensure a cleaner, healthier, and more resilient urban future.

2.4.4 Responses to poor waste management

The responses implemented by the municipality and partners to improve waste management in the municipality are presented below;

Acquisition of garbage truck: The municipality has initiated the purchase of a garbage truck to improve waste collection in the different divisions. The first truck was acquired using property

tax revenue, with plans to procure more, each designated for a specific ward. The garbage truck is used in the clean-up days of the different divisions and it is management by the municipal council.

Community clean-up days: In response to the mounting waste management challenges in Makindye Ssabagabo Municipality, community clean-up days have emerged as a practical and participatory strategy to promote environmental cleanliness, restore public spaces, and raise awareness about responsible waste disposal. These events have become a visible expression of citizen engagement and municipal leadership in tackling the pervasive problem of uncollected waste.

Enhancement of Water, Sanitation, and Hygiene (WASH) services: The municipality has collaborated with WaterAid to enhance WASH services. Through community forums known as "barazas", the communities are engaged in identifying and addressing WASH challenges. WaterAid also supports capacity-building for municipal leaders and introduces digital data management systems to improve service delivery. WASH services enhance waste management by integrating human waste handling with solid waste disposal, encouraging safe and hygienic behavior, building necessary infrastructure, and fostering community-led solutions.

Briquette making from organic waste: Collaborations between the Rotary Club of Lubowa and Mirembe Vocational Institute have resulted in the 'Mviringo Project,' which trains youth to produce biomass briquettes from organic waste. The project has also promoted sustainable consumption and circular economy practices, and built local capacity for community-led waste management. This initiative not only addresses waste management but also provides income-generating opportunities for young people.

Makindye Ssabagabo Municipality is actively transitioning toward a more sustainable and integrated waste management system. The growing collaboration between the municipal council, civil society, development partners, and the private sector is critical to building community resilience, improving public health, and enhancing environmental quality. Continued investment in infrastructure, enforcement, and community engagement will be essential in achieving a clean, safe, and sustainable urban environment.

2.4.5 Recommendations

The following recommendations are made to improve waste management in Makindye Ssabagabo Municipality;

Strengthen waste collection and transportation infrastructure: This includes increasing the number of waste collection vehicles and providing adequate waste bins and skips in strategic locations across all wards. Implementing a structured and predictable waste collection schedule will enhance efficiency and public trust in municipal services.

Promote waste segregation and recycling: Waste segregation at source should be institutionalized to promote recycling and resource recovery. The Municipality should develop and enforce guidelines for separating organic, recyclable, and hazardous waste, and support the establishment of community-based recycling centers and waste banks. Partnerships with private sector actors, particularly youth-led green enterprises, should be incentivized to create a vibrant recycling ecosystem.

Enforce waste management regulations: The Municipality should develop and implement a waste management by-law to enable effective enforcement of regulations against illegal dumping and unsanitary practices. Strengthening municipal inspection and enforcement capacity is critical for ensuring compliance among households, institutions, and commercial actors.

Improve public sanitation facilities: Public sanitation infrastructure should be expanded, particularly through the construction of public toilets in high-density and transit areas. Public-private partnerships (PPPs) can be leveraged for the operation and maintenance of these facilities, ensuring sustainability and accessibility. Construction of these facilities should incorporate climate-resilient and inclusive designs (e.g., accessible toilets, water-efficient technologies).

Institutionalize community clean-up days and expand wash education and behavior change campaigns: Community engagement should be institutionalized through regular clean-up days, formally integrated into the municipal calendar and budget. These events not only improve physical cleanliness but also foster civic responsibility and behavioral change. Schools, religious institutions, and youth groups should be key mobilization targets for such initiatives. Complementary to these efforts, WASH education and hygiene promotion campaigns should be intensified. These campaigns should use diverse communication tools, including radio, barazas, and school programs, to disseminate messages on proper waste disposal, public hygiene, and environmental protection.

Develop a sustainable financing strategy: Financial sustainability is essential for long-term waste management improvements. The Municipality should explore the introduction of a tiered waste collection fee system based on property use and waste generation volumes. In addition, development partners and climate finance opportunities should be tapped to support investments in circular economy initiatives and low-carbon waste infrastructure. A portion of local revenue particularly from property tax should be earmarked for environmental sanitation programs.

Strengthen partnerships and stakeholder coordination: Multi-stakeholder coordination remains a cornerstone for successful implementation. The Municipality should formalize collaborative frameworks with NGOs, community-based organizations (CBOs), and private waste management actors. Establishing a regular stakeholder engagement platform would support participatory planning, monitoring, and knowledge sharing. Partnerships with technical institutions and universities can also offer innovations and research-based solutions tailored to the local context.

Invest in waste treatment infrastructure: Feasibility studies should be undertaken to guide the development of composting facilities or material recovery facilities (MRFs). The Municipality should promote small-scale biogas digesters and briquette production from organic waste, particularly in schools, markets, and farming communities. The safe treatment and reuse of feacal sludge in agriculture should also be explored in alignment with national environmental and health standards.

2.5 Climate Change

Makindye Ssabagabo Municipality experiences significant climate-related challenges due to rapid urbanization and environmental pressures. This chapter presents an overview of the state, trend, drivers, impacts, responses, and recommendations of climate change in the municipality

2.5.1 State and Trends of Climate Change

State of climate in the municipality

Mean temperature and rainfall were used to describe climate variability within the municipality.

Mean temperature: In 2024, average mean annual temperature in the municipality was 22.9 °C, with Busabala and Mutungo parishes experiencing the highest mean temperatures and Mutundwe, Namasuba, and Masajja divisions experiencing the lowest mean temperatures (Figure 7). The highest mean temperature of 23.8 °C was experienced in February while the lowest mean temperature of 22.3 °C was experienced in July.

Rainfall: Mean annual rainfall in Makindye Ssabagabo Municipality in 2024 was 1,251.3 mm, with the parishes around the shores of Lake Victoria (Busabala and Mutungo) experiencing the highest rainfall as compared to areas of Seguku which received the lowest rainfall (Figure 7). Monthly rainfall in the Municipality ranged from 57.7 mm in February to 213.5 mm in April.

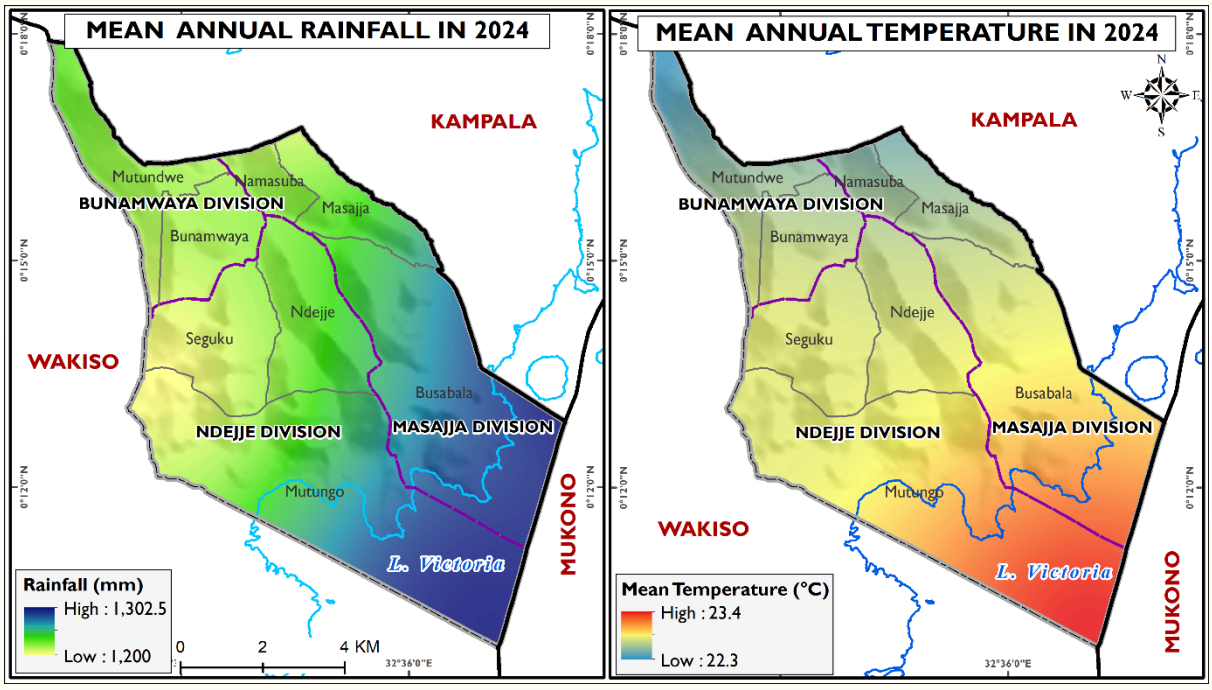


Figure 18: Mean annual rainfall and mean annual temperature in 2024

Trend of climate in the municipality

The communities were asked whether they have experienced climate change or variability in their areas. About 89.5% of the respondents reported experiencing climate change or variability in their respective wards. The indicators of climate change or variability reported by community members include increased temperatures (15.3%), increased extreme events such as floods and droughts (12.4%), prolonged dry spells (11.2%), unpredictable and erratic rainfall patterns (10.2%), increased amount of rainfall (8.6%), change of planting seasons (7.1%), and increased occurrence of strong winds (6.8%) (**Figure 8**). The key informants also revealed that climate change in Makindye Ssabagabo has manifested through increased prolonged dry spells, increased temperatures characterized by too much heat, and changes in planting seasons.

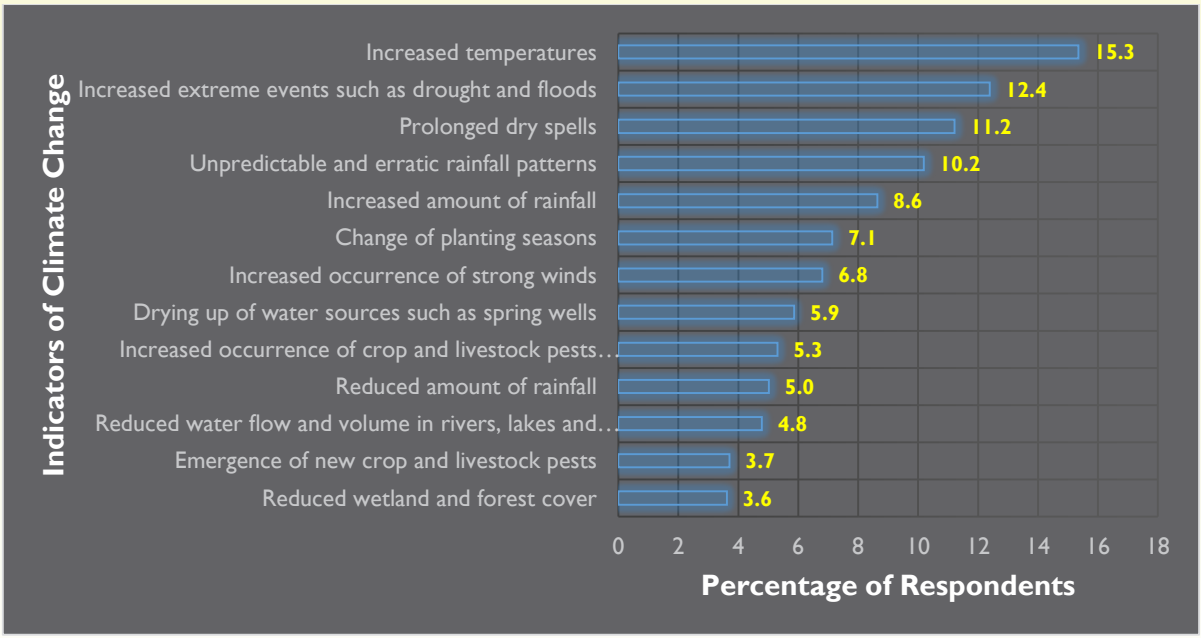


Figure 19: Indicators of climate change in Makindye Ssabagabo Municipality

Mean temperature

For the climatological period 1994–2024, the mean temperature in Makindye Ssabagabo municipality has increased at a rate of 0.22°C per decade. Between 1994 and 2024, the highest mean temperature of 23.1°C was recorded in 2024 whereas the lowest mean temperature of 22.0°C was recorded in 1996 (**Figure 9**). The projections of mean temperature (2025–2030) indicate an increase of 0.56°C and 0.65°C under moderate (RCP 4.5) and high (RCP 8.5) greenhouse gas (GHG) concentration scenarios, respectively, relative to the 1994–2024 average.

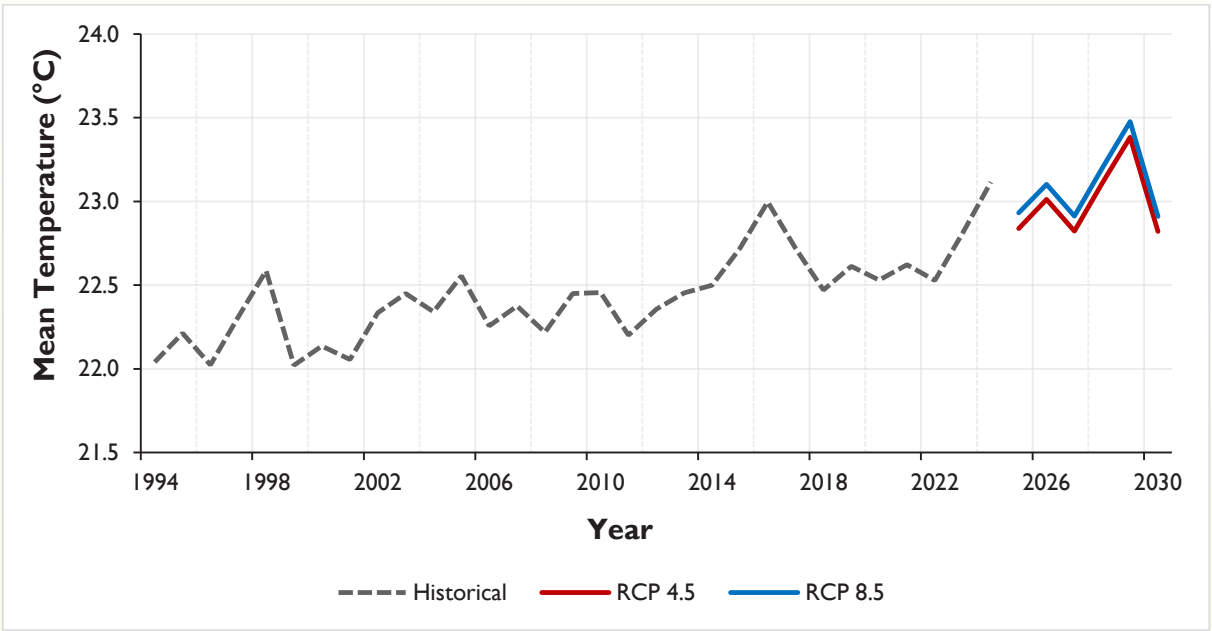


Figure 20: Observed and projected mean temperatures for Makindye Ssabagabo municipality from 1994 to 2030

Data source: Uganda National Meteorological Authority

Rainfall

Between 1994 and 2024, the total annual average rainfall for Makindye Ssabagabo Municipality was 1,411.4 mm, and mean monthly rainfall varied from 59.8 mm in February to 218.9 mm in April. For the climatological period 1994-2024, rainfall in Makindye Ssabagabo municipality has decreased at a rate of 24.2 mm per decade. Between 1994 and 2024, the highest annual rainfall of 1,756.4 mm was recorded in 1997 (**Figure 10**). The projections (2025-2030) indicate an increase in mean rainfall for Makindye Ssabagabo Municipality. Under RCP 4.5, mean rainfall is projected to increase by 271.1 mm (19.2%) relative to the 1994-2024 average. Similarly, under RCP 8.5, mean rainfall is projected to increase by 245.6 mm (17.4%) relative to the 1994-2024 average. These projections suggest a potential shift toward wetter conditions in the near future under both climate change scenarios.

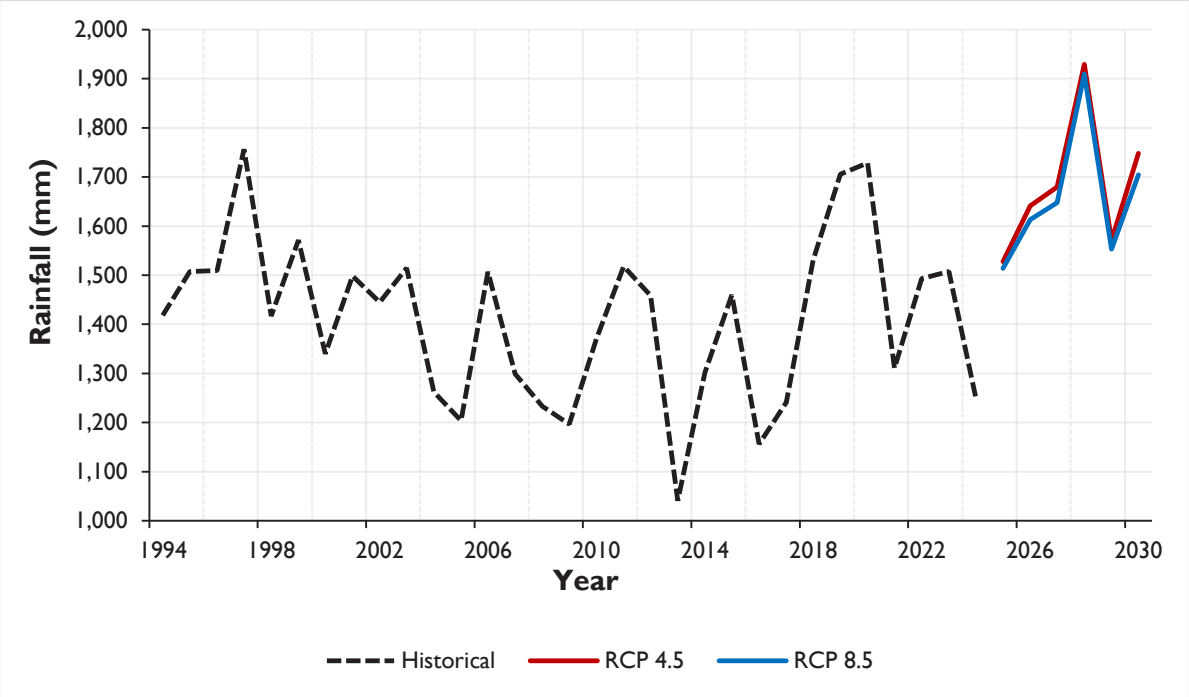


Figure 21: Observed and projected rainfall for Makindye Ssabagabo municipality from 1994 to 2030

Data source: Uganda National Meteorological Authority

2.5.2 Drivers of climate change

The local communities revealed that the changes in climate in Makindye Ssabagabo Municipality are caused by population increase (15.2%), cutting down trees (14.4%), rapid urbanization (14.1%), and wetland encroachment (13.3%) (**Table 1**). The rapid population growth in the Municipality has significantly contributed to environmental degradation and local climate change. As the demand for housing, transport, energy, and public services increases, natural vegetation, including wetlands, is being cleared to make way for urban development. This not only reduces carbon sequestration but also disrupts vital ecosystem services. Additionally, rising energy consumption often from non-renewable sources along with increased vehicle use, contributes to higher greenhouse gas emissions. Waste generation has also surged, with inadequate disposal systems leading to the release of methane and other pollutants. These combined pressures are intensifying the urban heat island effect and exacerbating the municipality's vulnerability to climate change impacts such as flooding, heatwaves, and reduced air quality.

Table 5: Perceived causes of climate change in Makindye Ssabagabo Municipality

Causes of climate change	Bunamwaya		Masajja		Ndejje		Total	
	N	%	N	%	N	%	N	%
Population increase	50	16.3	135	13.8	55	16.0	240	15.2
Cutting down trees	56	18.3	148	15.1	73	21.2	227	14.4
Rapid urbanization and infrastructure development	80	26.1	93	9.5	49	14.2	222	14.1
Encroachment on the wetland	52	17.0	110	11.2	48	14.0	210	13.3
Clearing of natural vegetation	12	3.9	114	11.7	57	16.6	183	11.6
Burning of solid waste	33	10.8	122	12.5	16	4.7	171	10.8
Industrial activities	20	6.5	58	5.9	35	10.2	113	7.2
Emissions from vehicles	3	1.0	99	10.1	7	2.0	109	6.9
Degradation of river banks	0	0.0	99	10.1	4	1.2	103	6.5
Total	**306**	**100**	**978**	**100**	**344**	**100**	**1578**	**100.0**

2.5.3 Impacts of climate change

Climate change has impacted both the local livelihoods and ecosystems in the municipality. The major reported impacts of climate change on local livelihoods in the different divisions of the municipality include increased food prices (16.8%), damage and/or loss of property during floods (15.7%), water shortages due to prolonged dry spells (13.1%), increased cost of water during the dry spells (11%), loss of income and unemployment due to floods (10.9%), and reduced crop yields for urban farmers (9.3%) (**Table 2**).

Table 6: Perceived impacts of climate change on local livelihoods in Makindye Ssabagabo Municipality

Impacts of climate change on local livelihoods	Bunamwaya		Masajja		Ndejje		MSMC	
	N	%	N	%	N	%	N	%
Increased food prices	63	31.0	131	12.3	63	24.0	257	16.8
Damage and/or loss of property during floods	58	28.6	138	13.0	45	17.1	241	15.7
Water shortages due to prolonged dry spells	55	27.1	105	9.9	40	15.2	200	13.1
Increased cost of water during the dry spells	9	4.4	124	11.6	36	13.7	169	11.0
Floods lead to loss of income and unemployment	8	3.9	128	12.0	31	11.8	167	10.9
Reduced crop yields for urban farmers	9	4.4	121	11.4	13	4.9	143	9.3
Increased occurrence of crop pests and diseases	1	0.5	111	10.4	31	11.8	143	9.3
Reduced livestock productivity	0	0.0	115	10.8	3	1.1	118	7.7
Increased occurrence of livestock pests, parasites and diseases	0	0.0	92	8.6	1	0.4	93	6.1
Total	**203**	**100**	**1,065**	**100**	**263**	**100**	**1,531**	**100**



Plate 3: Homes affected by floods in Namasuba ward, Makindye Ssabagabo Municipality

Climate change has not only impacted livelihoods but also ecosystems in Makindye Ssabagabo municipality. The major perceived impacts of climate change on ecosystems in Makindye Ssabagabo municipality include decline/loss of wetland benefits (21.6%), water pollution and decreased water quality (20.2%), reduced wetland biodiversity (14.8%), increased vulnerability of ecosystems to heat stress and drought (13.6%), and increased sedimentation in rivers and wetlands (12.6%) (**Table 3**). Decline/loss of wetland benefits was largely reported by the residents of the Bunamwaya division.

Table 7: Perceived impacts of climate change on ecosystems

Impact of climate change on ecosystems	Bunamwaya		Masajja		Ndejje		MSMC	
	N	%	N	%	N	%	N	%
Decline/loss of wetland benefits	59	42.4	133	16.8	42	27.3	234	21.6
Water pollution and decreased water quality	37	26.6	136	17.2	46	29.9	219	20.2
Reduced wetland biodiversity	12	8.6	124	15.7	25	16.2	161	14.8
Increased vulnerability of ecosystems to heat stress and drought	20	14.4	117	14.8	11	7.1	148	13.6
Increased sedimentation in rivers and wetlands	0	0.0	129	16.3	8	5.2	137	12.6
Emergence of invasive species	1	0.7	103	13.0	9	5.8	113	10.4
Increased habitat fragmentation	10	7.2	50	6.3	13	8.4	73	6.7
Total	**139**	**100**	**792**	**100**	**154**	**100**	**1,085**	**100**

2.5.4 Responses to climate change

The major responses to climate change implemented by households in Makindye Ssabagabo Municipality include rainwater harvesting (8.8%), participation in Savings and Credit Cooperative Organizations (SACCOs) and Village Savings and Loan Associations (VSLAs) (8.5%), digging channels to divert water during heavy rains (7.5%), participation in climate awareness programs (6.4%), participation in regular clean-up activities (6.3%), livelihood diversification (6.2%), elevation of homes and latrines (5.8%), planting of drought-resistant crop varieties (5.5%), and placing sandbags at doorsteps to block water during heavy rains (5.1%) (**Figure 11**).

Makindye Ssabagabo Municipal Council has adopted a range of strategic responses to address the growing impacts of climate change and environmental degradation. As part of the UN-Habitat-led

Climate Smart Cities Challenge, the municipality is spearheading initiatives to develop affordable, low-carbon housing solutions that align with sustainable urban growth. In addition, efforts have been made to improve solid waste management systems, supported by climate financing mechanisms such as the Gap Fund, to reduce environmental pollution and greenhouse gas emissions. Collaborations with development partners like WaterAid have enhanced the municipality's water, sanitation, and hygiene (WASH) infrastructure through community engagement and improved monitoring tools. Furthermore, the Physical Development Plan integrates environmental conservation into urban planning, ensuring the protection of natural resources amid rapid urbanization. These actions underscore the municipality's commitment to building resilience and promoting sustainable, climate-conscious development.

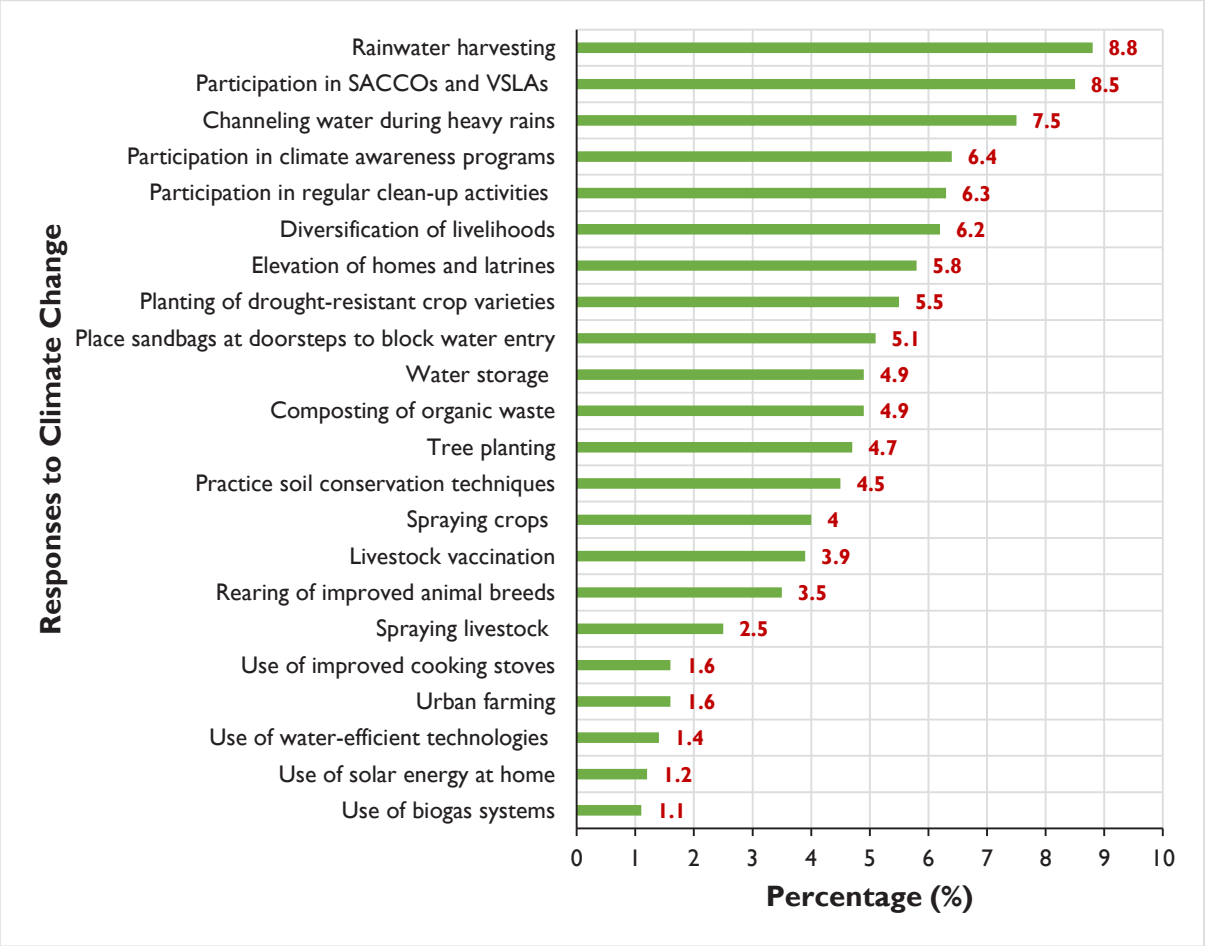


Figure 22: Household responses to climate change in Makindye Ssabagabo Municipality

The biggest barriers to climate change adaptation in Makindye Ssabagabo Municipality include limited government support (25.1%), limited awareness of climate change (16.9%), limited financial resources (16.2%), low education levels (13.7%), and limited access to information (12%) (**Figure 12**).

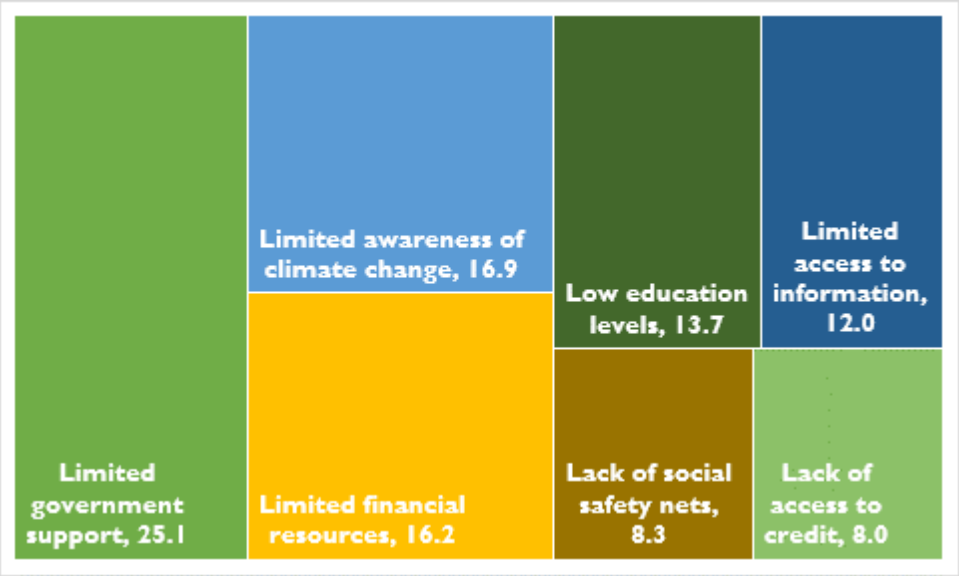


Figure 23: Barriers to climate change adaptation in Makindye Ssabagabo Municipality

2.5.5 Recommendations

The following recommendations are made to strengthen the resilience of communities to climate change in Makindye Ssabagabo Municipality;

Mainstream climate adaptation into urban and municipal development planning: Integrate climate risk assessments and adaptation priorities into the Municipality's Physical Development Plans, Strategic Investment Plans, and Municipal Development Plans. This includes enforcing climate-sensitive land use regulations, promoting risk-informed spatial planning, and ensuring that development control guidelines account for flood zones, erosion-prone areas, and heat island hotspots.

Upgrade and climate-proof critical infrastructure: Implement a phased program to retrofit and construct resilient infrastructure, particularly in flood-prone divisions such as Ndejje. Priority interventions include the development of sustainable urban drainage systems, reinforced embankments, permeable pavements, and elevated road networks. Promote the adoption of green infrastructure such as retention ponds, and vegetated buffers to complement grey infrastructure investments.

Establish a decentralized early warning and emergency preparedness system: Collaborate with the Uganda National Meteorological Authority (UNMA) and Office of the Prime Minister (OPM) to operationalize localized, real-time climate early warning systems. Strengthen institutional capacity within the Municipal Disaster Management Committee and community-based disaster preparedness teams through simulation exercises, risk communication training, and integration with national contingency plans.

Promote climate-smart livelihood diversification: Support the development of adaptive livelihoods by facilitating access to microfinance, capacity building, and market linkages for green enterprises. These should include urban agriculture using drought-resistant crops and vertical farming techniques, decentralized solar energy systems, eco-tourism ventures along Lake Victoria's fringe wetlands, and circular economy initiatives (e.g., plastic repurposing, organic waste composting).

Scale up environmental education and behavior change communication: Develop a multi-stakeholder Environmental Awareness Strategy targeting schools, local leaders, and civil society. Utilize culturally appropriate messaging and participatory tools to enhance knowledge on climate change impacts, waste reduction, and ecosystem conservation. Leverage platforms such as Local Council meetings and faith-based institutions to foster civic engagement and behavioral shifts.

Restore and protect natural ecosystems: Prioritize the restoration and protection of wetlands, forests, and riparian buffers, particularly in the catchments of River Mayanja and Lake Victoria's shoreline. Implement ecosystem-based adaptation (EbA) measures, including afforestation, bamboo planting for slope stabilization, and protection of ecological corridors. Strengthen enforcement of environmental laws (e.g., the National Environment Act, 2019) in collaboration with NEMA and community-based monitors.

Enhance integrated solid waste management systems: Scale up formal and informal sector engagement in waste segregation, recycling, and composting through Public-Private Partnerships (PPPs). Establish satellite waste transfer stations and Material Recovery Facilities (MRFs) to improve collection efficiency. Integrate climate mitigation co-benefits by capturing methane from organic waste and reducing landfill burdens in the fast-growing urban centers.

Foster multi-stakeholder climate governance and resource mobilization: Establish a Municipal Climate Resilience Coordination Task Force to lead multi-sectoral planning, monitoring, and reporting. Strengthen partnerships with NGOs, private sector actors, and development partners to leverage technical assistance and financial resources through frameworks such as the Local Climate Adaptive Living Facility (LoCAL) and the Green Climate Fund (GCF). Institutionalize participatory monitoring and accountability tools such as environmental scorecards and stakeholder feedback mechanisms.

2.6 Air Quality

2.6.1 State of air quality

Makindye Ssabagabo Municipality is undergoing rapid urbanization, which has led to increased concerns about air quality. Monitoring air quality is essential to identify pollution hotspots, assess compliance with national and international guidelines, and inform strategies to reduce exposure to harmful pollutants. Air quality in the Municipality was determined through sampling representative cells based on population density, level of activity, and proximity to suspected pollution sources. Particulate matter (PM2.5) was measured in each location using the Casella micro dust monitor. The air quality results were compared with the national air quality standards and the WHO air quality guidelines. Air quality in Makindye Ssabagabo Municipality was assessed based on NEMA air quality index (**Table 4**) and mapped using interpolation technique in GIS.

Table 8: Air quality index for describing air quality levels

Health Concern	$PM_{2.5}$ (μgm^{-3})	Precautions
Good	0 - 9.0	None: Air quality is considered satisfactory, and air pollution poses little or no risk.
Moderate	9.1 - 35.4	Unusually sensitive people should consider reducing prolonged or heavy exertion
Unhealthy for Sensitive Groups	35.5 - 55.4	Sensitive groups should reduce prolonged or heavy exertion
Unhealthy	55.5 - 150.4	Everyone should reduce prolonged or heavy exertion, take more breaks during outdoor activities
Very Unhealthy	150.5 - 250.4	Everyone should avoid prolonged or heavy exertion, move activities indoors or reschedule
Hazardous	250.5 & above	Everyone should avoid all physical activities outdoors.



Plate 4: Air quality monitoring in Makindye Ssabagabo Municipality

The air quality (PM2.5) for most of the sampled villages/cells in Makindye Ssabagabo Municipality was above the WHO and NEMA air quality permissible limits (**Figure 13**). Areas such as Lufuka, Masajja A, and Kibiri B had air quality above the NEMA air quality permissible limit. On the other hand, areas such as Lubowa, Nyanama, and the council offices had air quality lower than the WHO and NEMA air quality permissible limits.

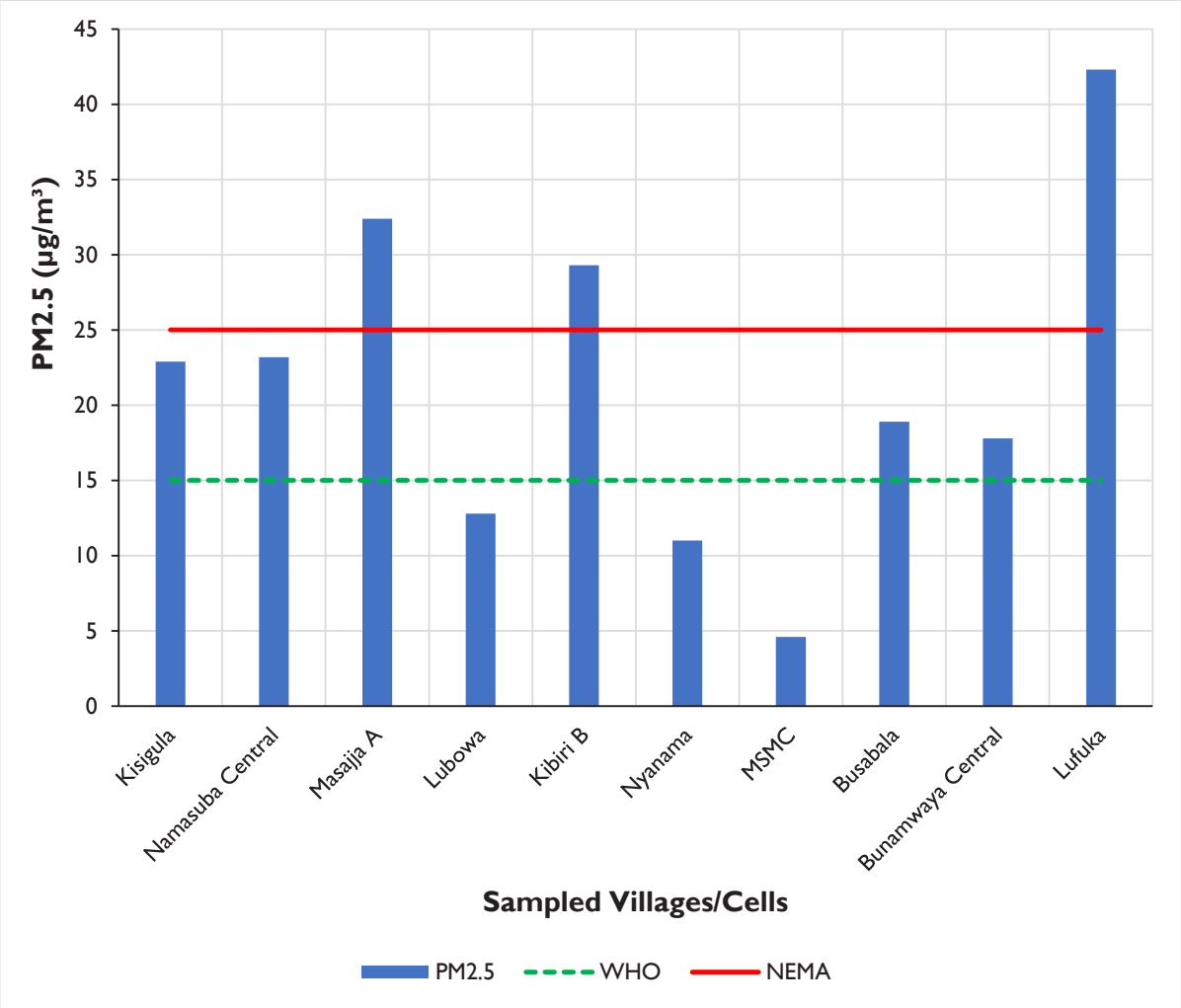


Figure 24: Air quality (PM2.5) in the sampled villages of Makindye Ssabagabo Municipality

The biggest part of the municipality is exposed to air quality that ranges between 9.1 and 35.4 µg/m³. The health concern for this air quality is moderate; and this suggests that air quality is generally acceptable, however, there may be a risk for some individuals, particularly those who are unusually sensitive to air pollution. Ndejje ward has air quality that is unhealthy for sensitive groups such as children, the elderly, and individuals with pre-existing health conditions. Exposure to elevated levels of particulate matter can lead to respiratory issues, cardiovascular diseases, and other health problems. The areas near the shores of Lake Victoria (Busabala and Mutungo wards) have good air quality, with a few pockets in Masajja, Ndejje, Mutundwe, and Namasuba wards.

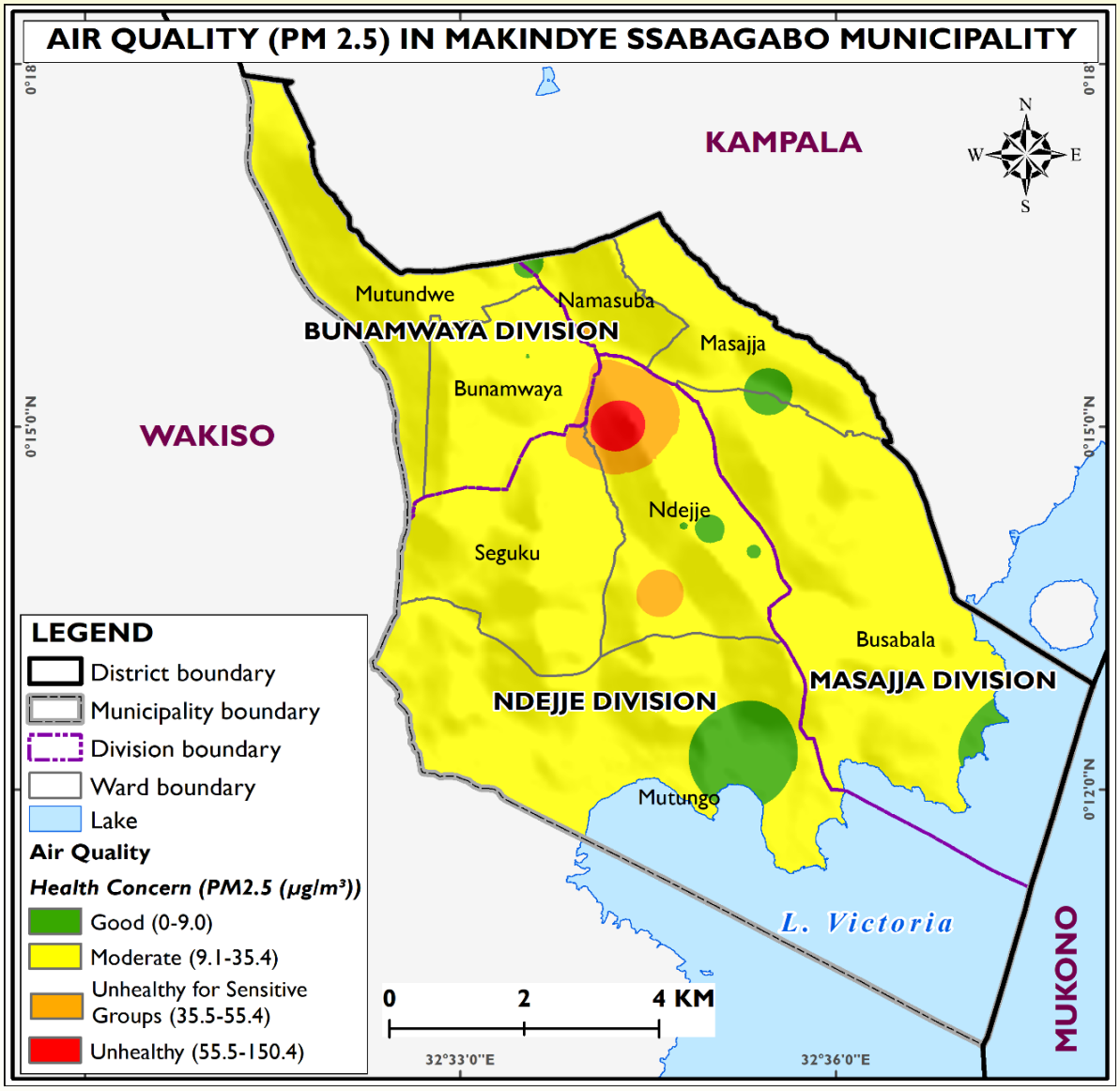


Figure 25: Air quality (PM2.5) in Makindye Ssabagabo Municipality in May 2025

2.6.2 Drivers of poor air quality

The drivers of poor air quality in Makindye Ssabagabo Municipality are presented below;

Vehicular emissions and transport-related pollution: The increase in motorized transport especially second-hand, fuel-inefficient vehicles lacking catalytic converters has led to elevated emissions of nitrogen oxides (NO_x), carbon monoxide (CO), black carbon, and fine particulate matter ($PM_{2.5}$ and PM_{10}). The absence of a municipal vehicle emission testing program and the proliferation of informal transport (boda-bodas and commuter taxis) further aggravate pollutant loading. Peak-hour traffic congestion on unpaved or poorly maintained roads generates significant resuspension of road dust, contributing to coarse particulate matter (PM_{10}) emissions.

Biomass and waste combustion: A substantial portion of the population relies on biomass fuels (charcoal and firewood) for cooking, both at the household level and in the informal food service sector. These fuels are combusted in poorly ventilated environments and emit high concentrations of $PM_{2.5}$, volatile organic compounds (VOCs), and carbon monoxide. Additionally, the open burning of mixed solid waste, particularly plastics, textiles, and organic matter due to insufficient

collection infrastructure introduces persistent organic pollutants (POPs) and heavy metals into the atmosphere, compounding the public health risks.

Industrial, artisanal, and construction emissions: The growing number of small-scale industries and informal workshops, including metal fabrication, paint application, welding, and poultry waste incineration, often operate without pollution control technologies. These facilities release particulate matter, sulfur dioxide (SO_2), lead, and other hazardous air pollutants (HAPs) into the local environment. Unregulated brick kilns and cement-based construction activities further contribute to ambient dust levels, especially during the dry season when dust suppression measures are minimal.

Land use change and ecosystem degradation: Widespread deforestation and wetland reclamation particularly in the catchments surrounding Lubowa, Ndejje, and Mutungo have led to a reduction in vegetative cover that would otherwise act as natural air filters. This degradation contributes to localized increases in air temperature, decreased atmospheric dispersion, and higher concentrations of near-ground pollutants, especially during thermal inversion events.

Inadequate urban waste management: Low solid waste collection coverage, especially in peri-urban and informal settlements, results in significant quantities of refuse being burned or dumped in open spaces. The absence of material recovery facilities and weak enforcement of waste handling regulations have allowed for the accumulation and incineration of hazardous and non-segregated waste, further worsening air quality.

Meteorological and seasonal factors: Climatic conditions, particularly prolonged dry spells with low wind speeds and high atmospheric stability, limit pollutant dispersion and promote the accumulation of airborne contaminants. During early mornings and evenings, temperature inversions are common, trapping pollutants close to the surface and intensifying human exposure, particularly for roadside traders, schoolchildren, and residents in low-lying areas.

2.6.3 Impacts of poor air quality

The sustained decline in ambient air quality across Makindye Ssabagabo Municipality has resulted in multifaceted impacts on human health, environmental systems, and socio-economic development. These impacts are increasingly being felt as the municipality undergoes rapid urbanization without commensurate investments in pollution control, waste management, or sustainable mobility. The effects of poor air quality are classified below across three major domains:

A. Public Health Impacts

Air pollution is a major environmental health threat, with disproportionate effects on vulnerable populations such as children, the elderly, and low-income households. Ambient concentrations of fine particulate matter ($PM_{2.5}$), nitrogen dioxide (NO_2), carbon monoxide (CO), and volatile organic compounds (VOCs) routinely exceed recommended health-based thresholds in localized hotspots, leading to:

Increased respiratory and cardiovascular morbidity: High exposure to $PM_{2.5}$ and NO_2 is strongly associated with asthma exacerbations, chronic obstructive pulmonary disease (COPD), bronchitis, ischemic heart disease, and stroke.

Premature mortality: Long-term inhalation of fine particulates is a recognized contributor to early death, particularly among the elderly and those with pre-existing conditions.

Reduced pulmonary development in children: Exposure during critical stages of growth impairs lung function and increases susceptibility to respiratory infections.

Adverse maternal and neonatal outcomes: Women exposed to indoor biomass smoke or roadside emissions are at elevated risk of low birth weight, preterm birth, and pregnancy complications.

B. Environmental Impacts

Air pollutants not only pose direct health threats but also degrade ecosystems and compromise environmental quality:

- Damage to vegetation and urban greenery: Sulfur dioxide (SO_2), ozone (O_3), and other reactive pollutants damage plant tissues, reduce crop yields, and impair the ecological functions of urban forests and wetlands.
- Soil and water degradation: Acidic pollutants such as SO_2 and NO_x contribute to acid deposition, which alters soil chemistry, affects nutrient availability, and threatens aquatic biodiversity through runoff into water bodies like Lake Victoria.
- Biodiversity stress: Pollution-sensitive species, particularly insects, amphibians, and flowering plants, are increasingly threatened by air and soil contamination in peri-urban ecosystems.
- Contribution to climate change: Black carbon and methane (CH_4) emissions from open waste burning and inefficient biomass use are short-lived climate pollutants (SLCPs) that accelerate local warming and alter rainfall patterns.
- Increased urban heat island effect: Pollutant build-up in densely built environments, combined with loss of tree cover, exacerbates temperature extremes, particularly during the dry season.

C. Socio-Economic Impacts

Air pollution carries significant economic consequences that undermine local development goals and urban resilience:

Reduced economic productivity: Illness linked to poor air quality results in absenteeism and lost earnings, particularly among informal sector workers, drivers, street vendors, and construction laborers.

Elevated healthcare costs: Households spend a growing share of income on medication, clinic visits, and long-term treatment of respiratory and cardiovascular diseases.

Education disruption: Children frequently absent from school due to air pollution-related illnesses experience learning loss, undermining education outcomes and future earning potential.

Reduced quality of life and urban liveability: Persistent smoke, odours, and dust reduce outdoor mobility, social interaction, and community well-being, particularly in densely populated areas like Masajja.

Impact on urban investment and tourism: Degraded environmental aesthetics and public health risks may deter tourism and investment in real estate, hospitality, and commercial services.

2.6.4 Responses to poor air quality

In response to the growing challenge of poor air quality, Makindye Ssabagabo Municipality has undertaken a range of interventions, both independently and in partnership with national agencies, academia, civil society, and development partners. One of the key institutional responses has been the integration of environmental sustainability principles into local development planning, particularly within the Physical Development Plan. This framework recognizes air pollution as a cross-cutting issue affecting health, transport, waste, and urban land use. At the national level, the municipality collaborates with the National Environment Management Authority (NEMA) to enforce compliance with the National Environment (Air Quality Standards) Regulations, 2024, while technical support on monitoring and forecasting is provided by the Uganda National Meteorological Authority (UNMA) and the Ministry of Water and Environment (MWE).

A significant milestone in evidence-based air quality management has been the deployment of real-time monitoring sensors in partnership with AirQo, a research initiative of Makerere University. These low-cost sensors have been installed in various strategic locations, including Seguku and Ndejje, enabling continuous tracking of $PM_{2.5}$ levels and improving spatial understanding of pollution hotspots. Data from these sensors is publicly accessible through mobile applications and is being used to guide municipal decision-making, inform public education campaigns, and raise awareness among residents. Additionally, sensitization programs often led in collaboration with environmental NGOs such as ACTogether, CIDI, and Youth Go Green are being implemented to promote clean cooking technologies, discourage open waste burning, and enhance community knowledge on the health impacts of air pollution.

At the community level, schools, local councils, and faith-based institutions have been mobilized to engage in air quality improvement initiatives. Environmental clubs in schools conduct awareness campaigns, tree planting, and waste segregation as part of a broader strategy to create an environmentally conscious youth demographic. The municipality has also expanded its waste management efforts by improving solid waste collection services in densely populated neighborhoods like Namasuba, and Mutundwe. Pilots on waste sorting and composting have been introduced with support from private sector actors and partners like the Deutsche Gesellschaft für Internationale Zusammenarbeit (GIZ), to reduce reliance on open burning as a method of waste disposal.

In the infrastructure domain, the municipality is working to promote green spaces and urban greening efforts, particularly along major roads, institutions, and public spaces. These initiatives are supported by environmental NGOs and community-based organizations under climate adaptation and beautification programs. Cleaner technologies are also being introduced in the informal sector, especially among small-scale industries and roadside workshops, where business owners are being trained on safe chemical handling, emissions reduction, and cleaner production practices. Such capacity building has been supported by international partners such as UN-Habitat and SNV Netherlands Development Organization, which provide both technical expertise and resources.

Despite these efforts, enforcement remains a challenge due to limited municipal capacity, financial constraints, and low public compliance. However, the municipality's natural resources department, working in collaboration with NEMA, has issued notices and penalties for open burning and illegal dumping in breach of the National Environment Act (NEA), 2019. Institutional efforts are ongoing to develop local bylaws to align more closely with national air quality standards and the NEA, 2019. Overall, while responses are still evolving, the municipality has laid a foundational framework for integrated air quality management, leveraging partnerships with stakeholders such as Makerere University (AirQo), NEMA, UNMA, MWE, ACTogether, CIDI, GIZ, UN-Habitat, and the private sector. Moving forward, sustained investment, capacity building, and multi-sectoral coordination will be essential to scale up these interventions and address the root causes of air pollution in Makindye Ssabagabo.

2.6.5 Recommendations

To address the rising threat of air pollution and its impacts on human health, ecosystems, and urban development, Makindye Ssabagabo Municipality must adopt a coordinated, multisectoral approach to air quality management. The following recommendations are proposed:

1. Develop a municipal air quality management plan (AQMP): Formulate and adopt a dedicated air quality management plan aligned with the National Environment (Air Quality Standards) Regulations, 2024. The AQMP should define emission reduction targets, priority sectors, implementation timelines, monitoring mechanisms, and roles of various stakeholders.

2. Scale up air quality monitoring infrastructure: Expand the network of real-time air quality monitoring stations across the municipality, prioritizing pollution hotspots such as Ndejje and Lufuka cells. Strengthen partnerships with AirQo, NEMA, and UNMA to enhance data quality, coverage, and accessibility.

3. Integrate air quality into urban planning and transport policy: Embed air quality considerations into the Physical Development Plan, zoning regulations, and municipal infrastructure projects. Encourage mixed-use planning to reduce travel distances and promote non-motorized transport (NMT) through dedicated pedestrian and cycling infrastructure. Regulate roadside vending and open-air markets in congested areas to reduce exposure.

4. Promote clean cooking and renewable energy access: Accelerate the adoption of clean cooking solutions, such as LPG, electric cookers, and improved biomass stoves, particularly in low-income and peri-urban areas. Collaborate with energy companies and NGOs to support subsidies, awareness campaigns, and last-mile distribution.

5. Regulate and formalize informal sector activities: Support the transition of informal garages, welding shops, and artisanal industries to cleaner technologies through training, equipment upgrades, and compliance incentives. Introduce emission control standards for fuel-burning processes and spray-painting operations.

6. Improve municipal solid waste management: Strengthen the coverage and efficiency of solid waste collection, especially in high-density zones. Enforce the ban on open burning of waste, and invest in decentralized composting, recycling hubs, and waste-to-energy pilots. Introduce fines and rewards to promote household-level waste sorting.

7. Enforce vehicle emission standards and promote public transport: In collaboration with the Ministry of Works and Transport and the Uganda Revenue Authority, implement regular inspection and maintenance programs for vehicles operating within the municipality. Promote the use of low-emission vehicles and invest in safe, affordable, and efficient public transport systems.

8. Enhance green infrastructure and urban forestry: Expand tree planting initiatives along roadsides, schools, public parks, and wetlands to filter air pollutants and mitigate the urban heat island effect. Promote the protection and restoration of vegetative buffers around construction sites and major roads.

9. Raise public awareness and community participation: Implement sustained behaviour change communication campaigns to educate residents on the causes, risks, and prevention of air pollution. Use community radios, schools, faith-based platforms, and digital tools to disseminate targeted messages and promote civic responsibility.

10. Institutionalize multi-stakeholder coordination: Establish an Air Quality Coordination Task Force comprising representatives from the municipality, NEMA, MWE, UNMA, Makerere University (AirQo), NGOs, and the private sector. This platform should guide joint planning, resource mobilization, and policy advocacy.

11. Mobilize sustainable financing: Seek financial support through climate resilience funds, green bonds, and public-private partnerships to finance air quality improvement interventions. Advocate for increased budget allocation to the municipal environment department and invest in staff training.

The implementation of these recommendations will significantly strengthen the municipality's capacity to manage air quality proactively. A shift from reactive enforcement to preventive, data-driven, and community-centered planning will be essential in ensuring clean air, safeguarding public health, and promoting sustainable urban growth in Makindye Ssabagabo.

2.7 Land Use and Soils

2.7.1 State and trend

Between 2015 and 2025, Makindye Ssabagabo experienced notable land use and cover changes, with built-up areas expanding dramatically from 29.8 sq km (35%) to 52.8 sq km (62.2%). This growth, predominantly in areas like Bunamwaya and Masajja, has occurred at the expense of croplands and grasslands, which declined from 14.6 sq km to 3.4 sq km and from 14.4 sq km to 2.3 sq km, respectively. Figure 26 visually illustrates this spatial shift, emphasizing urban sprawl across the municipality.

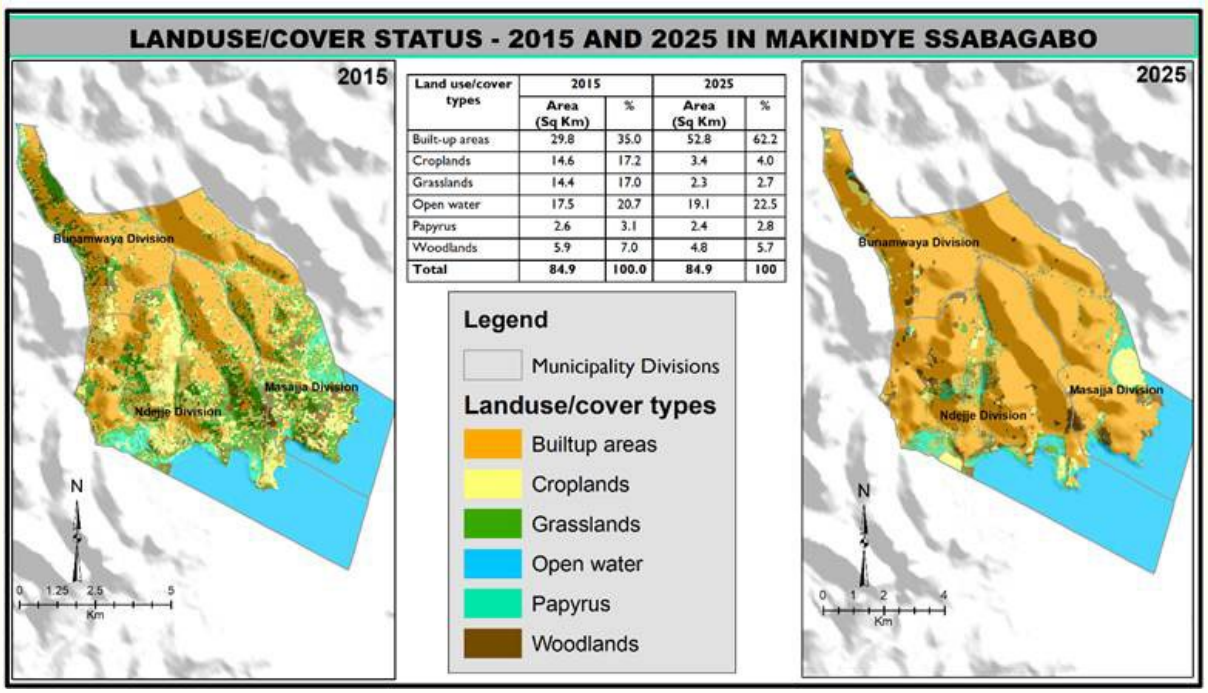


Figure 26: State and trend of land use/cover in Makindye Ssabagabo (2015-2025)

Meanwhile, natural ecosystems have seen minor but significant declines. Woodlands reduced from 5.9 sq km to 4.8 sq km, and papyrus wetlands dropped from 2.6 sq km to 2.4 sq km. Open water bodies slightly increased from 17.5 sq km to 19.1 sq km, potentially due to improved wetland retention or flood dynamics. These trends are depicted in Figure 26, which shows a rising trajectory for built-up areas and a declining trend for vegetative and agricultural land. These changes, clearly visualized in Figure **6,** underscore the pressing need for integrated land management approaches that balance urban development with the conservation of natural ecosystems.

2.7.2 Drivers of land cover/use changes

The development of roads, schools, and health facilities has significantly altered land cover in Makindye Ssabagabo, particularly over the period 2014 to 2025, as the municipality urbanizes rapidly (NEMA, 2021). A prominent example is the construction of the Entebbe Expressway, completed in 2018, which passed through key areas like Kigo and Kajjansi, resulting in the conversion of wetlands and farmlands into transportation corridors (Ministry of Works and Transport, 2019). This project not only physically altered the landscape but also stimulated further land use change, with increased real estate activity and commercial development emerging near access points and interchanges. Beyond transportation, the construction of public infrastructure such as schools and health centers in expanding neighborhoods like Bunamwaya East, Ndejje, and Mutundwe has replaced vegetated zones and agricultural fields with built-up institutional land use (Makindye Ssabagabo Municipal Council, 2022).

Urbanization and rapid population growth are the main drivers of land use and land cover change in Makindye Ssabagabo. The Municipality has become an ideal location for settlement and commercial development due to its proximity to Kampala, the country's economic and administrative Centre. Over the past two decades, former agricultural and peri-urban areas such

as Namasuba, Ndejje, and Bunamwaya have undergone significant transformation into densely populated residential, commercial, and mixed-use urban areas. This change is mainly due to the increasing demand for housing, transportation, and public services as a result of the growing urban population seeking affordable and easily accessible accommodation close to the capital city (National Planning Authority, 2022; NEMA, 2021). According to the National Population and Housing Census (UBOS, 2017), the city experienced a sharp increase in population between 2002 and 2014, leading to land fragmentation, informal settlements, and loss of green spaces. The expansion has also led to land pressure for infrastructure construction, including roads, schools, and health centers, which has further accelerated the conversion of wetlands, forests, and agricultural land into built-up areas (UBOS, 2017; National Planning Authority, 2022).

Unregulated sand and clay mining have emerged as key drivers of land degradation and land cover transformation in Makindye Ssabagabo, particularly in Kajjansi and Seguku Wards. These areas are rich in clay and sand deposits that support the production of tiles, bricks, and construction materials. However, due to weak enforcement of extraction regulations, extensive mining has resulted in the formation of open pits, destabilized land, and the widespread loss of natural vegetation cover (NEMA, 2021).

The extraction process often involves clearing topsoil and vegetation to access sand or clay deposits, which permanently alters the landscape. In Kajjansi, formerly green zones have been transformed into barren, excavated sites, with visible gullies, exposed soil, and abandoned pits that collect stagnant water, posing both environmental and public health risks (Makindye Ssabagabo Municipal Council, 2022). In Seguku, clay mining has caused siltation of nearby wetlands and streams, reducing their ecological function and contributing to downstream flooding, especially during the rainy season.

Weak enforcement of land use regulations and inadequate urban planning frameworks have played a significant role in driving uncontrolled land use changes in Makindye Ssabagabo. Despite the presence of national policies such as the National Land Use Policy (2007) and environmental regulations enforced by agencies like NEMA, implementation at the municipal level remains inconsistent and under-resourced (NEMA, 2021; Ministry of Lands, Housing and Urban Development, 2020). As a result, there has been widespread illegal conversion of protected wetlands, forested buffers, and riparian zones into residential and commercial plots, often without proper environmental impact assessments or development approvals. In areas such as Kyetinda, Lubigi Extension, and Kajjansi, land designated as ecologically sensitive has been sold and subdivided for real estate development, sometimes with informal approval or political interference (Makindye Ssabagabo Municipal Council, 2022).

In peri-urban areas of Makindye Ssabagabo, particularly in Kigo, Mutungo, and parts of Kabuuma, agricultural expansion and intensification have significantly contributed to land use and land cover changes. Driven by increasing food demand from the growing urban population and limited access to formal employment, many households have turned to farming as a livelihood strategy (NEMA, 2021). This trend has led to the clearing of natural vegetation, including grasslands and woodland patches, as well as the conversion of wetland margins into agricultural plots. Crops such as maize, vegetables, and bananas are commonly grown along streams and wetlands such as Mayanja–Kato

and Kyetinda, often without observing ecological buffer zones. In Kigo, swampy areas previously dominated by papyrus and wetland grasses have been drained or reclaimed for seasonal cultivation, undermining their ecological function (Makindye Ssabagabo Municipal Council, 2022). The proliferation of informal settlements in Makindye Ssabagabo, particularly in rapidly growing peri-urban areas such as Kikajjo, and parts of Bunamwaya East, has significantly contributed to land cover transformation and environmental degradation. These unplanned residential areas have emerged due to population growth, housing shortages, and the relatively low cost of land in previously undeveloped zones (Makindye Ssabagabo Municipal Council, 2022). In the absence of proper land use regulation and enforcement, many of these settlements have been established in ecologically sensitive areas, including wetlands, riparian buffers, and forested hillsides. For example, in Masajja, residents have cleared sections of the Kaliddubi Wetland and nearby woodlands to construct semi-permanent houses and small-scale subsistence farms. Similarly, in Kikajjo, forested slopes have been cleared for makeshift housing, leading to deforestation, loss of biodiversity, and increased soil erosion (NEMA, 2021).

2.7.3 Impact of Land use/cover changes

Land use change has intensified water pollution, particularly through increased runoff and poor sanitation practices. The expansion of settlements and agriculture in areas like Ndejje and Mutundwe has led to widespread use of fertilizers and the construction of unlined pit latrines close to water points. As a result, boreholes tested in 2020 revealed elevated levels of nitrate and iron, which were directly linked to fertilizer leaching and leakage from pit latrines. This contamination poses a serious threat to drinking water safety and has heightened the risk of waterborne diseases, especially among children and other vulnerable groups (WaterAid Uganda, 2021).

Land cover change has led to significant habitat destruction. In areas like Kikajjo, Kabuuma and Ndejje, Lufuka deforestation caused by informal settlements and clay mining has destroyed wildlife habitats. Between 2015 and 2021, forested areas near Kaliddubi Wetland were cleared, reducing sightings of wetland bird species such as the Grey Crowned Crane *(Balearica regulorum),* Hadada Ibis *(Bostrychia hagedash),* and Black-headed Heron (*Ardea melanocephala),* as well as small mammals like the African marsh mongoose *(Atilax paludinosus)* and Gambian rat (*Cricetomys gambianus)* (NEMA, 2021). These species are indicators of healthy wetland and forest-edge ecosystems, and their decline reflects the broader loss of biodiversity caused by unchecked land conversion.

Wetland degradation caused by land use change has intensified flooding in Makindye Ssabagabo. The conversion of natural wetlands into residential and agricultural land has severely reduced the area's ability to absorb and regulate rain water. In particular, wetlands such as Kyetinda and Mayanja–Lufuka have been reclaimed for housing and farming, leading to blocked natural drainage paths and a loss of ecological flood control functions. As a result, frequent flooding has become a major urban challenge. In April 2023, heavy rainfall caused River Lufuka to overflow, displacing households in Namasuba and Bunamwaya, and triggering emergency relief efforts from the municipal council (National Planning Authority, 2022).

Land cover changes due to urbanization have contributed to rising surface temperatures in Makindye Ssabagabo. The widespread replacement of natural vegetation with concrete surfaces,

paved roads, and iron-roofed structures in urbanizing areas such as Bunamwaya Central, Mutundwe, and Kajjansi has significantly altered the local microclimate. Between 2018 and 2023, this transformation led to a recorded 3°C increase in land surface temperatures, as green cover that once provided natural cooling was removed. The result is worsened urban heat stress, particularly affecting vulnerable populations like the elderly and young children during dry seasons (NEMA, 2021).

Conversion of agricultural land to urban uses has significantly reduced local food production in Makindye Ssabagabo. Rapid residential expansion in areas like Mutungo and Kigo has led to the loss of formerly productive farmlands, undermining local food self-sufficiency. By 2022, more than 60% of farmland in Mutungo Ward had been converted into residential estates, reducing the availability of land for cultivation. This shift has increased reliance on external food supplies and led to a rise in household food costs, especially for low-income families (Makindye Ssabagabo Municipal Council, 2022).

Communities in Makindye Ssabagabo recommend practical responses to land use and cover change impacts based on local experiences. In areas like Kyetinda and Zana, residents recommend forming community wetland patrols to monitor encroachment, similar to successful efforts around Lubigi Wetland, where local surveillance reduced illegal reclamation (NEMA, 2020). Along degraded rivers such as Lufuka and Walugogo, residents propose tree planting and riverbank restoration, inspired by reforestation success in River Nabajjuzi, Masaka, which stabilized banks and improved water clarity (NEMA, 2020). In Bunamwaya and Kikajjo, locals call for inclusion in land use planning through parish-level committees, a strategy that reduced land disputes in Mukono District using participatory mapping methods (MLHUD, 2019). Farmers in Mutungo and Seguku suggest sustainable farming like raised-bed gardening, drawing lessons from Kiswa parish in Kampala, where sack gardening improved food production while conserving land (KCCA, 2020). In Kikajjo and Ndejje, communities advocate for affordable land titling and legal education, reflecting success in Makindye Ssabagabo Municipality, where land registration helped curb informal wetland encroachment (UBOS, 2020). Lastly, in Masajja, locals recommend community waste collection groups to reduce water pollution, modeled after Kyebando, Kampala, where cooperative waste management led to cleaner drains and job creation (NEMA, 2018). These examples show that community-led strategies, when supported with proper structures, can play a pivotal role in sustainable land use and resource protection.

2.7.5 Responses

Short term

To quickly halt environmental degradation, Makindye Ssabagabo Municipal Council intensified its enforcement of environmental regulations. In collaboration with NEMA and Wakiso District officials, the municipality conducted joint inspections and demolished illegal structures erected in wetland zones such as Zana, Kikajjo, and Bunamwaya. These rapid interventions aimed to deter further encroachment, protect fragile ecosystems, and reassert legal boundaries of protected areas.

In response to the increasing land use/cover changes, particularly the illegal conversion of wetlands into settlements and farms, the Makindye Ssabagabo Municipal Council, in collaboration with the Ministry of Water and Environment (MWE) and the National Environment Management Authority (NEMA), initiated the physical demarcation of ecologically sensitive wetlands. Key sites like Kyetinda and Kaliddubi were marked using permanent beacons and signage to clearly define boundaries, deter encroachment, and support enforcement of land use regulations (NEMA, 2021). To complement these measures, the municipality supported community-based wetland restoration activities aimed at reversing land degradation. These included the replanting of reeds, papyrus, and native grasses to stabilize eroded soils, rebuild buffer zones, and restore ecosystem services lost due to uncontrolled land conversion.

In a proactive response to the encroachment of Kyetinda Wetland, some residents voluntarily vacated illegally occupied wetland buffer zones following targeted environmental education and awareness campaigns led by the municipal council and local leaders. This approach emphasized the ecological importance of wetlands and the associated risks of settlement in flood-prone areas (Makindye Ssabagabo Municipal Council, 2022). The voluntary relocation allowed for the immediate restoration of wetland buffers, enabling replanting of native vegetation and improving the wetland's capacity for flood control, water filtration, and biodiversity regeneration.

In response to the increasing impact of unregulated urban expansion and the associated rise in solid waste pollution, community groups in Kikajjo and Lubowa launched informal waste collection and recycling programs. These locally organized initiatives were aimed at reducing illegal dumping of waste into streams and drainage channels, which has been a major contributor to water pollution and clogged infrastructure (Makindye Ssabagabo Municipal Council, 2022).

Long Term

As part of its response to rapid and unregulated land use changes, Makindye Ssabagabo Municipal Council adopted a Physical Development Plan in 2019 to guide orderly urban growth and promote sustainable land management. This strategic plan outlines long-term frameworks for zoning, infrastructure planning, and wetland protection, particularly in high-growth wards like Bunamwaya, Kikajjo, and Seguku (Makindye Ssabagabo Municipal Council, 2019). The plan focuses on regulating settlement patterns, preserving environmentally sensitive areas, and coordinating land use across sectors.

In response to the effects of deforestation and land degradation caused by unregulated land use changes, community-based tree planting initiatives have been implemented in Namasuba and Mutundwe. Organized groups, including youth and local environmental associations, have carried out tree planting along riverbanks and hill slopes to rehabilitate areas affected by vegetation loss and soil erosion (Makindye Ssabagabo Municipal Council, 2022). These efforts aim to restore forest cover, enhance soil stability, and rebuild degraded ecosystems. The planted trees also contribute to the regeneration of riparian buffers, the reduction of surface runoff, and the improvement of microclimatic conditions in formerly deforested zones.

With support from NGOs and the municipal council, residents participated in mapping flood-prone and degraded zones across the municipality. The data collected is being used for future restoration planning and zoning, forming a foundation for long-term land management.

To address the growing pressure on open spaces and wetlands caused by land use change, the Makindye Ssabagabo Municipal Council has promoted sustainable urban farming techniques, particularly in densely populated areas such as Mutundwe and Kabuuma. One notable practice has been the introduction of sack gardening and container-based agriculture, aimed at reducing the need for cultivating wetlands and undeveloped land (Makindye Ssabagabo Municipal Council, 2022). This intervention is part of a broader strategy to shift long-term land use patterns, improve household food security, and preserve fragile ecosystems.

In response to sensitization efforts, several households around Kyetinda Wetland voluntarily dismantled semi-permanent structures built within the wetland buffer. This community-led action enabled partial restoration of natural drainage and water retention capacity, and prevented further degradation during seasonal floods.

2.7.6 Recommendations

Strengthening land use planning is crucial in controlling the unregulated expansion of settlements in ecologically sensitive zones such as Kikajjo and Zana. Implementing the Makindye Ssabagabo Physical Development Plan can help prevent further encroachment into wetlands and agricultural lands. Similar success was observed in Wakiso District, where the adoption of a district-level physical development plan improved zoning compliance and reduced land conflicts through structured land allocation (Gipea Africa Ltd., 2019). By enforcing proper zoning laws and equipping local authorities with GIS tools to monitor real-time land use changes, Makindye Ssabagabo can better regulate development.

Encouraging vertical and infill development in built-up areas like Namasuba and Lubowa can reduce pressure on greenfield areas such as Kigo and Mutungo. Vertical housing provides more units per plot and helps accommodate a growing population without expanding into agricultural or wetland zones. In Kampala, vertical housing projects have enabled more efficient land use and improved service delivery in high-density areas (ArchDaily, 2016). Infill development utilizes underused plots within existing urban zones, reducing urban sprawl and infrastructure costs. Incentives such as tax waivers or streamlined building permits can motivate developers to adopt this sustainable urban form.

Sand and clay mining, particularly in Masajja, and Kigo, has significantly altered land cover and destabilized riverbanks. Establishing strict mining regulations and environmental impact assessments can reduce land degradation and promote site restoration. In Namibia, the implementation of a new regulatory framework helped control illegal mining and rehabilitate degraded riverbeds (The Namibian, 2023). In Makindye Ssabagabo, local councils should designate specific mining zones, issue licenses, and enforce post-extraction rehabilitation. This will protect water sources from sedimentation and help restore ecological balance in mined areas.

Restoration of degraded wetlands and riverbanks such as Kaliddubi, Kyetinda, and River Lufuka is essential for improving ecological stability and water quality in Makindye Ssabagabo. Wetlands

serve as natural filters, absorbing pollutants and moderating floods. The International Water Management Institute (IWMI) successfully implemented restoration projects in Uganda, demonstrating improved biodiversity, reduced flood risk, and cleaner water sources (IWMI, 2023). In Makindye, a similar approach involving replanting native vegetation and reestablishing wetland boundaries could rejuvenate its ecosystem services.

Engaging local communities in these restoration efforts ensures sustainability and helps reduce continued encroachment and degradation. Raising community awareness about sustainable land use and environmental conservation is critical for long-term success. In Northern Uganda, sensitization campaigns have led to behavior change in hygiene, waste management, and conservation practices (Protect Your Environment Now, 2013). A similar approach in Makindye Ssabagabo can educate residents about the importance of protecting wetlands, following zoning regulations, and practicing responsible farming. Using platforms such as radio programs, community meetings, and local leaders to relay messages ensures grassroots involvement and ownership of conservation initiatives. This collective action is vital for reducing encroachment and supporting enforcement efforts.

2.8 Urbanization and settlement

2.8.1 Introduction

Makindye Ssabagabo Municipality is one of the fastest growing cities in the Greater Kampala Metropolitan area, with a population density of 7,251 people per km^2 and a 10% per annum population growth[3]. Informal settlements are rapidly growing in peri-urban areas, driven by a lack of affordable housing and migration from Kampala.

2.8.2 State and trend

Urbanization and settlement refer to the process of people moving from rural areas to cities and towns, resulting in the growth and development of urban centers. Makindye Ssabagabo is currently experiencing a high rate of urban expansion influenced by both population migrations from different parts of the country and the spillover effects of the Kampala Capital City operations. The population is characterized by a mix of urbanized, affluent, less affluent, and slum areas. Unfortunately, the urbanization process has not been adequately controlled and guided leading to sprawl and poor environment (Physical Development Plan,2019). The municipality borders Kampala, Uganda's capital city, and a has a projected population of about 563,900 people, of these,262,200 people (46.5%) are males whereas 301,700 people (53.5%) are females, according to UBOS 2025.

[3] Uganda Bureau of Statistics, 2021

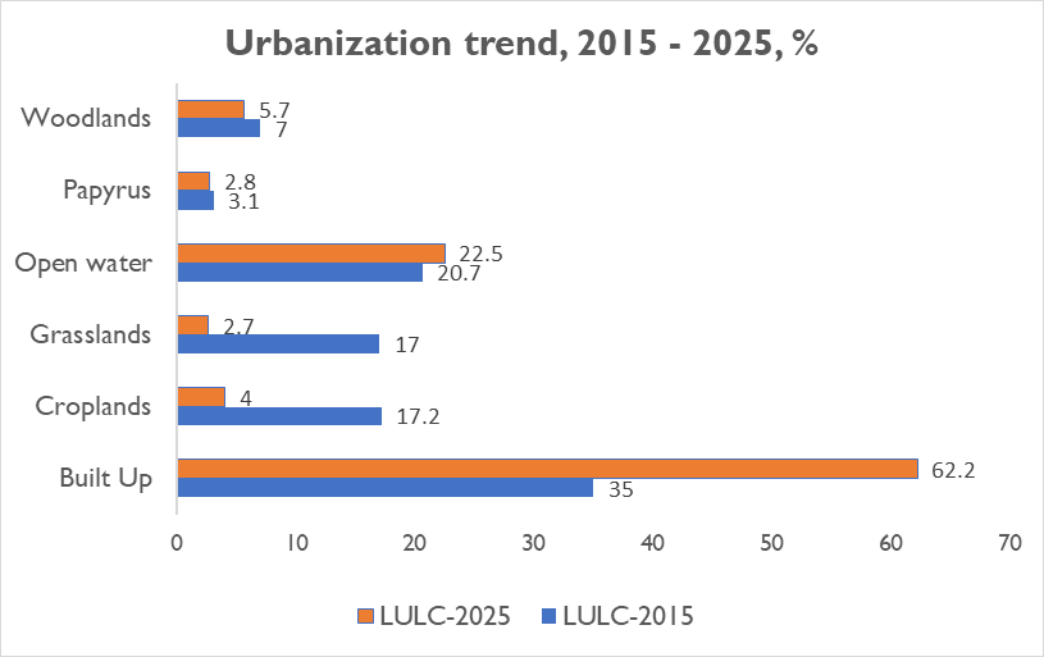


Figure 27: Urbanization trends between 2015 and 2025 in Makindye Ssabagabo

2.8.3 Drivers of urbanization and settlement

Makindye Ssabagabo municipality is experiencing rapid urbanization, driven by a combination of factors that are transforming the socio-economic landscape of this urban center. Understanding the drivers of urbanization and settlement in Makindye Ssabagabo is crucial for effective urban planning, infrastructure development, and service delivery.

Proximity to Kampala: Makindye Ssabagabo location, bordering Kampala, Uganda's capital city, contributes to its urbanization trend. The municipality's proximity to economic opportunities, infrastructure, and services in Kampala attracts people seeking better living standards.

Economic Opportunities: The municipality's growing economy, driven by local businesses, industries such as roofings Ltd, and services, creates job opportunities, attracting people from rural areas and other parts of the country.

Infrastructure Development: The municipality's Investments in infrastructure, such as roads, transportation, and utilities, enhance the municipality's attractiveness to residents, businesses, and investors. The arterial road network and access to public transport has led to significant growth and expansion of settlements in the various parts of the municipality with a centric origin of urban corridor concentrations that have occurred along the major traffic corridors of Entebbe Road, Busabala Road, and other arterials such as Bunamwaya, Lubowa, Seguku, Ndejje-Lugubumu, Masajja-Kibira, among others.

Demographic Factors: Makindye Ssabagabo has experienced a high population growth, which is influenced by the movement of people from rural areas in search of better economic services and living standards (Richmond et al., 2018). Changes in household structures, such as smaller family sizes and increased single-person households, influence housing demands and settlement patterns, coupled with a significant number of youths with high demand for services, housing, and employment, influence urbanization trends in Makindye Ssabagabo.

Environmental factors: The municipality's natural environment, including its climate, topography, and natural resources such as streams, rivers, Lake Victoria, and its catchments, provides a unique natural landscape that shapes settlement patterns and urban development. The settlement pattern is in preference of the gentle, sloppy land, utilizing the valley for farming and preservation. The areas Naziba, Ndikutumada, Ngube, Nfuufu, Kanaala, Zone 4&5, Mutungo Central, and Ziranumbu are the most raised but not steep and have been degraded for unplanned development (Physical Development Plan, 2019)

Social Factors: Access to education, healthcare, and social services, as well as cultural and recreational opportunities, attract people to urban areas.

Government Policies and Initiatives: Decentralization policies and initiatives, such as the Operation Wealth Creation Program, promote economic growth, agricultural development, and social services, contributing to urbanization.

2.8.4 Impacts of urbanization and settlement

Makindye Ssabagabo Municipality is experiencing rapid urbanization, which has significant environmental implications. The impact of urbanization and settlement in Makindye Ssabagabo can be seen in several areas:

Climate Change

Increased Greenhouse Gas Emissions: Urbanization and population growth in Makindye Ssabagabo leads to increased energy consumption, transportation, and industrial activities, resulting in higher greenhouse gas emissions. This contributes to climate change, exacerbating temperature increases and altering precipitation patterns.

Urban heat island: Urbanization has led to significant urban heat island effect that interacts with the local lake -land breeze circulation (Brousse et al.,2020).

Flooding: According to Sliuzas et al. (2013), flooding is a major concern expected to worsen due to climate change and increased surface sealing from construction. Communities adjacent to wetlands, such as Bunamwaya, experience frequent flooding due to heavy rainfall, which has resulted in loss of property, displacement, and interruption of businesses and education.

Resource depletion

Resource Consumption: Urbanization leads to increased consumption of resources, including water, energy, and materials, which puts a strain on the existing resources and infrastructure (Lwasa et al,2007).

Loss of ecosystem services: Urbanization can result in the loss of ecosystem services such as air and water filtration, soil formation and climate regulation.

Land Use/Land Change and Biodiversity

Habitat destruction – As the municipality experiences rapid urban growth, natural habitats, agricultural lands are being destroyed and converted into urban land uses, such as residential, commercial and industrial areas. This is evident in Nankinga in Bunamwaya parish, Mirembe and Wabiyinja in Mutundwe parish, and Lweza A in Mutungo (Physical Development Plan,2019). Habitat destruction and loss of biodiversity have affected ecosystems and ecosystem services (White et al.,2015).

Land Degradation: Urban development has resulted in soil erosion, loss of fertile land, and increased risk of landslides (Lwasa et al.,2005).

Waste Management

Increased waste generation: Urbanization in Makindye Ssabagabo leads to increased waste generation, which strains waste management infrastructure and pose environmental and health risks, requiring effective waste management strategies to maintain environmental sustainability.

Inadequate waste disposal: Inadequate waste disposal can lead to environmental pollution, including soil and water pollution.

Air and water pollution

Air pollution: Urbanization leads to increased emissions from vehicles, industries and energy consumption, contributing to air pollution and negatively impacting human health.

Water pollution: Urban runoff, industrial activities, and inadequate waste management pollute water sources, affecting human health and ecosystems.

Increase in informal settlements

The housing situation for low-income earners has further deteriorated, with urban poor living in substandard dwellings lacking in both quality and quantity (Mukiibi,2011). The increased housing demand, rising land prices and growing urban poverty, has reduced access to decent housing and seen an increase in informal (slums) development, posing challenges for municipal service provision and environmental management

2.8.5 Responses

- **Proper urban planning:** Makindye Ssabagabo municipal council in response to the impact of urbanization and poor land management is implementing effective urban planning strategies to manage growth and development, reducing negative environmental impacts. The Makindye Ssabagabo Municipal Physical Development Plan 2019- 2024 was developed to guide and control the urbanization process.
- **Climate Smart Cities Challenge:** The municipality's participation in the Climate Smart Cities Challenge highlights efforts to address environmental concerns and promote sustainable development.
- **Solid waste management; T**he municipal council is focusing on garbage collection. The collection and transportation of solid waste to landfills have been fully privatized. Households now pay between 10,000/= and 15,000/= per month for door-to-door waste removal services. The development of a municipal waste management integrated strategic plan is under underway.
- **Densification of existing low-density plots,** medium density, reasonable and high standard single and semi-detached house plots by subdivision and/or addition of residential units on the plot.
- Providing and/or upgrading infrastructure (water, sewage, electricity, roads, lighting, etc.).

- **Restructured Markets System** including, Extension, upgrading and establishment of commercial markets in residential zones integrated with the transport system and extending, upgrading and developing new tourism attractors and facilities
- **Land use planning** and servicing in urban and peri-urban areas; expulsion and banishment of wetland encroachers.
- **Tree planting:** Initiatives to plant trees are being considered to help manage environmental impacts.
- **Regulation of commercial activities:** The council is regulating commercial cyclists and park fees, and valuing properties in the municipality to improve revenue.
- **Community Mobilization and empowerment:** The municipal council is working with development partners, including NGOs to mobilize communities to participate in socio-economic and political development programs, such as the Youth Livelihood Program and Uganda Women Entrepreneurship Programme.
- **Revenue management:** The municipal council is working to improve revenue performance through decentralization, enabling it to better fund development projects and services.

2.8.6 Recommendations

- Restrict further urban sprawl and investing in brownfields development for mixed-use residential and commercial use will bring people closer to economic opportunities and services.
- Implementing sustainable urban planning practices is crucial to address infrastructure development, service delivery, and environmental sustainability challenges in Makindye Ssabagabo.
- Implementing waste reduction and recycling programs: Minimize waste generation through raising awareness about means to reduce wasteful consumption, re-use resources and recycle.
- The municipal council to collaborate with implementing partners and conduct sensitization initiatives that focus on socialization structures, involvement of women, youth and marginalized groups to promote better waste management practices
- Intensifying the carrying out of environmental impact assessments for the development projects before they are carried out, which will enable achievement of sustainable development.
- Tree planting and green infrastructure: Planting and incorporating green infrastructure such as parks and green roofs in Makindye Ssabagabo can help manage stormwater runoff, reduce urban heat islands effects, and promote biodiversity.
- Environmental education and awareness: Promoting environmental education and awareness can help individuals and communities understand the importance of environmental protection and take action to reduce their environmental footprint.
- Creating awareness and improving access to family planning clinics and contraceptives municipality. While population growth is not necessarily bad for the environment, when accompanied by insufficient housing, limited land tenure, services and employment, it can have an increasingly negative impact on the environment.
- The urgent need for developing a comprehensive piped sewage system for all urban areas (existing and planned).

- Improve accessibility, availability and affordability of quality health services in the municipality, especially reproductive health services.
- Design and implement awareness programs on reproductive health rights and emphasize the role of men in reproductive health issues.
- Improve public education services and ensure that established population groups have functional and competitive skills and participate in training and functional adult literacy programs.
- Empower women through education and promote retention of the girl child in school to stifle early marriages and teenage pregnancies.
- Intervention at the national level by integration of urban and rural development programs: both the urban and rural should be accorded similar attention in infrastructure development, social services and livelihood programs.
- Local governments should make byelaws stipulating requirements to register all new settlers in the municipality and there must be a legal requirement making it mandatory for NIRA to share information generated on registration of death and birth with the relevant local authorities for planning purposes.

2.9 Socio-economic status

2.9.1 State and trend

Makindye Ssabagabo Municipality has experienced rapid growth, driven by its proximity to Kampala and economic opportunities. The population has been increasing steadily, with a notable growth rate. According to UBOS 2024, compared to the national population projected at 48 million (49% male,51% female) in 2024 from 37.8 million in 2015 (27% increase), Makindye Ssabagabo municipality has a projected population of about 563,900 people (Table 1). Of these, 262,200 people (46.5%) are males whereas 301,700 people (53.5%) are females. Between 2015 and 2025, the population in Makindye Ssabagabo municipality is projected to have increased by 256,100 people (88.7%), exerting pressure on the already scarce natural resources such as land, wetlands, and forests. Masajja division has the highest population of 233,400 people in Makindye Ssabagabo municipality (Table 1). The municipality has a population density of about 6,640 people per square kilometer. Bunamwaya division has the highest population density of 11,521 people per square kilometer.

Table 9: Projected population in Makindye Ssabagabo Municipality[4]

Divisions	2015			2025				
	Male	Female	Total	Male	Female	Total	Area (Sq.km)	Population density
Bunamwaya	35,600	40,900	76,500	66,700	77,700	144,400	12.5	11,521

[4] **Data Source:** UBOS, 2025

Masajja	57,800	65,900	123,700	108,300	125,100	233,400	28.3	8,252
Ndejje	46,500	52,100	98,600	87,200	98,900	186,100	44.1	4,220
Total	139,900	158,900	298,800	262,200	301,700	563,900	84.9	6,640

Gender, Youth, and Social Groups

In Makindye Ssabagabo Municipality, different gender and social groups contribute to and are impacted by environmental changes in various ways.

Women: Women comprise over 53.5 percent of Makindye Ssabagabo population (UBOS, 2025). Their domestic activities, such as waste generation and water usage contribute to environmental changes. However, they also play a crucial role in promoting sustainable practices and environmental conservation within their households and communities. Women often manage household waste and can influence family members' environmental behaviors. Providing women with the opportunity and empowerment to participate and influence decision making is key for environmental management.

Youth: According to UBOS National Population and Housing Census of 2024, Wakiso has 849,857 youths,424,186 in employment,87,208 are unemployed, 511,394 youths are in active labor resulting in an unemployment rate of 17.1 percent slightly higher than the national rate of approximately 13-15%, while 338,463 youth (39.8%) are neither employed nor unemployed (not in the labor force for instance students and trainees, discouraged workers and home makers and care givers).

Makindye Ssabagabo Municipality is part of Wakiso, the municipality has a significant youth demographic, with many young people driving socio-economic dynamics and demanding services. Youth in the municipality contribute to environmental changes through their consumption patterns, including the use of single-use plastics and other non-biodegradable materials. Young people can also be powerful advocates for environmental conservation and sustainability, influencing their peers and communities to adopt more sustainable practices.

Marginalized Groups: Marginalized groups, including people with disabilities, refugees and the elderly, have limited access to environmental services and infrastructure, which contributes to environmental changes. These groups often face significant challenges in accessing basic services, including clean water, sanitation, and healthcare, which can exacerbate environmental health issues.

Livelihoods derived from natural resources in Makindye Ssabagabo municipality

Makindye Ssabagabo Municipality, with its proximity to Kampala and Lake Victoria, supports diverse livelihoods reliant on natural resources. These activities drive socio-economic development but also exert pressures on the environment, such as resource depletion and pollution. Based on stakeholder consultations (e.g., workshops in divisions like Masajja and Bunamwaya) and data from UBOS (2024 Census) and municipal vulnerability assessments under the GKMA-UDP, approximately 40-50% of the municipality's ~563,900 residents depend on natural resource-based livelihoods. This includes agriculture (engaging ~25% of households),

fishing (~10-15%), eco-tourism (emerging, ~5%), and extractive activities like sand mining and bricklaying (~10%).

- **Agriculture and farming**: A significant sector in Makindye Ssabagabo municipality, engaging residents in crop (e.g., maize, vegetables) and livestock (e.g., poultry, cattle) production, leveraging the municipality's fertile soils and location near Lake Victoria. It supports households, particularly in rural fringes like Ndejje Division, providing food security and income. However, intensification (e.g., chemical use) has led to soil erosion and water pollution.
- **Fishing:** Is a cornerstone livelihood in Makindye Ssabagabo Municipality, especially in lakeshore areas such as Busabala, rivers, and wetlands such as Lufuka Wetland**.** It supports thousands of residents, by providing direct employment to fishermen and indirect jobs to fish processors, traders, transporters, and boat builders, contributes to food security (fish as a major protein source) and drives local economic activity, many women and youth are involved in fish processing and trading, making fishing a key sector for inclusive economic development. However, sustainable management is essential to ensure the long-term viability of this sector and the health of Lake Victoria's ecosystem (MAAIF,2022).
- **Eco-tourism:** The area's natural resources, such as forests, wetlands, wildlife (e.g., bird species in Lubigi), and cultural heritage (e.g., local fishing traditions), support emerging eco-tourism initiatives. This includes guided wetland tours and community homestays in Busabala, generating income for residents (e.g., through lodges and crafts). Potential is high but underexploited due to pollution.
- **Income-generating activities such as bricklaying and sand mining:** These extractive activities provide quick incomes, especially for unemployed youths (~17.1% rate, UBOS 2024) in areas like Masajja. Sand mining from wetlands and rivers supports construction booms, employing ~5,000-7,000 people, while bricklaying uses local clays. However, they cause severe environmental damage, such as wetland degradation (20% loss) and erosion.

2.9.2 Drivers of Socio-economic factors

The socio-economic landscape of Makindye Ssabagabo Municipality is shaped by a complex interplay of factors that influence the lives of its residents. Understanding the drivers of these socio-economic factors is crucial for developing effective strategies to promote sustainable development, improve the well-being of the community and realize sustainable environmental management.

- **Rapid population growth and Urbanization:** Makindye Ssabagabo is among the fastest-growing urban centers in GKMA, experiencing intense population growth due to its proximity to Kampala City. This demographic shift is driven by rural-urban migration in search of employment opportunities in and around Kampala, better services, and improved living standards. This has put pressure on existing infrastructure and services, contributing to challenges like dense settlements, poor sanitation, habitat encroachment for instance, informal settlements on wetlands in Masajja division have expanded rapidly to accommodate incoming populations, while Informal settlements emerging in Ndejje ward have strained urban infrastructure and services. Population density strains land, water, and waste systems, and the urban sprawl converts green spaces into informal settlements, disrupting ecosystems.

- **Poverty and livelihood needs:** A significant portion of the population in Makindye Ssabagabo relies on informal sector activities such as small-scale trade, artisanal work like brick-making, furniture, and urban agriculture for survival. Poverty (affecting 30% of households per 2022 UBOS data) forces people into unsustainable resource use such as charcoal production and wetland encroachment for agriculture/fuelwood and illegal sand mining in rivers for income and construction, degrading water bodies. For example, brick-making in wetlands is a common livelihood because the clay soil and water are free raw materials.
- **Economic development and industrialization**: Makindye Ssabagabo has seen growth of small-scale industries, commercial centers, and informal businesses, ranging from metal fabrication workshops to food processing units, often operating with minimal environmental oversight, especially in areas like Namasuba, Ndejje, and Bunamwaya. High unemployment rates, particularly among youth and women, drive economic growth but at environmental costs.
- **Gender and Social Equity Dynamics:** Women, youth, and marginalized communities often have different relationships with the environment based on their social roles, economic activities, and access to resources. Traditional gender roles frequently position women as primary managers of household resources (water, fuel, food), making them disproportionately affected by environmental degradation while also being key agents of change in sustainable resource management.
- **Land market dynamics:** Proximity to Kampala drives up land prices in formally planned areas. This has pushed lower and middle-income earners to seek cheaper, often untitled land in environmentally sensitive areas like wetlands and steep slopes. Additionally, the rising land values have incentivized the conversion of wetlands and green spaces into commercial and residential developments. For instance, areas like Lweza and Seguku have witnessed extensive wetland encroachment for real estate development.
- **Policy and governance gaps:** Weak environmental governance structures and limited institutional capacity at the municipal level hinder effective environmental management. While the National Environment Act, 2019 mandates urban councils to prepare State of Environment Reports every two years, implementation challenges persist due to limited resources, technical capacity, and coordination among different sectors. The lag between the pace of development and the ability of the Municipal Council to enforce the Physical Development Plan, zoning regulations, and environmental laws fuels unplanned settlements. For example, inadequate zoning allows urban sprawl into green spaces.

2.9.3 Impacts of Socio-economic factors

The socio-economic factors in Makindye Ssabagabo Municipality have significant impacts on key sectors such as agriculture, tourism, and natural resource management, influencing the livelihoods of residents and the municipality's economic development. These impacts can be both positive and negative, shaping opportunities and challenges faced by the community in these critical areas.

- **Public Health Burden:** Waterborne diseases like cholera and typhoid show periodic outbreaks, particularly in informal settlements. Respiratory infections linked to air pollution have increased, with children and elderly populations most affected. The municipality's health facilities report treating over 30% more respiratory cases compared to a decade ago.

- **Economic Losses:** Flooding damages property and disrupts business operations, with annual losses estimated in billions of shillings. Small businesses in flood-prone areas like Kitebi market lose inventory and income during flood events. The cost of water treatment has increased due to deteriorating raw water quality.
- **Livelihood Disruption:** Fisherfolk along Lake Victoria shores report declining catches due to pollution and habitat degradation. Urban farmers face reduced yields from contaminated soils and irrigation water, while Informal waste collectors face health risks from unmanaged dump sites.
- **Reduced Ecosystem Services**
- Degraded ecosystems such as wetlands and forests provide fewer services such as flood regulation, water purification, climate regulation, and recreational opportunities, reducing overall quality of life and increasing municipal costs for artificial alternatives.
- **Impacts on Urban Resilience**
- The cumulative environmental degradation reduces the municipality's overall resilience to shocks and stresses, making recovery from disasters like floods more difficult and costly, and limiting adaptive capacity to future climate change impacts
- **Social Inequity:** Environmental degradation disproportionately affects low-income communities who cannot afford to relocate from hazardous areas or access clean water alternatives. Women are disproportionately affected by environmental changes, including water pollution and lack of access to clean water and sanitation facilities. They spend more time collecting water from distant sources when local sources become contaminated and bear the brunt of caring for family members affected by environmental-related illnesses, which can increase their workload and stress levels (Makindye Ssabagabo Municipal Council Physical Development Plan,2019).
- Youth are affected by environmental changes, including limited opportunities for sustainable livelihoods and increased exposure to environmental health risks. Young people may face challenges in finding employment or starting businesses in environmentally sustainable sectors, which can impact their economic prospects and well-being (Kamara et al.,2019).
- Marginalized groups are particularly vulnerable to environmental changes, including limited access to environmental services and infrastructure, and increased exposure to environmental health risks. These groups often lack the resources and support needed to adapt to environmental changes, making them more susceptible to negative impacts.

2.9.4 Responses

Makindye Ssabagabo municipality is actively addressing socio-economic factors impacting the environment through various initiatives including:

- Waste management: The municipality has introduced a waste management by-law that mandates waste collection containers, introduces fines for littering, and regulates garbage collection companies to improve efficiency and accountability.
- Formation of environmental committees at parish level
- Attempts to implement local environment ordinances and periodic clean-up campaigns in markets and public spaces
- Urban Planning: Development of a physical development plan and a detailed plan for the central business district to guide infrastructure development and ensure it aligns with

environmental sustainability goals, enabling enforcement of building regulations in some areas.

- Community involvement: The municipality is working with NGOs and CBOs through community sensitization campaigns to raise awareness about natural resource management and involve the community in development programs.
- Climate Smart Cities Challenge: Makindye Ssabagabo is involved in this challenge, focusing on developing solutions for zero-carbon neighborhoods, freight logistics management and green buildings.
- Partnership with NEMA for wetland restoration pilots
- Participation in the GKMA-UDP program.

However, these responses remain inadequate given the scale of challenges.

2.9.5 Recommendations

The environmental challenges facing Makindye Ssabagabo Municipality are complex and interconnected, requiring comprehensive responses that address underlying socio-economic drivers while building resilience for future sustainability.

1. Institutional Strengthening

- **Establish Environmental Management Unit**: Create a dedicated unit within the municipality with adequate staffing and resources to coordinate environmental management activities. This unit should include environmental officers, planners, and enforcement personnel.
- **Develop Environmental Information System:** Implement a digital platform for tracking environmental indicators, citizen reporting, and data management. This system should integrate with national databases and support evidence-based decision-making.
- **Strengthen Enforcement Capacity:** Train and equip environmental police and municipal enforcement officers to effectively implement environmental regulations. Establish environmental courts or specialized tribunals for expedited case handling.

2. Infrastructure Development

- **Green Infrastructure Investment:** Prioritize nature-based solutions such as constructed wetlands for wastewater treatment, bioswales for stormwater management, and urban forests for air quality improvement. Pilot projects could start in areas like Lweza and Seguku.
- **Waste Management Systems:** Establish material recovery facilities in each division, promote source segregation through incentive programs, and support community-based composting initiatives. Partner with private sector for recycling value chains.
- **Sustainable Transportation:** Develop non-motorized transport infrastructure including dedicated bicycle lanes and pedestrian walkways. Support transition to cleaner fuels through partnerships with boda-boda associations.

3. Economic Instruments

- **Environmental Fiscal Reforms:** Introduce graduated environmental levies based on pollution levels, with revenues ring-fenced for environmental management. Provide tax incentives for businesses adopting cleaner production technologies.

- **Eco-jobs and green enterprises:** Create employment opportunities in environmental management through waste collection cooperatives, waste recycling, urban farming initiatives, and ecosystem restoration projects. Target youth and women's groups for these opportunities.
- **Payment for Ecosystem Services:** Develop mechanisms to compensate landowners who maintain wetlands and green spaces. Partner with water utilities to fund upstream conservation activities.

4. Community Engagement

- **Environmental Education:** Integrate environmental education in schools and conduct regular community sensitization programs. Use local languages and culturally appropriate messaging to enhance understanding.
- **Participatory Planning:** Involve communities in environmental planning through parish-level consultations and citizen science programs. Establish environmental action groups in each ward.
- **Traditional Knowledge Integration:** Document and incorporate traditional environmental management practices into modern conservation strategies. Engage cultural leaders as environmental ambassadors.

5. Regulatory Framework

- **Environmental Ordinances:** Develop and enforce municipal environmental ordinances addressing specific local challenges such as wetland protection, noise pollution, and waste management.
- **Strategic Environmental Assessment:** Require environmental impact assessments for all major developments and integrate environmental considerations into the municipal physical development plan.
- **Building Standards:** Update building codes to include green building requirements such as rainwater harvesting, energy efficiency, and proper waste management facilities.

6. Monitoring and Evaluation

Environmental Quality Monitoring Establish permanent monitoring stations for air and water quality with real-time data transmission. Partner with research institutions for regular biodiversity assessments.

Performance Indicators Develop and track key performance indicators including:

- Percentage of solid waste collected and properly disposed
- Water quality index for major water bodies
- Air quality index for major centers
- Wetland coverage and condition
- Green space per capita

Annual Reporting: Produce annual state of environment reports with clear targets and progress tracking. Make reports publicly available and use findings to adjust strategies.

3. Chapter Three: Integrated Assessment

3.1 Environment and Agriculture

Agriculture plays a vital role in the economy and livelihood options in Makindye Ssabagabo Municipality, particularly in peri-urban areas and on undeveloped land. Despite the urban expansion, which has reduced the amount of land available for farming, many residents continue to engage in small-scale crop cultivation, such as maize, beans, vegetables, bananas (matooke), cassava, and sweet potatoes, poultry rearing, and piggery for household consumption and local markets. Agriculture employs approximately 8% of the total workforce in the GKMA region, indicating its continued significance. In Makindye Ssabagabo Municipality specifically, about 26% of households are involved in either crop farming or animal husbandry. Of these, 12.5% focus solely on crop production, while 19.2% are engaged exclusively in livestock keeping (MSB, 2019). These agricultural activities primarily support food security, especially among vulnerable households.

3.1.1 Impacts of Agriculture on the Environment in Makindye Ssabagabo

Increased greenhouse gas emissions. Poultry rearing, contributes significantly to GHG emissions. Exacerbated by poor feed quality and inefficient breeds.

Wetland Encroachment and Degradation. Expanding cultivation, particularly in peri-urban wetlands such as Kabuuma, and Kaliddubi wetlands, has led to the loss of wetland vegetation, reduced ecosystem functioning, and declining water quality due to increased runoff and sedimentation.

Agrochemical Pollution and Fertilizer Emissions. Indiscriminate use of chemical fertilizers, herbicides, and pesticides without proper guidance leads to nutrient loading in water sources, contamination of wetlands, and potential health hazards for both people and wildlife. Methane and Nitrogen can contribute to GHG emissions.

Deforestation and Loss of Biodiversity. Clearing of natural habitats such as Kabuuma wetland for farming reduces vegetation cover, destroys habitats for flora and fauna, and contributes to loss of biodiversity.

Livestock Waste and Water Quality. Unregulated livestock grazing and inadequate manure management contribute to nutrient pollution and faecal contamination of local streams and wetlands.

Urban-Agriculture Interface Conflicts. Increasing pressure from residential, commercial, and infrastructure developments often leads to land fragmentation, disputes, and incompatibilities with agricultural uses, further threatening the sustainability of the sector.

3.1.2 Recommendations to environmental challenges from Agriculture in Makindye Ssabagabo

Protect Wetlands and Sensitive Ecosystems through sensitization and protecting wetlands from encroachment. Community wetland users should be supported with livelihood options that minimize wetland degradation, such as switching to ecotourism.

Improve Agrochemical Use and Waste Management

- Strengthen farmer education and extension services on safe and judicious use of fertilizers and pesticides.
- Encourage organic farming methods and integrated pest management to reduce reliance on chemicals.

- Introduce proper farm waste management practices, including the establishment of manure pits and composting.

Secure Urban-Agriculture Interface

- Integrate agricultural land considerations in urban physical planning to minimize land use conflicts and preserve productive zones within the municipality.
- Support the formation of farmer organizations and advocacy groups for effective engagement in urban development processes.

Strengthen Stakeholder Engagement and Capacity Building

- Conduct regular participatory planning sessions with farmers, local leaders, and technical officers to review and adapt environmental management strategies.
- Build local capacity for monitoring, early warning, and response to environmental issues related to agriculture.

3.2 Environment and Industry

The industrial sector in Makindye municipality accounts for 6% of businesses and 4% of employment. Roofings Group, located in Lubowa, Makindye Ssabagabo, is the largest manufacturer of steel construction materials in Uganda with an installed production capacity of 72,000 metric tonnes annually. Local manufacturing primarily focuses on carpentry (furniture), metal fabrication (window frames, doors, gates), and agro-processing (grain milling, animal feeds, milk processing). While industries provide significant benefits to the economy through employment, they also present potential risks to the environment that require appropriate management and regulation. Industrial activities are a source of pressure on the environment, mainly in the form of waste generation, resource consumption, and emissions to the atmosphere, soil, and water ecosystems. Air pollution from industrial facilities has significant costs associated with its impacts on ecosystems and climate. Some industrial sites cause local issues with water and air quality and can create noise and odors problems, which impact a community's ability to enjoy its local environment.

3.2.1 Impacts of Industrialization on the Environment in Makindye Ssabagabo

- Air Pollution; Emissions and Fumes: Small-scale industries emit untreated fumes containing particulate matter (PM2.5/PM10) and greenhouse gases.
- Occupational Health Risks; Lack of Safety and protective gear leads to respiratory illnesses, hearing loss, and injuries.
- Odor Nuisance.
- Impacts on amenity; When not appropriately controlled, emissions from industry may directly reduce the amenity of the environment and the local communities nearby.

3.2.2 Recommendations to environmental challenges associated with industrialization in Makindye Ssabagabo

- Regular Monitoring: Conduct quarterly environmental audits for industries, focusing on wastewater discharge, air emissions, and waste management.
- Relocate Non-Compliant Industries: Move industries operating in wetlands or residential areas to designated zones, as per the Makindye Ssabagabo Physical Development Plan.
- Adopt Cleaner Production Technologies

- Incentivize Green Practices: Offer tax breaks or subsidies to industries adopting renewable energy (e.g., solar), water recycling systems, or waste-to-energy technologies.
- Phase Out Polluting Processes: Ban open burning of waste and encourage alternatives (e.g., biogas digesters for organic waste).
- Hazardous Waste Collection: Partner with private firms to collect and safely dispose of chemicals, oils, and electronic waste.
- Restore Degraded Ecosystems
- Wetland Rehabilitation: Launch community-led initiatives to replant native vegetation in encroached wetlands
- Enforce Buffer Zones: Maintain 100-meter buffer zones around all wetlands, as per the National Environment Act, 2019.
- Enhance Worker and Community Safety
- Mandate PPE Use: Enforce provision of protective gear (masks, gloves, earplugs) for workers i n high-risk industries (e.g., welding, chemical processing).
- Community Sensitization: Train residents near industrial zones to monitor pollution and report violations through toll-free hotlines.
- Strengthen Stakeholder Collaboration
- Public-Private Partnerships (PPPs): Collaborate with industry associations to fund cleaner production training programs.
- Community Liaison Committees: Establish forums for industries, residents, and local leaders to address grievances and co-design mitigation plans.

3.3 Environment and Energy

3.3.1 Introduction

Energy includes both direct use of biomass, oil, liquid, and wood fuels and use of electricity. The generation of useful energy for mobility, heating, and electricity through the combustion of oil products and natural gas releases polluting by-products that have significant adverse impacts on the climate and environment. Fossil fuel use is a significant source of key air pollutants and greenhouse gases (GHGs).

3.3.2 State and Trends

Makindye Ssabagabo Municipality depends on a variety of energy sources, including the national electricity grid, petroleum products, and traditional biomass. In recent years, electricity access has expanded, enabling more homes and businesses to connect to the grid, contributing to urban development, industrialization, and improved living conditions. However, several challenges remain, including inconsistent power supply, frequent power outages, and high connection costs, which limit access for many low-income residents, especially in Busabala, Ndejje, and Mutungo parishes. In informal and peri-urban areas, many households and small businesses still rely heavily on firewood and charcoal for cooking and heating, resulting in indoor air pollution. Charcoal and firewood are the most widely used cooking fuels, though other sources like LPG (liquefied petroleum gas), biogas, and briquettes are used to a lesser extent. Much of the charcoal is imported from the North and Northwestern regions of Uganda, as local supplies can't meet

energy demands. Lighting needs are mostly met through electricity from the national grid, solar power, rechargeable lamps, and batteries.

Petroleum products such as petrol, diesel, and LPG are commonly used for transportation, small-scale industrial activities, and increasingly for domestic purposes as household incomes grow. The presence of numerous fuel stations and LPG distribution points across the municipality reflects a rising demand and gradual transition toward cleaner energy options. Nonetheless, high and volatile petroleum prices, coupled with limited LPG adoption for cooking, hinder widespread usage, particularly among low-income groups.

Briquettes and energy-efficient stoves have been promoted to reduce dependency on firewood and charcoal and minimize the carbon footprint on the environment but remain expensive alternatives. Despite these initiatives, biomass remains the predominant energy source for most households and small businesses.

3.3.3 Drivers

The drivers of energy demand in Makindye Ssabagabo municipality vary across the energy-consuming sectors, but economic growth and population growth are the most significant drivers.

- Electricity prices have a key impact on the energy demand. The high dependency on imported fossil fuels makes the municipality highly vulnerable to fuel price variations, which in turn are influenced by geopolitical events.
- Transport fuel demand; Transport GHG emissions have seen the greatest growth in energy-related emissions. This growth can be directly linked to the increased number of passenger cars and freight transport.

3.3.4 Impacts of Energy resources on the environment in Makindye Ssabagabo

Air pollution: Poor air quality results from the emission of pollutants into the atmosphere. The major pollutants include particulate matter, Sulphur dioxide, nitrous oxides, and volatile organic compounds. These pollutants pose health risks to human and animal respiratory systems, and in some cases can trigger various cancers. Indoor air pollution from small-scale combustion of solid and liquid generates high levels of particulate matter, carbon monoxide and, in some instances, Sulphur dioxide, depending on the fuel choice increasing the risk of respiratory diseases.

Habitat destruction and loss of biodiversity; Biomass is a typical fuel used for household, commercial and industrial purposes. Unsustainable cutting of trees for firewood and charcoal burning can lead to biodiversity loss and ecosystem degradation.

GHG emissions; Carbon dioxide and methane from vehicles, burning of charcoal and wood fuel are the most common GHGs linked to the energy value chain. Their increased levels in the atmosphere prevent heat from escaping into space, thereby contributing to global climate change. These impacts may include increases in ambient temperatures and extreme weather events, and variations in rainfall patterns.

Human settlements; Charcoal and Electricity are the largest sources of energy used by urban households for lighting, cooking and heating. This illustrates the resource intensity of human settlements, resulting in environmental impacts such as light pollution in urban areas, loss of habitat as a result of development and increased GHG emissions.

3.3.5 Recommendations to environmental impacts of energy resources in Makindye Ssabagabo

- The government of Uganda, through the Uganda Electricity Generation Company Limited (UEGCL) UEGCL is encouraged to reduce tariffs on HEP to encourage households to use HEP for cooking other than wood fuel
- The government of Uganda, through the Petroleum Authority of Uganda is encouraged to reduce tariffs on LPG to encourage households to use it for cooking and minimize tree cutting for wood fuel
- Mass sensitization and capacity building on the use of alternative and renewable energy sources like solar and biogas.
- Enforcement of regulations on tariffs and environmental levies on charcoal
- Energy decarbonization alternatives include energy demand reduction
- Increased electrification
- Deployment of market-ready renewables and low-carbon heating options

3.4 Environment and Transport

A sustainable, accessible, and efficient transport system is important for trade, the economy, and human well-being, characterized by low environmental pressures, cleaner, quieter towns and city centers. The transport sector is a major consumer of energy and material resources and a source of environmental pollution, particularly greenhouse gases, other pollutants, and noise.

3.4.1 State and Trend

The high population density of Makindye Ssabagabo requires a well-functioning, frequent, and reliable transport system to serve the population. Boda-bodas and commuter taxis are the most widely used modes of public transport due to their time-saving nature and affordability, respectively. However, their increased usage has contributed to environmental pollution and high vehicle maintenance costs. Meanwhile, non-motorized modes of transport, such as walking and cycling, face significant challenges due to inadequate infrastructure and a lack of designated space on the existing road network.

There is a challenge of a lack of integrated public Transport infrastructure limiting intermodal linkages (between road and water transport). Water Transport Services (Ferry services) don't exist in the municipality despite the presence of landing sites (Busabala and Nakabugo). As an alternative, Private boat operators provide informal services on waterways, mainly using open canoes with onboard or outboard engines. The canoes carry passengers, goods, and livestock. The vessels are often overloaded, with the risk of capsizing.

Makindye Ssabagabo Municipality lacks a transport hub, where passengers and cargo are exchanged between vehicles or between transport modes. Lack of a planned taxi park, commuter taxi parks are thus unregulated.

3.4.2 Impacts from the Transport Sector on the Environment

- Energy consumption; The 2018 Physical Development Plan for Makindye Ssabagabo highlighted that continued growth in energy consumption in the sector, with a significant proportion of road transport energy met by fossil fuels

cars. The transport sector has substantial negative impacts on the environment and human health. Air pollution arises from diesel and petrol consumed by private cars and Heavy Goods Vehicles.

- Environmental noise from transport sources especially Boda and commuter taxis, is a health risk, has effects on sleep and cardiovascular and metabolic function.
- Altering the quality and connectivity of habitats, especially wetlands and creating physical barriers to the movement of animals between habitat areas.

3.4.3 Recommendations to Environmental Challenges from the Transport sector

- Reducing demand and encouraging a major modal shift to walking, cycling, and using public transport is a long-term goal as we move to more integrated spatial and transport planning.
- Ensure communities' access to affordable transport options.
- Reduce noise and air pollution, greenhouse gas emissions, and energy consumption. Noise action plans include the designation of quiet areas.

3.5 Environment and Education

3.5.1 Introduction

Education in Makindye Ssabagabo is critical in environmental management from its role in promoting awareness among people on energy conservation, environmental protection, and preservation. The municipality has a total of 379 education institutions comprising 248 primary schools, 47 secondary schools, and 84 pre-school institutions (PSI, inequitably distributed. Although PSIs were accessible to 98.4%, utilization was at 41.5%. A good number of PSIs in the municipality were not suitably located, in appalling condition, and not aptly designed to handle the delicate learners who are in the formative years that lay the foundation for subsequent learning.

The best-provided service amongst the above categories was primary schools, totaling 248. According to the socioeconomic survey, these were accessible to 96.5% of the households, and yet only 65.2% of the households utilize them. These were located within 1 km of 77.1 % of households. However, 93.5% of these were privately owned, and they are considered expensive to the majority of the municipality's residents.

3.5.2 Distribution and Access to Primary Education Services

Mutundwe and Namasuba, lack public primary schools despite the fact that these hosted 14.1% and 10.6% respectively of the municipal population in 2014. And many of the public primary schools in the municipality were in a deplorable infrastructural state, and services offered were, in some instances, Poor, whereas most private schools were located on small land parcels with no room for expansion, in addition to a lack of facilities for extracurricular activities.

3.5.3 Distribution and Access to Secondary Education Services

The municipality had 47 secondary schools, unevenly distributed, with the majority in Masajja and Bunamwaya wards. Secondary schools are accessible to 82.3%, but utilization was very low at 24.3%, and the municipality has only 2 public secondary schools. Much as vocational and tertiary institutions are vital for accelerated skilling relevant to the job market, there were no tertiary

institutions to cater to the municipality's rapidly swelling population. The general trend is that the majority of the educational institutions in the municipality (95.3%) were privately owned and regarded as expensive.

Table 10:The Demand for Education in Makindye Ssabagabo[5]

Type	Standard Capacity	Current	Deficit	Notes
Primary school	5,000	248	80	The municipality appears to have an oversupply of primary schools but only 16 were public so 10,00093.5% were private and expensive.
Secondary school	10000	47		Although the municipality seems to have no deficit in secondary schools, majority (45) were private, and the owners can easily convert the status of the school any time.
Vocational or Tertiary Institutions	50,000	0	8	The municipality has to plan for vocational institutions in the short term since they are basic for skilling the population.

3.5.4 Environmental protection opportunities by educational entities in Makindye Ssabagabo

- Schools should inculcate environmental knowledge, awareness, positive attitudes and behavior at all levels.
- The need to redevelop, redistribute, and develop new education facilities according to the land use plan.
- The need to plan for the establishment of vocational and tertiary institutions in the
- municipality for accelerated skilling relevant to the job market, enhancement of the competitiveness of the municipal residents, and building and strengthening the municipality's human capital.
- Waste recycling

[5] Source: PDP, Makindye Ssabagabo

4. Chapter Four: Environment and Planning

4.1 Stakeholder Involvement in Environment and Planning

The active participation of diverse stakeholders in environmental management in Makindye Ssabagabo Municipality can promote a more inclusive and transparent approach in addressing environmental challenges and proactive planning. Key stakeholders include government ministries, departments, agencies, the private sector, civil society organizations, and the local community. The key roles are described below.

Table 11: Stakeholder framework for Environmental management in Makindye Ssabagabo

Entity	Roles and responsibilities
National Environment Management Authority	- To ensure the integration of environmental concerns in overall national planning through coordination with the relevant ministries, departments and agencies of the government. - To liaise with the private sector, inter-governmental organizations, non-governmental agencies and governmental agencies of other states on issues relating to environment. - To propose environmental policies and strategies to the policy committee. - To litigate legislative proposals, standards and guidelines on the environment in accordance with the Act. - To review and approve environmental impact assessments and environmental impact statements submitted in accordance with the NEA or any other law. - To promote public awareness through formal, non-formal and informal education about environmental issues - To undertake such studies and submit such reports and recommendations with respect to environment as government or policy committee may consider necessary. - To ensure observance of proper safeguards in the planning and execution of all development projects, including those already in existence, that have or likely to have significant impact on the environment, determined in accordance with the act. - To undertake research and disseminate information about the environment. - To mobilize, expedite and monitor resources for environmental management.
Makindye Ssabagabo Municipal Council [1]	- Conduct due diligence on applications for titles within the municipality to ensure titles are issued according to the municipal land use plan. - With the approval of the environmental officer, order an Environment and Social Impact Assessment (ESIA) for projects that are likely to negatively impact the environment. - Set long-range development goals for the municipality and ensure the integration of environmental action plans and concerns into the planning process at the municipality and local levels.

	- Act as a forum for community members to discuss and recommend environmental policies and bylaws. - Collect and disseminate data. - Coordinate activities of local environmental committees. - Mobilize the public to initiate and participate in environmental and natural resource management activities through self-help. - Ensure that the local people, NGOs, the private sector, CBOs, etc., participate in environmental planning and the implementation of environmental programs.
Civil society organizations and community-based organizations[2]	- Raise awareness, consciousness of society about existing and emerging environmental issues. - Filing environmental non/compliances on behalf of the public. - Conduct policy analysis, advocacy and monitoring and evaluation of municipal performance in managing environment. - Participation in environmental management policy processes. - Influence community participation in environment management at the grassroots level. - Resource mobilization (such as human resources, financial resources, etc.). - Lobbying - Community empowerment and building capacity among local communities to take charge in managing their environment. - Advocacy
Private sector	- Promote environment protection through social corporate responsibility and sustainability programs
Local community	- Local environmental knowledge, and attitudes - Participate in environmental programs - Participate in decision making processes.

4.2 Environmental policy implementation and performance

4.2.1 Makindye Ssabagabo's performance on waste and circular economy

Policies: Electronic Waste (E-Waste) Management Policy for Uganda, 2012; Guidelines for E-Waste Management in Uganda, 2016; National Environment (Waste Management) Regulations, 2020; National Strategy for Management of Plastic Pollution (2023-2028); NEMA Guidelines for the Management of Landfills in Uganda, 2020

Despite regulations on waste management, there has been little tangible progress on transitioning to a circular economy. Waste often ends up in open dumpsites, water bodies, or informal landfills, where it poses significant risks to public health, water quality, and the environment. Despite ongoing efforts to improve waste management infrastructure, many areas still lack consistent waste collection services, and waste disposal is largely inefficient and unsustainable. Rapid urbanization in the municipality has further widened the gap between waste generation and management capacity. A significant proportion of the population lives in informal settlements with

limited access to waste disposal services. These challenges are exacerbated by the absence of a national waste management policy that incorporates modern principles like waste segregation and recycling. There is a need to improve coordination and enforcement of legislation on waste management. Additionally, there is a need to incorporate waste management in local land use planning.

4.2.2 Makindye Ssabagabo's performance on air quality

Policies: National Climate Change policy, 2015, National Environment (Air Quality Standards) Regulations, 2024

Air pollution is on the rise despite the existence of air quality regulations. Sources include fumes from vehicles and motor bikes, burning of waste, industrial emissions, indoor pollution, and dust from marram roads. the regulation sets limits for pollutants such as particulate matter (PM), nitrogen oxides (NOx), sulfur dioxide (SO2), and ozone (O3). For instance, the permissible limit for PM10 is set at 150 $\mu g/m^3$, while PM2.5 has a limit of 35 $\mu g/m^3$. There is still limited coordination between Makindye Ssabagabo municipality and NEMA to ensure compliance with set limits. Furthermore, Community engagement initiatives are insufficient to raise awareness about air pollution and its health impacts.

4.2.4 Makindye Ssabagabo's performance on climate change

Policies: National Climate Change Policy, 2015, Uganda Green Growth Development Strategy (UGGDS) 2017/28-2030/31. Despite efforts to achieve climate ambitions, Makindye Ssabagabo municipality needs to decarbonize all sectors of the economy by implementing urgent adaptation actions to manage floods and achieve a climate-resilient future inline with National Adaptation Plans of Action for the relevant sectors. Greenhouse Gas emissions in the municipality are still significantly high. There is a need for the municipality's climate change plan to be integrated across all municipality departments to address climate change.

4.3 Improving environmental performance and policy implementation

- Policy coherence; Municipality plans should be aligned with national environmental goals as a prerequisite to effective implementation. Specific plans, such as the Climate Action Plan, should cut across all departments. For example, high-level integration is needed between land use planning and transport planning to support the move to more sustainable modes of transport
- Capacity building; Continued training of municipal staff to expand the knowledge base on evolving areas such a climate change.
- Regulation and enforcement; There is a need for more rigorous enforcement of existing legislation to protect nature, water, and air quality and to control industrial emissions, noise, and waste management.
- Research Data and evidence: There is also significant focus on enhancing connectivity and engagement between research institutions and the municipality.
- Effective monitoring: Monitoring and reporting on the implementation of plans and programmes has been shown to support better implementation. Publishing monitoring

reports improves public access to information and increases the accountability of implementing bodies.

- Public engagement: promotes engagement, support, and behavioral change from businesses, communities, and citizens. Overcoming the barriers to individual and collective environmental action. For example, local community beliefs and attitudes, and the challenges are incorporated in planning.
- Investment and infrastructure

4.4 Challenges in Environmental Management

4.4.1 Structural challenges

- The limited coordination between the leaders at the different administrative levels, MDAs, and sectors creates disparities in the prioritization of the environmental interventions.
- Poor enforcement of the existing environmental policies and guidelines was cited as a key driver of environmental degradation in the municipality.
- Limited management support and commitment.
- Difficulties in identifying suitable environmental performance indicators.
- Limited enforcement and monitoring.
- Inadequate governmental support.
- Political interference, corruption.

4.4.2 Financial-related

- Limited and inequitable distribution of financial resources to the Natural Resources Department at the municipality affects the effective implementation and monitoring of environmental interventions of action plans and legal provisions.
- The high cost of environmental implementation and maintenance.

4.4.3 Equipment related

- Challenges in collecting accurate and real-time environmental data

4.5 Framework for integrating environmental considerations into local development planning and decision-making processes

4.5.1 Monitoring and Evaluation Framework

The environmental monitoring and evaluation framework is essential for tracking the progress and impact of environmental projects, programs, and policies, highlighting progress, gaps, opportunities and weaknesses. The framework provides a systematic way to collect, analyze, and interpret data, enabling informed decision-making and adaptive management.

The key steps include:

- Identification of the scope and objectives of the project, program or policy.
- Indicator identification: Status, trends of environmental conditions, project activities and achievement of objectives.

- Set targets and Baselines: Targets are specific, measurable, achievable, relevant and time bound to define the desired level of performance for each indicator
- Development of the monitoring plan: The plan outlines, the methods, procedures for collecting, storing and analyzing data specifying the frequency of data collection, the responsible parties and resources required.
- Evaluation: the systematic assessment of the effectiveness, efficiency, relevance, and sustainability of the program including its impact and lessons learned.
- Adaptive management strategy development: This process uses monitoring and evaluation data to adjust program activities and strategies. It allows for flexibility and learning, ensuring that the project remains relevant and effective in the face of changing conditions.
- Stakeholder participation: Stakeholder participation is essential throughout the monitoring and evaluation process. Engaging stakeholders, including local communities, government agencies, and NGOs, ensures that the framework is relevant, transparent, and responsive to their needs and concerns. This can be through participatory workshops to gather local knowledge about environmental conditions and involve community members in data collection activities and share monitoring data with relevant government agencies to inform policy decisions and coordinate conservation efforts.
- Communication: Effective communication of monitoring and evaluation findings is crucial for influencing decision-making and promoting accountability. Reports, presentations, and other communication materials should be clear, concise, and tailored to the general public, and policy makers.
- Data Management and Analysis: Proper data management and analysis ensure the quality and reliability of monitoring and evaluation results. This involves establishing clear data collection protocols, using appropriate statistical methods, and ensuring data security and accessibility.
- Capacity building: Investing in capacity building for municipality staff and stakeholders is crucial for the long-term sustainability of monitoring and evaluation efforts. Training programs, workshops, and mentoring can enhance skills in data collection, analysis, interpretation, and reporting. For example, training workshops, Mentoring and support.

4.5.2 Indicator development and regular assessments

- The Municipality sectoral committees, the technical working group and the environment and natural resources department is mandated with the preparation of annual workplans and budgets and allocate resources for the planned activities and actions.
- The M&E team will be required to monitor and produce reports quarterly to feed into the annual reports
- The Municipality environment and natural resources department is tasked with producing annual progress reports on implementation of environmental actions and activities.
- Production of the MSOER biannually to give updates on the on the status of environment.
- Civil society organizations and the private sector are informed and included in participatory processes that reflect priorities and composition of the local community.
- Recommend a timeline for periodic assessments (annually or bi-annually) to track environmental changes and evaluate the effectiveness of management strategies.
- Facilitate community participation in monitoring efforts, empowering residents to take ownership of local environmental conditions.

4.5.3 Feedback and Adaptive Management

- The feedback and adaptation process will involve gathering and analyzing data from SOER monitoring and feedback systems to support collaborative, timely, and well-informed decision-making to respond to emerging environmental issues effectively. This approach will help achieve meaningful outcomes for stakeholders in a way that is timely, context-specific, and cost-effective.
- The feedback from the monitoring and evaluation will be used to reinforce the implementation of the SOER.
- Measures will be established to ensure environmental compliance with established standards and management plans, including environmental audits in all sectors such as industry, mining, and transport.
- Budgetary instruments such as fines, taxes, tariffs, and environmental levies will be introduced to support environmental management efforts.

5. Chapter Five: Future Outlook

5.1 Outlook components

5.1.1 Identifying trends

Preliminary results and emerging trends on the knowledge, technology, institutions and finances status needed for generating and implementing adaptation measures for the current and future environmental challenges in Makindye Ssabagabo municipality is presented in Table 18). They deal with the available human resources and socio-economics, structural, institutional and technological characteristics of the environmental governance system. Knowledge entails levels of education and awareness about the environment or climate change, its impact, dissemination of information. Technology includes the availability and access to technological choices for adaption including either improvement of an old technology or the introduction of new one. Institutions include governance systems, institutional and legal concerns such as the efficiency and capacities of vital institutions, enforcement of environmental laws, openness of procedures and decision making. Economy includes finances and budget appropriations for the environmental sector.

Table 12: Emerging aspects on the existing and future scenarios for adaptive capacities on environmental management

Adaptive Component	Business as usual scenario (Current situation)	Increased environmental sustainability (Desired future situation)
Knowledge of the technical persons and community members on environmental issues	• Limited knowledge and skills of • Poor attitude and mindset of community members • Low awareness of community members • Low of integration of community perspectives in environmental planning • Limited information on key environmental issues I	• Enhanced environmental information and knowledge sharing • Integration of indigenous knowledge in NRMs • Integration of environmental and climate change curriculum institutions of learning • Co-development and co-implementation of interventions • Knowledge sharing and skills development among actors • Awareness creation and mindset change programs
Technology	• Limited equipment for monitoring • Use of outdated technology and equipment • Low technological knowledge • Low access to technology	• Equipping of sectors and administrative units air, water & noise quality tools • Proper maintenance of tools and technological installations • Improvement of old technology to new technology -
Institutions	• Low enforcement of existing policies • Shortfalls in existing policies	• Effective monitoring and law enforcement • Amendments to the legal and regulatory framework

Adaptive Component	Business as usual scenario (Current situation)	Increased environmental sustainability (Desired future situation)
	• Non-existent or unfunctional by-laws and ordinance • Non-existent or unfunctional local environmental committees' by-laws and ordinance • Bottom-up approaches • Collaboration • Political influence and corruption	• Participatory monitoring and law enforcement • Collaborative and environmental action across actors • Formulation and implementation of by-laws at LC1 levels • Establishment and empowering • Capacity building of stakeholders
Economics	• Low finance resources and budgetary allocation	• Increase finance resources and budgetary allocation • Timely release of funds • Funds mobilization from development partners • Mobilization of revenue through environmental levies and penalties

5.1.2 New developments, and potential risks or opportunities

Potential risks

Some of the potential risks include;

1. **Weak enforcement** due to high levels of impunity undermines the enforcement of laws, regulations, and policies. This can be mitigated by implementing and enforcing environmental penalties and strengthening institutional accountability of the resources under their control. Empower Civil Society and the media to hold public officials accountable.
2. **Contradictory policies both within and across sectors.** This constrains effective enforcement of policies, laws, and regulations on natural resources. This can be mitigated through the harmonization of policies and the strengthening of coordination among MDAs to align national policies with local government frameworks.
3. **Invasion of ecosystems by invasive and alien species** reduces biodiversity and undermines efforts in environmental sustainability. This can be addressed by enhancing monitoring, surveillance, and early warning systems.
4. **Urban refugee influxes** put pressure on the environment. The risk can be mitigated through strengthening partnerships with international organizations such as the IOM, ensuring funding and resettlement of refugees.
5. **The adverse impacts of climate change**, including extreme weather events such as floods, may result in loss of biodiversity, natural resources productivity, and worsen environmental degradation. Strengthen early warning systems and disaster preparedness;

6. **Procurement-related risk** may result in persistent delays in the execution of environmental projects, inefficiencies, and cost overruns. Build capacity in procurement planning and contract management.
7. **Land tenure systems** may result in natural resource conflicts, which may limit access and effective utilization and productivity of natural resources. Land tenure reforms to enforce the cancellation of titles in affected ecosystems. Strengthen the conflict resolution mechanism of Natural resources.

Opportunities

1. High and increasing levels of industrialization
2. The high and increasing number of industries presents an opportunity for a holistic socioeconomic transformation of both livelihoods, communities, and the modernization of agriculture through value addition for both the domestic and regional markets.
3. Tourism, entertainment, and the leisure industry.
4. There is great potential for the tourism sector in the municipality, especially the natural landscape along the shores of Lake Victoria. Energy resources
5. Providing incentivizing clean energy options, such as solar and HEP, has the potential to transition households to clean energy alternatives. However, extension of rural electrification to every household and reduction of HEP tariffs are crucial for abandoning the overuse of wood fuel.
6. Population
7. The high and increasing in number of people in Makindye Ssabagabo municipality will offer an able workforce to steer socio-economic transformation. Creation of funding opportunities and establishment of Technical Vocational Education and Training (TVETs) will be key.
8. Partnerships and regional cooperation
9. Transboundary environmental challenges such as solid waste management, water pollution, and air pollution call for cooperation of stakeholders across the different urban councils and MDAs, in line with SDG 16 which emphasizes cooperation between actors at national, regional and international levels.
10. Infrastructure development
11. The demand for improved infrastructural development, including roads, proper drainage channels, and residential apartments, stems from several factors, including increased population, urbanization, and industrialization. Calls for increased resource allocation from the central government and municipal council administration for the improvement of public infrastructure, such as roads and drainage channels. road points, small culverts, and bridges that flood and become impassable during the rainy season can be replaced with burrow boxes and box culverts.
12. Land tenure and ecosystem degradation
13. The high levels of degradation of natural resources from mining, agriculture, industrialization and settlements call for strategies that will promote the conservation of natural resources on private land tenure, including collaborative forest management and private-public partnerships.
14. Environmental and climate change-related disasters

15. Makindye Ssabagabo Municipality is affected by climate-related disasters such as floods, heatwaves, and windstorms, worsened by rapid urbanization, poor drainage, and inadequate waste management. Strengthening early warning systems, restoring wetlands, and improving community awareness are essential to reducing vulnerability and building resilience. NDP IV priorities emphasize the need for well-planned, resilient urban development. This includes capacity building for the technical persons, including engineers, planners, and the enforcement of urban plans.

6. Chapter Six: Recommendations

6.1 Conclusion

The immense value of Makindye Ssabagabo municipality's environment cannot be taken for granted, and societal progress will be severely hampered if our actions damage this essential asset on which we all depend for our air, water, and resources. There is only one environment, and this integrated assessment of the state of Makindye Ssabagabo municipality's environment shows that the challenges facing it are closely linked and interrelated. Our understanding of these issues is greater than ever before and is underpinned by comprehensive monitoring systems and research.

Some progress is being made. We need to build momentum to deliver not only on our climate goals but also on our biodiversity, circular economy, and pollution reduction goals. Delivering on these goals will require transformational change in many of the core systems of our society and, consequently, will involve everyone in providing a healthy environment for future generations. In Makindye Ssabagabo municipality, we need to seek to be a generation that changes the path we are on and delivers a municipality where our major environmental challenges have been solved.

6.2 Policy recommendations

Priority actions on Climate change

- Promote integrated urban planning: Incorporate climate adaptation and disaster risk reduction into municipal development plans to guide sustainable land use and infrastructure development.
- Invest in climate-resilient infrastructure: Construct flood control systems, green spaces, and sustainable drainage networks to reduce the impact of heavy rains and urban heat.
- Enhance early warning and emergency preparedness: Establish local early warning systems and provide community training on disaster response and risk management.
- Support sustainable livelihoods: Promote climate-smart income-generating activities such as urban farming, recycling businesses, and renewable energy enterprises to reduce socio-economic vulnerability.
- Strengthen environmental education and public awareness: Conduct continuous community sensitization on climate change impacts and environmental stewardship to encourage behavior change and participation.
- Restore and protect natural ecosystems: Rehabilitate wetlands, plant trees, and enforce environmental regulations to preserve natural buffers against climate shocks.
- Improve waste management systems: Expand solid waste collection and recycling initiatives to reduce pollution and methane emissions, especially in informal settlements.
- Foster multi-stakeholder partnerships: Engage NGOs, private sector actors, and development partners to mobilize technical and financial resources for climate resilience initiatives.

Priority actions on biodiversity

Effective conservation requires robust and enforceable legal instruments. In Makindye Ssabagabo, the priority lies in strengthening the legal framework and resolving longstanding land use conflicts.

- Review and Regularize Land Titles: Many land titles within wetlands predate the 1995 National Environment Management Authority (NEMA) Act, creating legal ambiguities that complicate enforcement. A review process is needed to identify, audit, and potentially revoke or compensate titles that contravene current environmental laws. This legal clarity is essential for effective restoration and enforcement.

Priority actions on waste management

- Introduce new policy instruments, including economic instruments, to promote prevention and make reusing and recycling more economically attractive.
- Increase waste collection and recycling rates by involving more private sector players in waste management.
- Shift reusable and recyclable waste away from incineration and landfilling.
- Prepare a waste management plan in line with national legislation.

Priority actions on Water resources

- Urgently ensure appropriate controls over industrial water abstraction permits and hydro morphological changes in compliance with national legislation.
- Follow up on monitoring of hotspot areas that show increasing nitrate concentrations.
- Construct the necessary infrastructure to protect water courses and wetlands from urban waste accumulation.
- Improve the coordination of the implementation of water and nature policies and plans in the municipality.
- Improve monitoring efforts of hydro-morphological conditions, of surface, wetlands, and groundwater quality and quantity.

References

Oosterveer, P., & Van Vliet, B. (2010). Environmental systems and local actors: decentralizing environmental policy in Uganda. Environmental Management, 45, 284-295.

Tukahirwa, J. T., Mol, A. P., & Oosterveer, P. (2010). Civil society participation in urban sanitation and solid waste management in Uganda. Local Environment, 15(1), 1-14.

Makindye Ssabagabo Municipal Council. (2019). *Physical Development Plan 2019*. Retrieved from https://gipeafricaltd.com/gipeafricaltd.com

Uganda Bureau of Statistics 2024: The National Population and Housing Census 2024 – Final Report - Volume 1 (Main), Kampala, Uganda.

https://www.ubos.org/wpcontent/uploads/statistics/Revised_Subcounty_population_2015_2030_146_Districts.xlsx

Uganda Bureau of Statistics 2024: The National Population and Housing Census 2024 – Final Report - Volume 1 (Main), Kampala, Uganda.

Makindye Ssabagabo Municipal Council. (2019). *Physical Development Plan 2019*. Retrieved from https://gipeafricaltd.com/gipeafricaltd.com

Makindye Ssabagabo Municipal Council https://msabagabo.go.ug/community-based-services

G.Pearson,C.Barratt,J. Seeley,A. Ssetaala,Georgina Nabbagala,G. Asiki,2013. Making a livelihood at the fish-landing site:Exploring the pursuit of economic independence amongst Ugandan women.

J. K Kamara,Barbara Mirembe Namugambe,R.Egessa,Gilbert Kamanga,A.Renzaho, 2019.The socioeconomic and Sexual Health Status of Young People Living in Urban Slum Areas of Kampala, Uganda

Uganda Bureau of Statistics, 2021

Uganda Bureau of Statistics 2024: The National Population and Housing Census 2024 – Final Report - Volume 1 (Main), Kampala, Uganda.

A. Richmond, Ian Myers, Hasifa Namuli,2018. Urban Informality and Vulnerability: A case Study in Kampala, Uganda.

O. Brousse, H. Wouters, M. Demuzere, W.Thiery,Jonas Van de Walle,N.Lipzig ,2020.The local climate impact of an African city during clear sky conditions-Implications of the recent urbanization in Kampala (Uganda)

R. White, Chyi-Yun Huang, Herbert Oule, Martin Onyach-Olaa, Jan Bachmann, D. Dale, Brian Goldberg, Maritza..., 2015.Promoting green urban development in African cities: Kampala, Uganda-urban environmental profile

R. Sliuzas, J. Flacke, V. Jetten,2013. Modelling urbanization and flooding in Kampala,Uganda.

S Lwasa,Shuaib J.B Nyakaana,S. Hannington,2005. Assessment of Environmental Impacts of Land Use Land Cover Change: A challenge for Urban Planning in Metropolitan Kampala.

S Mukiibi, 2011.The Effect of Urbanization on the Housing conditions of the Urban Poor in Kampala,Uganda.

ArchDaily. (2016). *Vertical Housing in Kampala*. https://www.archdaily.com

Gipea Africa Limited. (2019). *Wakiso District Physical Development Plan 2018–2040*. https://gipeafricaltd.com

International Water Management Institute. (2023). *Revive and Restore Degraded Wetlands*. https://www.iwmi.cgiar.org

Kampala Capital City Authority (KCCA). (2020). *Urban Agriculture and Waste Management Initiatives Report*. Kampala, Uganda.
Makindye Ssabagabo Municipal Council. (2022). *Environmental and Agricultural Land Use Trends Report*. Kampala: MSSMC.
Makindye Ssabagabo Municipal Council. (2022). *Physical Planning and Informal Settlement Management Report*. Kampala: MSSMC.
Ministry of Lands, Housing and Urban Development (MLHUD). (2019). *Participatory Land Use Planning Manual*. Kampala, Uganda.
Ministry of Lands, Housing and Urban Development. (2017). *National Urban Policy*. Retrieved from https://mlhud.go.ug/
Ministry of Works and Transport. (2019). *Annual Infrastructure Sector Performance Report*. Kampala: MoWT.
National Environment Management Authority (NEMA). (2018). *State of the Environment Report*. Kampala, Uganda.
National Environment Management Authority (NEMA). (2020). *Restoration and Protection of Wetlands in Uganda*. Kampala, Uganda.
National Environment Management Authority (NEMA). (2020). *State of the Environment Report for Uganda 2018–2019*. Retrieved from https://www.nema.go.ug/
National Environment Management Authority (NEMA). (2021). *State of the Environment Report for Uganda*. Retrieved from https://www.nema.go.ug
National Planning Authority. (2022). *Physical Development Plan for Makindye Ssabagabo Municipality*. Retrieved from https://nppb.go.ug
National Planning Authority. (2022). Uganda's Urban Resilience and Disaster Risk Management Report. Kampala: NPA.
Protect Your Environment Now! (2013). *Community Sensitization on Hygiene and Sanitation*. https://environment-sanitation.blogspot.com
The Namibian. (2023). *New Sand Mining Regulations*. https://www.namibian.com.na
Uganda Bureau of Statistics (UBOS). (2017). *National Population and Housing Census 2014 – Urbanization Analytical Report*. Retrieved from https://www.ubos.org
Uganda Bureau of Statistics (UBOS). (2020). *National Population and Housing Census 2014 – Area Specific Profiles*. Retrieved from https://www.ubos.org/
WaterAid Uganda. (2021). Water, Sanitation, and Hygiene Assessment in Urban Uganda. Kampala: WaterAid.
AirQo. (n.d.). *AirQo air quality monitoring platform*. Makerere University. https://www.airqo.net

AirQo. (n.d.). *Resources and tools for air quality management*. https://airqo.net/resources

Challenge Works. (2024). *Makindye Ssabagabo: Climate Smart Cities Challenge*. https://challengeworks.org/challenge-prizes/makindye-ssabagabo-climate-smart-cities-challenge

IQAir. (2024). *Seguku, Makindye Ssabagabo air quality index (AQI)*. https://www.iqair.com/uganda/central-region/kajansi/seguku-makindye-ssabagabo

National Physical Planning Board. (2023). *Makindye Ssabagabo Physical Development Plan Report*. https://nppb.go.ug/docs/makindye-ssabagabo-pdp-report

UN-Habitat. (2024). *World Cities Report 2024: Case study annex*. Shift Cities. https://www.shiftcities.org/sites/default/files/2024-12/world_cities_report_2024_case_study_annex.pdf

Intergovernmental Panel on Climate Change (IPCC). (2022). Climate Change 2022: Impacts, Adaptation and Vulnerability. Retrieved from https://www.ipcc.ch/report/ar6/wg2/

UN-Habitat. (2021, November 3). New city-focused climate innovation competition launched. Retrieved from https://unhabitat.org/news/03-nov-2021/new-city-focused-climate-innovation-competition-launched

UN-Habitat. (2024). World Cities Report 2024: Cities and Climate Action. Retrieved from https://unhabitat.org/sites/default/files/2024/11/world_cities_report_2024_case_study_annex.pdf

Uganda Ministry of Finance, Planning and Economic Development. (2024/25). Makindye Ssabagabo Municipal Council Local Government Quarterly Performance Report Quarter 1. Retrieved from https://budget.finance.go.ug/sites/default/files/Indivisual%20LG%20Budgets/Makindye%20%20Ssebagabo%20Mc.pdf

WaterAid Uganda. (n.d.). Empowering Change - Sarah Namutebi Transforming Community WASH in Makindye Ssabagabo. Retrieved from https://www.wateraid.org/ug/stories/empowering-change-sarah-namutebi-transforming-community-wash-in-makindye-ssabagabo

GIZ (2022). Pre-feasibility study of municipal organic waste recovery alternatives in Makindye Ssabagabo, Kira, Nansana, and Entebbe. GFA Consulting Group and Resources & Waste Advisory Group

Global Green Growth Institute. (2025). Greater Kampala Metropolitan Area Waste Situational Analysis. Retrieved from https://gggi.org/wp-content/uploads/2025/04/GKMA-Waste-Situational-Analaysis.pdf

Ministry of Water and Environment. (2014). The National Environment Management Policy for Uganda. Government of Uganda.

WaterAid Uganda. (n.d.). Empowering change: Sarah Namutebi transforming community WASH in Makindye Ssabagabo. Retrieved from https://www.wateraid.org/ug/stories/empowering-change-sarah-namutebi-transforming-community-wash-in-makindye-ssabagabo

Godoi, F. C., Prakash, S., & Bhandari, B. R. (2021). Final report Final report. Review of 3D Printing and Potential Red Meat Applications, 23 February 2021, 1–61. http://libdcms.nida.ac.th/thesis6/2010/b166706.pdf

UN-HABITAT. (2021). Makindye Ssabagabo challenge brief. 1, 1–42.

NEMA. (2000). The National Environment (Wetlands, Riverbanks, and Lake Shores Management) Regulations, No. 3/2000. Statutory Instruments, 2000(3), 24. http://www.nemaug.org/images/laws/Wetlands, Riverbanks and Lakeshore Management Regulations.pdf

Kadoma, A., Perry, M., & Renaud, F. G. (2025). Stakeholders' perceptions of wetland conservation and restoration in Wakiso District, Uganda. Environment, Development and Sustainability, 27(2), 3177-3200.

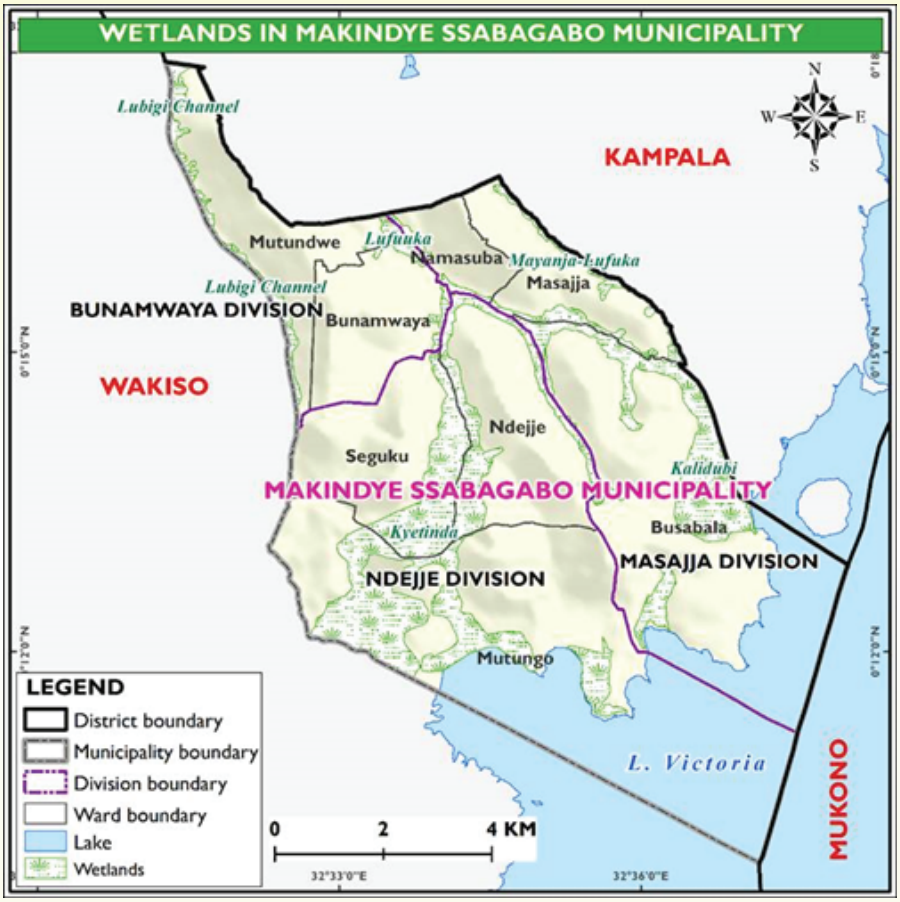
Warsame, A., Luyiga, S., & Akiyode, O. (2022). Assessing Wetland Degradation in a Growing Urban Area: The Case of Nsooba in Kampala, Uganda. J. Eng. Sci. Technol., 1(1), 1-6.

Kamukama, E. G., Okrah, A., Genesis, M., Mensah, C., Donnata, A., Prempeh, N. A., ... & Sarfo, I. (2024). Comprehensive analysis and quantitative assessment of land use/land cover dynamics in Wakiso and Kampala, Uganda: a multi-decadal remote sensing study.

Abel, W., & Moses, W. The Challenges and Way Forward of Pollution: A Case Study of Makindye Division, Kampala Capital City.
Tumwine, F. R., Bamutaze, Y., & Opedes, H. (2018). Demographic Shifts Influencing the Political Landscape of the Lake Victoria Basin in Uganda.
Makindye Ssabagabo Municipal Council. (2022). *Physical Planning and Informal Settlement Management Report*. Kampala: MSSMC.
Kampala Capital City Authority (KCCA). (2022). Environmental and social management report: Urban drainage and wetland encroachment in Kampala Metropolitan. Kampala: KCCA.
Kampala Capital City Authority (KCCA). (2022). Solid Waste Management Strategy for Greater Kampala Metropolitan Area. https://www.kcca.go.ug
Kampala Capital City Authority. (2022). Sanitation and Drainage Challenges in Kampala's Urban Settlements. KCCA Press.
Makindye Ssabagabo Division Council. (2021). Annual Local Government Performance and Resource Use Report. Kampala: MSSDC.
Makindye Ssabagabo Municipal Council. (2023). Municipal Infrastructure and Environmental Response Report 2023.
Mugerwa, R., Kansiime, F., & Ntale, M. (2021). Impacts of urbanization and solid waste management on water quality in Kampala wetlands. African Journal of Environmental Science and Technology, 15(3), 92–102. https://doi.org/10.5897/AJEST2021.2984
National Environment Management Authority (NEMA). (2021). State of the Environment Report for Uganda 2020/2021. https://www.nema.go.ug
National Environment Management Authority (NEMA). (2021). Urban environmental challenges: Solid waste and pollution in Wakiso District. Kampala: NEMA.
National Environment Management Authority (NEMA). (2021). Wetland degradation and restoration framework in urban areas: Case studies from Wakiso District. Kampala: NEMA.
National Environment Management Authority (NEMA). (2023). State of the Environment Report for Uganda 2022/2023. https://www.nema.go.ug
National Environment Management Authority. (2021). Environmental Monitoring Report on Urban Pollution.
National Planning Authority. (2022). Physical Development Plan for Makindye Ssabagabo Municipality. Retrieved from https://nppb.go.ug
National Planning Authority. (2022). Uganda's Food and Nutrition Security Report 2022. Kampala: NPA.
National Planning Authority. (2022). Uganda's Urban Resilience and Disaster Risk Management Report. Kampala: NPA.
National Water and Sewerage Corporation. (2022). Wastewater Management Progress and Infrastructure Expansion Plan 2020–2022. Kampala: NWSC.
Tumwebaze, H. K., & Nyenje, P. M. (2020). Urban solid waste and pollution dynamics in peri-urban wetlands of Kampala. Water Practice and Technology, 15(2), 347–359. https://doi.org/10.2166/wpt.2020.030
Wandiga, S. O., Owuor, D. T. O., & Omondi, B. (2021). Water pollution trends and impacts in East Africa: Case studies from Uganda and Kenya. International Journal of Environmental Research and Public Health, 18(5), 2713. https://doi.org/10.3390/ijerph18052713
WaterAid Uganda. (2020). Agricultural runoff and water quality: A case study of peri-urban farming zones in Wakiso District. Kampala: WaterAid Uganda.

WaterAid Uganda. (2020). Sanitation and waste management impacts on water quality in urban wetlands: A case of Makindye Ssabagabo. Kampala: WaterAid Uganda.
WaterAid Uganda. (2020). Water quality and sanitation baseline study in Wakiso District. Kampala: WaterAid Uganda.
WaterAid Uganda. (2021). Water, Sanitation, and Hygiene Assessment in Urban Uganda.
WaterAid Uganda. (2023). The Sanitation Problem: A Glance at Community-led Solutions. Retrieved from https://www.wateraid.org/ug
WaterAid Uganda. (2023). Urban WASH and Resilience Report: Addressing Water and Sanitation Challenges in the Greater Kampala Metropolitan Area. Kampala: WaterAid Uganda.
Wetlands Management Department. (2020). Impacts of farming and extraction activities on wetland systems in Greater Kampala. Ministry of Water and Environment.
Wetlands Management Department. (2022). Annual Wetland Status Report. Ministry of Water and Environment, Uganda.
World Health Organization. (2020). Guidelines on Sanitation and Health. Geneva: WHO of Water and Environment.
Langdale-Brown, I., Osmaston, H.A. and Wilson, J.G. 1964. *The Vegetation of Uganda, and its Bearing on Landuse.* Government Printer, Entebbe.

Godoi, F. C., Prakash, S., & Bhandari, B. R. (2021). Final report Final report. Review of 3D Printing and Potential Red Meat Applications, 23 February 2021, 1–61. http://libdcms.nida.ac.th/thesis6/2010/b166706.pdf
UN-HABITAT. (2021). Makindye Ssabagabo challenge brief. 1, 1–42.
NEMA. (2000). The National Environment (Wetlands, Riverbanks, and Lake Shores Management) Regulations, No. 3/2000. Statutory Instruments, 2000(3), 24. http://www.nemaug.org/images/laws/Wetlands, Riverbanks and Lakeshore Management Regulations.pdf
Kadoma, A., Perry, M., & Renaud, F. G. (2025). Stakeholders' perceptions of wetland conservation and restoration in Wakiso District, Uganda. Environment, Development and Sustainability, 27(2), 3177-3200.
Warsame, A., Luyiga, S., & Akiyode, O. (2022). Assessing Wetland Degradation in a Growing Urban Area: The Case of Nsooba in Kampala, Uganda. J. Eng. Sci. Technol., 1(1), 1-6.
Kamukama, E. G., Okrah, A., Genesis, M., Mensah, C., Donnata, A., Prempeh, N. A., ... & Sarfo, I. (2024). Comprehensive analysis and quantitative assessment of land use/land cover dynamics in Wakiso and Kampala, Uganda: a multi-decadal remote sensing study.
Abel, W., & Moses, W. The Challenges and Way Forward of Pollution: A Case Study of Makindye Division, Kampala Capital City.
Tumwine, F. R., Bamutaze, Y., & Opedes, H. (2018). Demographic Shifts Influencing the Political Landscape of the Lake Victoria Basin in Uganda.
Makindye Ssabagabo Municipal Council. (2022). *Physical Planning and Informal Settlement Management Report*. Kampala: MSSMC.



for more information please contact us
Makindye Ssabagabo Municipal Council Resource Pool
P.O.Box 1872, Kampala

GREATER KAMPALA METROPOLITAN AREA URBAN DEVELOPMENT PROGRAM

THE WORLD BANK